Engineering Mathematics - I · Unit_04 · Unit 4

DRAFT — UE26MA141A · Unit_04 — Unit 4 · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26MA141A · Unit_04: Unit 4
2. 2 · Gamma functions - definition, graph and properties
3. 3 · Definition of Beta function and its properties
4. 4 · Relation between Beta and Gamma functions, Duplication formula
5. 5 · Problems on Beta an Gamma functions
6. 6 · Problems on Beta and Gamma Function_continued
7. 7 · Series solution of Besselâ??s differential equation, Bessel functions
8. 8 · Source slide deck: introduction to Series solution
9. 9 · Source slide deck: Power series Continued
10. 10 · Source slide deck: Frobenius Method
11. 11 · Source slide deck: Frobenius method Continued
12. 12 · Source slide deck: Bessel functions of order n and n
13. 13 · Historical question-source pages · unit mapping candidates
14. Appendix A · Symbol glossary
15. Appendix B · Sources, coverage and gaps
16. Appendix C · Revision and retrieval practice

M1

Unit_04 · Unit 4

1 UE26MA141A · Unit_04: Unit 4

This unit master organizes source material for Engineering Mathematics - I, Unit_04, officially titled Unit 4. It includes 6 video transcripts and 11 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26MA141A, Unit_04 (Unit 4). Bucket folder present: yes. Videos: 6; slide decks: 11.

EV0000001

2 Gamma functions - definition, graph and properties

This lesson is indexed from the source video titled Gamma functions - definition, graph and properties. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams

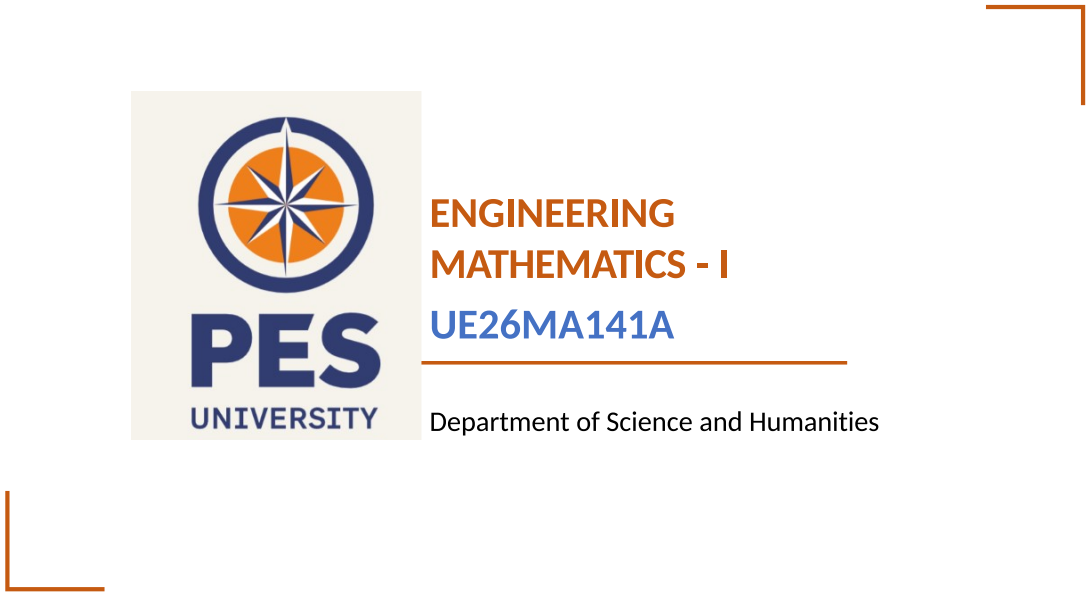


Figure G000001. Gamma Function · source slide page 1.

EV0000095

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

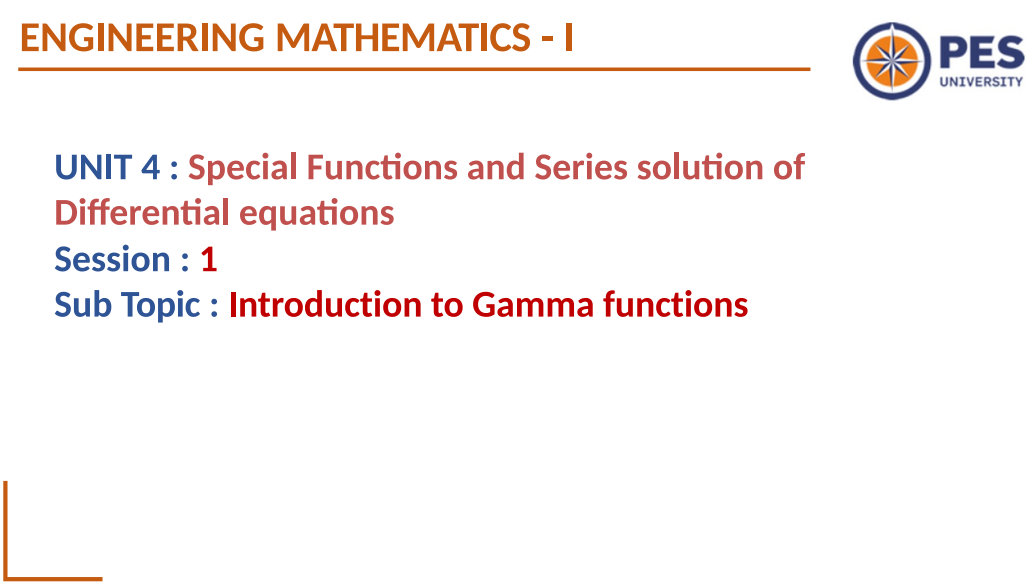


Figure G000002. Gamma Function · source slide page 2.

EV0000098

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

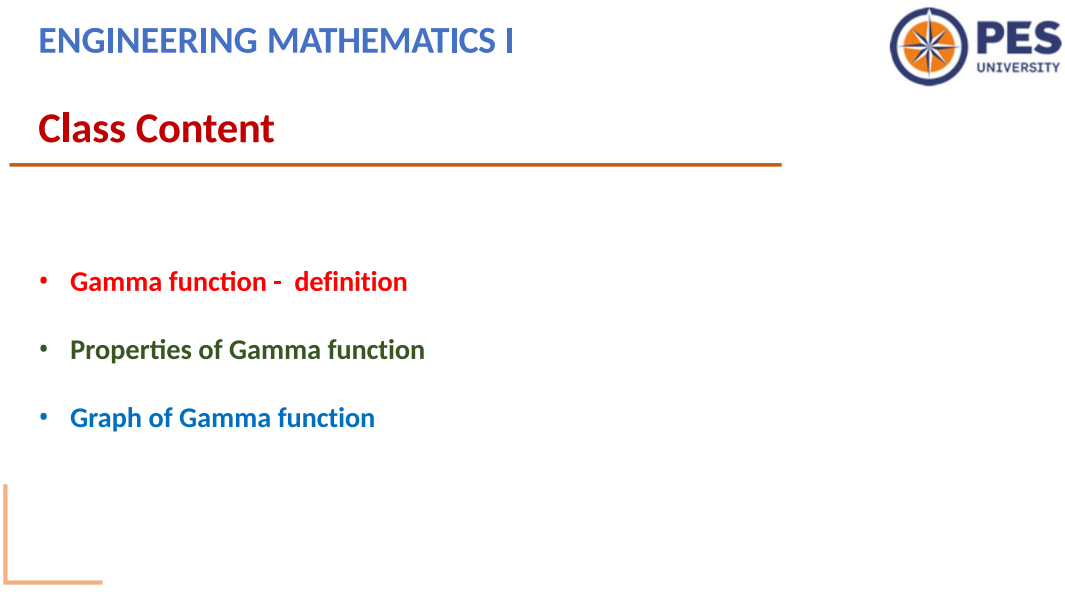


Figure G000003. Gamma Function · source slide page 3.

EV0000101

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

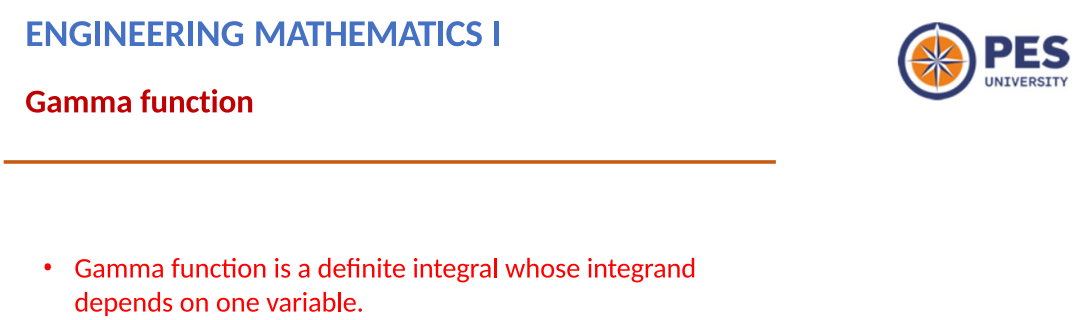


Figure G000004. Gamma Function · source slide page 4.

EV0000104

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

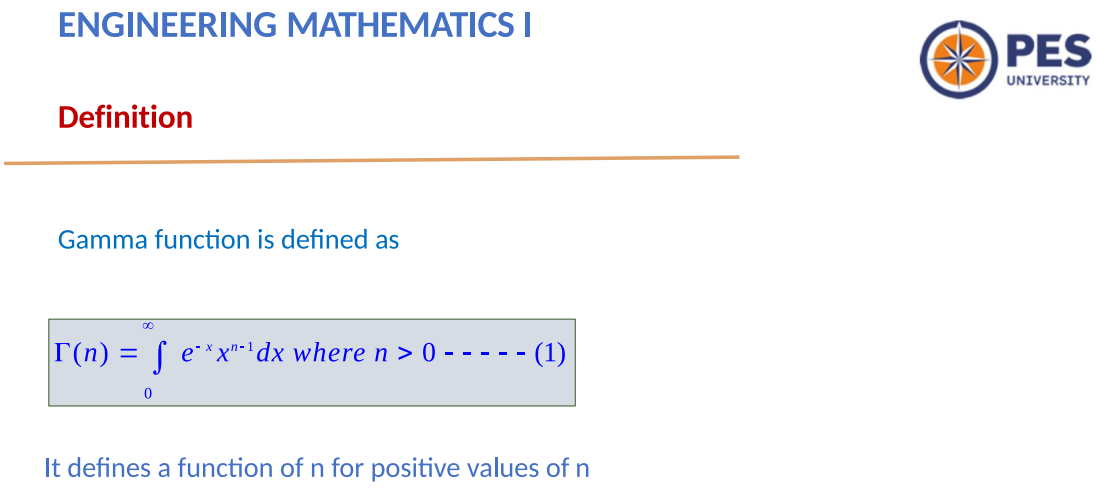


Figure G000005. Gamma Function · source slide page 5.

EV0000107

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

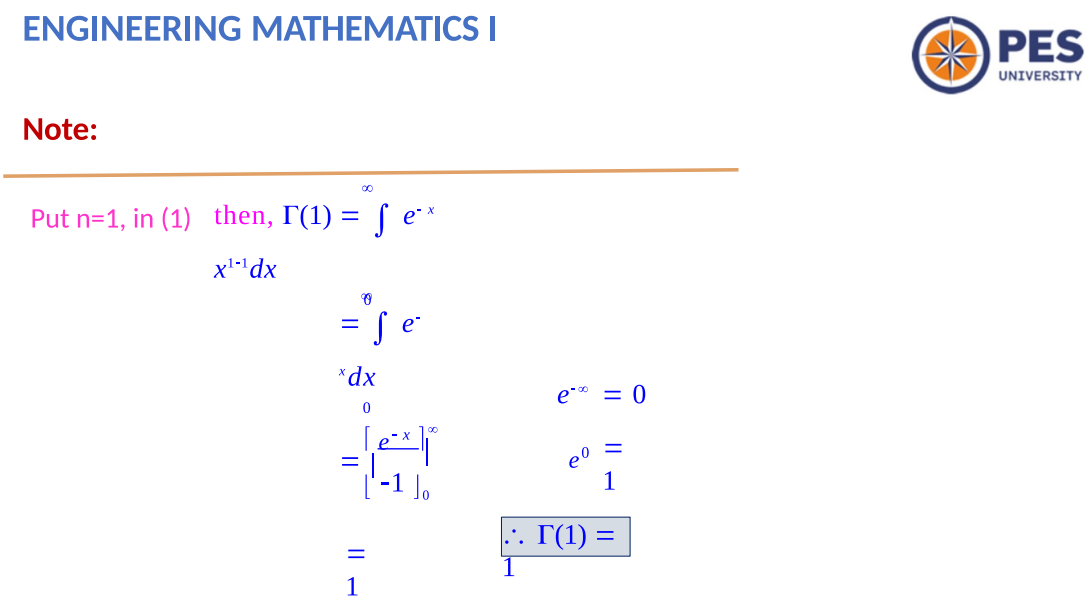


Figure G000006. Gamma Function · source slide page 6.

EV0000110

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

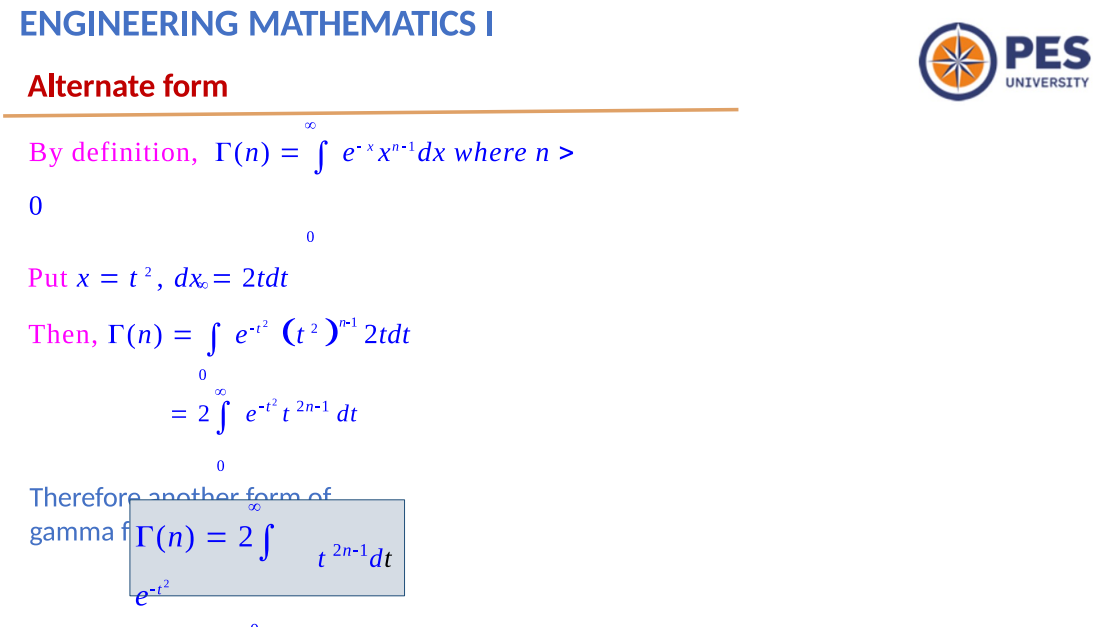


Figure G000007. Gamma Function · source slide page 7.

EV0000113

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

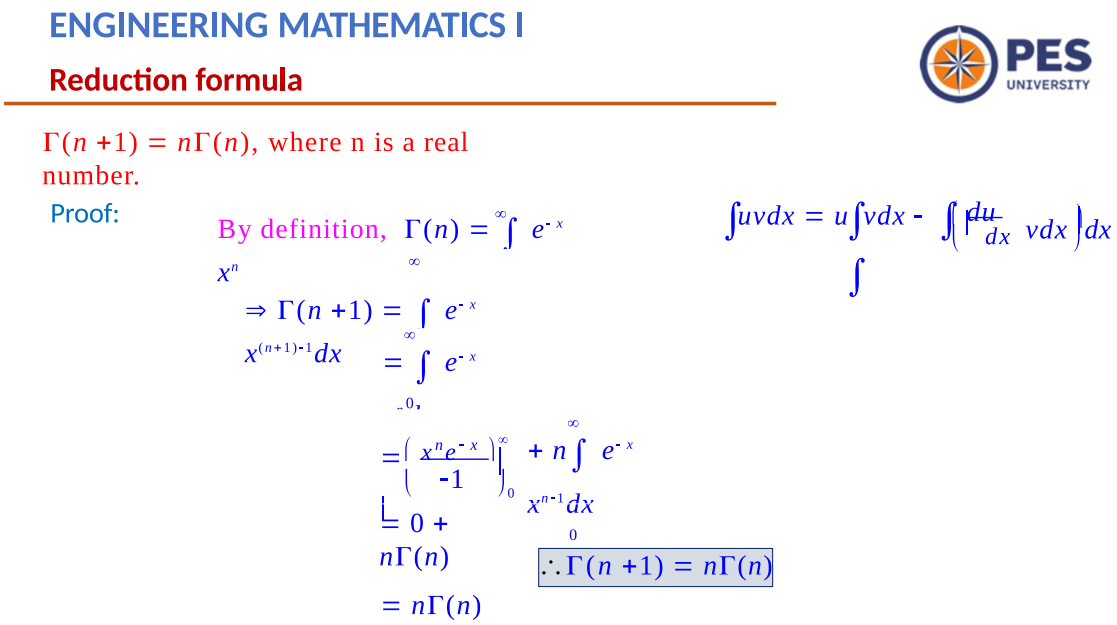


Figure G000008. Gamma Function · source slide page 8.

EV0000116

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

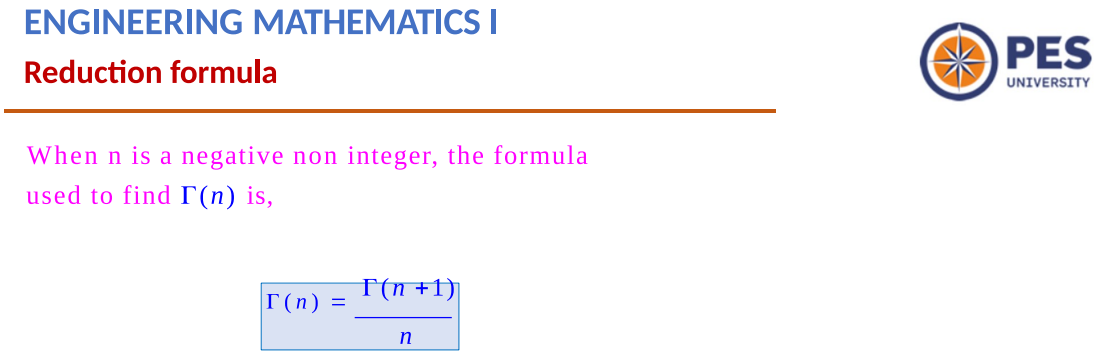


Figure G000009. Gamma Function · source slide page 9.

EV0000119

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

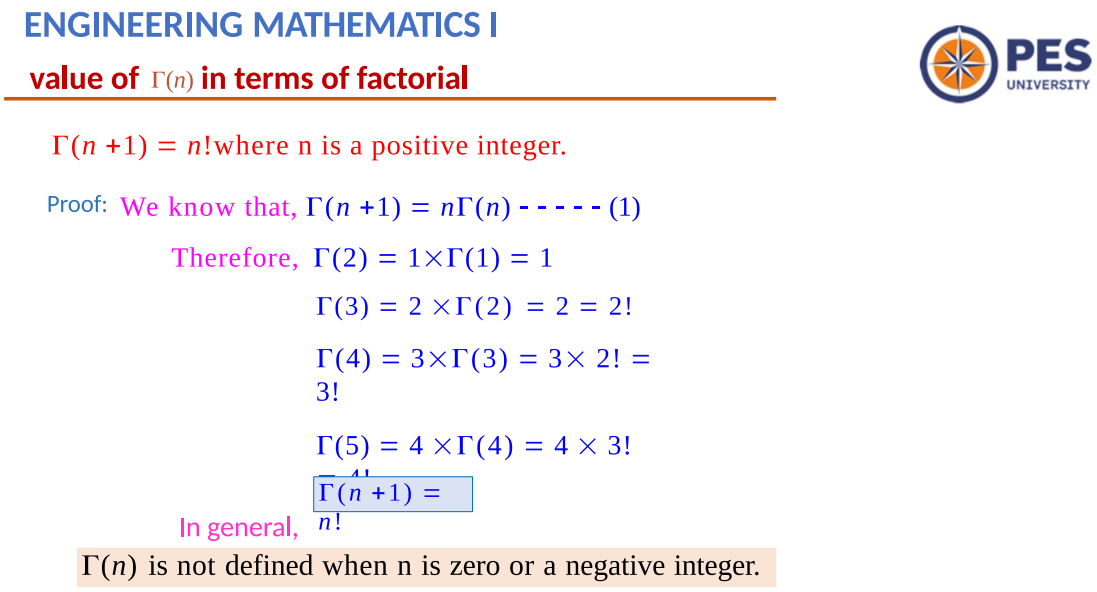


Figure G000010. Gamma Function · source slide page 10.

EV0000122

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

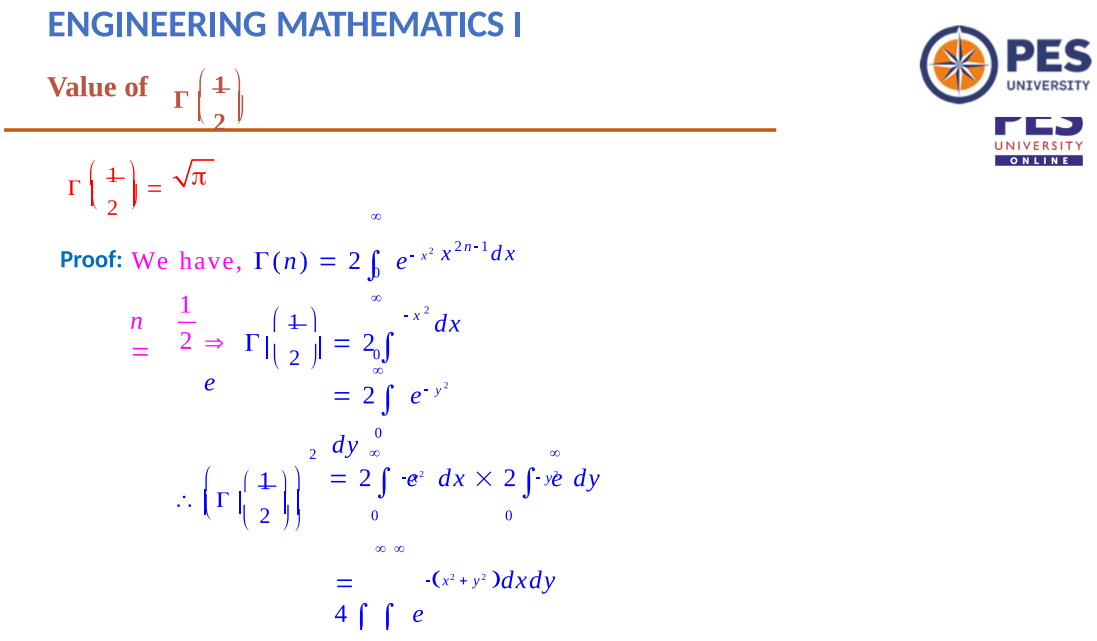


Figure G000011. Gamma Function · source slide page 11.

EV0000125

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

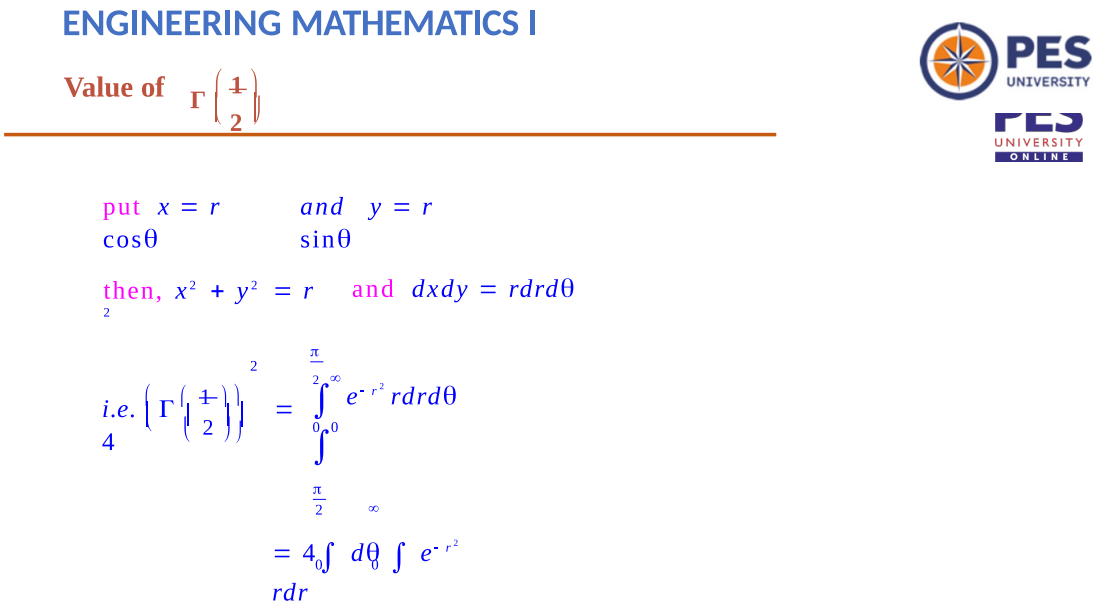


Figure G000012. Gamma Function · source slide page 12.

EV0000128

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

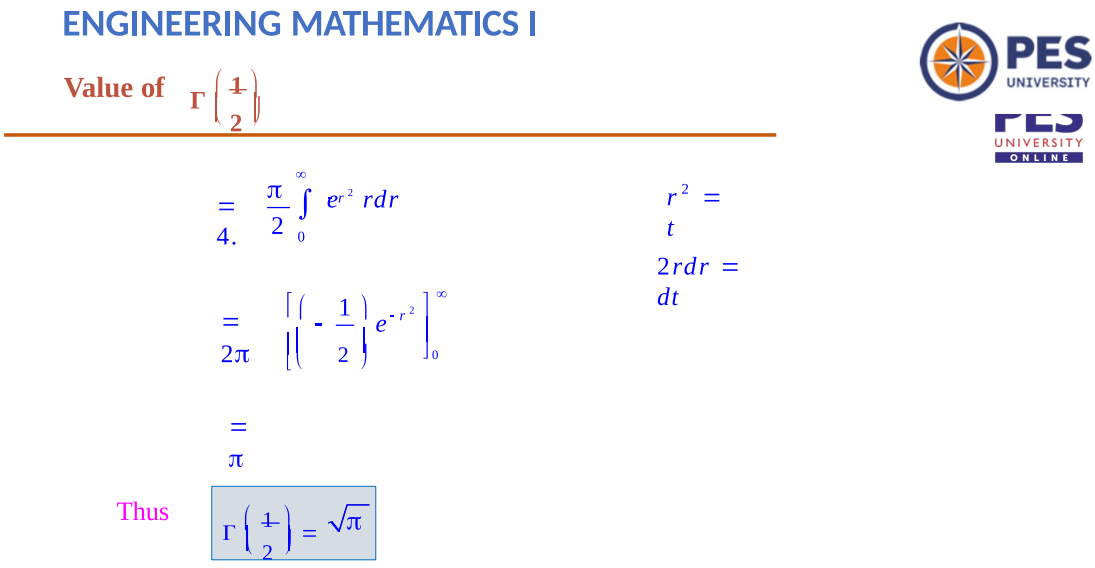


Figure G000013. Gamma Function · source slide page 13.

EV0000131

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS I

Table for Gamma Function

n	$\Gamma(n)$	n	$\Gamma(n)$	n	$\Gamma(n)$	n	$\Gamma(n)$
1.00	1.00000	1.25	0.90640	1.50	0.88623	1.75	0.91906
1.01	0.99433	1.26	0.90440	1.51	0.88659	1.76	0.92137
1.02	0.98884	1.27	0.90250	1.52	0.88704	1.77	0.92376
1.03	0.98355	1.28	0.90072	1.53	0.88757	1.78	0.92623
1.04	0.97844	1.29	0.89904	1.54	0.88818	1.79	0.92877
1.05	0.97350	1.30	0.89747	1.55	0.88887	1.80	0.93138
1.06	0.96874	1.31	0.89600	1.56	0.88964	1.81	0.93408
1.07	0.96415	1.32	0.89464	1.57	0.89049	1.82	0.93685
1.08	0.95973	1.33	0.89338	1.58	0.89142	1.83	0.93969
1.09	0.95546	1.34	0.89222	1.59	0.89243	1.84	0.94261
1.10	0.95135	1.35	0.89115	1.60	0.89352	1.85	0.94561
1.11	0.94739	1.36	0.89018	1.61	0.89468	1.86	0.94869
1.12	0.94359	1.37	0.88931	1.62	0.89592	1.87	0.95184
1.13	0.93993	1.38	0.88854	1.63	0.89724	1.88	0.95507
1.14	0.93642	1.39	0.88785	1.64	0.89864	1.89	0.95838
1.15	0.93304	1.40	0.88726	1.65	0.90012	1.90	0.96177
1.16	0.92980	1.41	0.88676	1.66	0.90167	1.91	0.96523
1.17	0.92670	1.42	0.88636	1.67	0.90330	1.92	0.96878
1.18	0.92373	1.43	0.88604	1.68	0.90500	1.93	0.97240
1.19	0.92088	1.44	0.88580	1.69	0.90678	1.94	0.97610
1.20	0.91817	1.45	0.88565	1.70	0.90864	1.95	0.97988
1.21	0.91558	1.46	0.88560	1.71	0.91057	1.96	0.98374
1.22	0.91311	1.47	0.88563	1.72	0.91258	1.97	0.98768
1.23	0.91075	1.48	0.88575	1.73	0.91466	1.98	0.99171
1.24	0.90852	1.49	0.88595	1.74	0.91683	1.99	0.99581
						2.00	1.00000

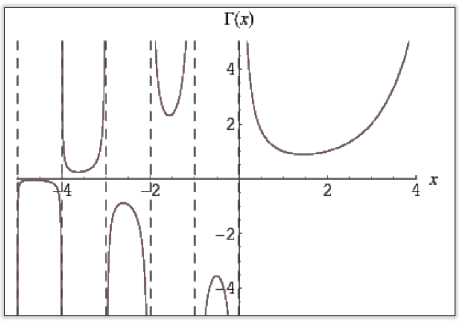
Figure G000014. Gamma Function · source slide page 14.

EV0000134

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS I

Graph Of Gamma Function



Graph of Gamma Function

$$\Gamma(n+1) = n\Gamma(n)$$

$$\Gamma(n) = \overline{\Gamma(n+1)}$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Figure G000015. Gamma Function · source slide page 15.

EV0000137

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000016. Gamma Function · source slide page 16.

EV0000140

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000094 · EV0000095 · EV0000096
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 4 : Special Functions and Series solution of Differential equations Session : 1 Sub Topic : Introduction to Gamma functions EV0000097 · EV0000098 · EV0000099
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I Class Content Gamma function - definition Properties of Gamma function Graph of Gamma function EV0000100 · EV0000101 · EV0000102
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I Gamma function Gamma function is a definite integral whose integrand depends on one variable. It is also known as Euler's integral of second kind. It is the generalization of factorial notation from integer values to real numbers. EV0000103 · EV0000104 · EV0000105
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I Definition Gamma function is defined as ☒ ☒(n) ☒ ☒ e☒ x xn☒1dx where n ☒ 0 ☒ ☒ ☒ ☒ ☒ (1) 0 It defines a function of n for positive values of n EV0000106 · EV0000107 · EV0000108
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I Note: ☒ then, ☒(1) ☒ ☒ e☒ x x1☒1dx 0 Put n=1, in (1) ☒ ☒ ☒ e☒ xdx 0 e☒☒ ☒ 0 ☒ e☒ x ☒☒ e0 ☒ ☒ ☒ 1 ☒ ☒ ☒1 ☒0 ☒ ☒(1) ☒ 1 ☒ 1 EV0000109 · EV0000110 · EV0000111

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I Alternate form ☒ By definition, ☒(n) ☒ ☒ e☒ x xn☒1dx where n ☒ 0 0 Put x ☒ t 2 , dx ☒ 2tdt Then, ☒(n) ☒ ☒ e☒t2 ☒t 2 ☒n☒1 2tdt 0 ☒ ☒ 2☒ e☒t2 t 2n☒1 dt 0 Therefore another form of gamma function is, ☒ ☒ ☒(n) ☒ 2☒ e☒t2 0 t 2n☒1dt EV0000112 · EV0000113 · EV0000114

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I Reduction formula ☒(n ☒1) ☒ n☒(n), where n is a real number. ☒ By definition, ☒(n) ☒ ☒ e☒ x xn☒1dx 0 ☒ du ☒ ☒ ☒☒ dx ☒ Proof: uvdx ☒ u vdx ☒ vdx ☒dx ☒ ☒ ☒ ☒ ☒(n ☒1) ☒ ☒ e☒ x x(n☒1)☒1dx 0 ☒ ☒ ☒ ☒ e☒ x xndx 0 ☒ xne☒ x ☒☒ ☒ ☒ n☒ e☒ x xn☒1dx 0 ☒ ☒ ☒ ☒1 ☒ ☒0 ☒ 0 ☒ n☒(n) ☒ n☒(n) ☒☒(n ☒1) ☒ n☒(n) EV0000115 · EV0000116 · EV0000117

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I Reduction formula When n is a negative non integer, the formula used to find ☒(n) is, ☒(n) ☒ ☒(n ☒1) n This formula is used to compute☒(n), when n is a negative non integer. EV0000118 · EV0000119 · EV0000120

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I value of ☒(n) in terms of factorial ☒(n ☒1) ☒ n!where n is a positive integer. Proof: We know that, ☒(n ☒1) ☒ n☒(n) ☒ ☒ ☒ ☒ ☒ (1) Therefore, ☒(2) ☒ 1☒☒(1) ☒ 1 ☒(3) ☒ 2 ☒☒(2) ☒ 2 ☒ 2! ☒(4) ☒ 3☒☒(3) ☒ 3☒ 2! ☒ 3! ☒(5) ☒ 4 ☒☒(4) ☒ 4 ☒ 3! ☒ 4! In general, ☒(n ☒1) ☒ n! ☒(n) is not defined when n is zero or a negative integer. EV0000121 · EV0000122 · EV0000123

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I Γ ☒ 1 ☒ Value of ☒ 2 ☒ ☒ ☒ ☒☒ 1 ☒ ☒ ☒ ☒ 2 ☒ ☒ ☒ ☒ Proof: We have, ☒(n) ☒ 2☒ e☒ x2 x2n☒1dx 0 1 2 ☒ ☒ 1 ☒ ☒ ☒☒ ☒ ☒ 2☒ e 2 n ☒ ☒ x dx ☒ 2 ☒ 0 ☒ ☒ 2☒ e☒ y2 dy 0 ☒ ☒ ☒ 2☒ e dx ☒ 2☒ e dy 2 ☒ ☒ 1 ☒☒ ☒ x2 ☒ y2 ☒☒☒☒ 2 ☒☒ ☒ ☒☒ ☒ 0 0 ☒ ☒ ☒ 4☒☒ e 0 0 ☒☒x2 ☒ y2 ☒ dxdy EV0000124 · EV0000125 · EV0000126

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I Γ ☒ 1 ☒ Value of ☒ 2 ☒ ☒ ☒ put x ☒ r cos☒ and y ☒ r sin☒ and dxdy ☒ rdrd☒ then, x2 ☒ y2 ☒ r 2 ☒ 2 ☒ 2 ☒☒ e☒ r2 rdrd☒ i.e. ☒☒☒ 1 ☒☒ ☒ 4 ☒ ☒ 2 ☒☒ ☒ ☒☒ ☒ 0 0 ☒ ☒ 2 ☒ 4☒ d☒ ☒ e☒ r2 rdr 0 0 EV0000127 · EV0000128 · EV0000129

13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I Γ ☒ 1 ☒ Value of ☒ 2 ☒ ☒ ☒ ☒ 2 ☒ ☒ e rdr 0 2 r ☒ t 2 ☒ r ☒ 4. 2rdr ☒ dt ☒ ☒ 1 ☒ ☒ ☒ 2 ☒ 2☒ ☒ r ☒ e ☒☒ 2 ☒ ☒ ☒☒ ☒ ☒0 ☒ ☒ ☒☒ 1 ☒ ☒ Thus ☒ ☒ 2 ☒ ☒ ☒ EV0000130 · EV0000131 · EV0000132

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I Table for Gamma Function [Image text, OCR] m T(n) m T(m) r(n) r(n) 1.00 1.00000 1.25 0.90640 1.50 0.88623 1.75 0.91906 1.01 0.99433 1.26 0.90440 1.51 0.88659 1.76 0.92137 1.02 0.98884 1.27 0.90250 1.52 0.88704 1.77 0.92376 1.03 0.98355 1.28 0.90072 1.53 0.88757 1.78 0.92623 1.04 0.97044 1.29 0.89904 1.54 0.88818 1.79 0.92877 1.05 0.97350 1.30 0.89747 1.55 0.88887 1.80 0.93198 1.06 0.916874 1.31 0.89600 1.56 0.88964 1.81 0.93408 1.07 0.96415 1.32 0.89464 1.57 0.89049 1.82 0.93685 1.08 0.95979 1.3.3 0.89338 1.58 0.09142 1.83 0.93969 1.09 0.95546 1.34 0.89222 1.59 0.89243 1.84 0.94261 1.10 0.95135 1.35 0.89115 1.60 0.89352 1.85 0.94561 1.11 0.94739 1.36 0.89018 1.61 0.89468 1.86 0.94869 1.12 0.94359 1.37 0.88931 1.62 0.89592 1.87 0.95184 1.13 1.38 0.88854 1.63 0.89724 1.88 0.93993 0.95507 1.14 0.93642 1.39 0.88785 1.64 0.89864 1.89 0.95838 1.15 0.93304 1.40 0.88726 1.65 0.90012 1.90 0.96177 1.16 0.92980 1.41 0.88676 1.66 0.90167 1.91 0.96529 1.17 0.92670 1.42 0.80636 1.67 0.90330 1.92 0.96878 1.18 0.92373 1.43 0.88604 1.68 0.90500 1.93 0.97240 1.19 0.92088 1.44 0.88580 1.69 0.90678 1.94 0.97610 1.20 0.91817 1.45 0.88565 1.70 0.90864 1.95 0.97988 1.21 1.46 0.88560 1.71 0.91550 0.91057 1.96 0.98374 1.22 1.47 0.88563 1.72 0.91311 0.91258 1.97 0.98768 1.23 0.91075 1.48 0.88575 1.73 0.91466 1.98 0.99171 1.24 0.90852 1.49 0.88595 1.74 0.91683 1.99 0.99581 2.00 1.00000 EV0000133 · EV0000134 · EV0000135
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS I Graph Of Gamma Function ☒(n ☒1) ☒ n☒(n) ☒(n) ☒ ☒(n ☒1) [Image text, OCR] T(x) 4 2 44 2 2 4 — 2 A n ☒ ☒☒ 1 ☒ ☒ ☒ 2 ☒ ☒ ☒ Graph of Gamma Function EV0000136 · EV0000137 · EV0000138
16. Source slide page 16 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000139 · EV0000140 · EV0000141

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.548] Hello everyone, I am Kavishri, Assistant Professor in the Department of Science and Humanities. I welcome all of you to special functions, which is your fifth unit in engineering mathematics one. [00:00:14.820] In this unit, you are going to learn gamma function, beta function, and basis function. In this particular class, we are going to discuss definition of gamma function. [00:00:29.092] Properties of gamma function and graph of gamma function. Gamma function is a definite integral whose integrant depends on one variable. It is also known as Euler's integral of second kind. [00:00:46.436] It is the generalization of factorial notation from integes to real numbers. We can define gamma function as [00:00:56.868] Gamma of n is equal to integral zero to infinity t power minus x power n minus one dx n greater than c. Thus gamma n defines a function of n for positive values of n. [00:01:17.636] From this result we can obtain the value of gamma. So let us put n is equal to 1 in the definition of gamma function. Then we get gamma 1 is equal to integral 0 to infinity e power minus x x power 1 minus 1 dx. [00:01:38.020] x power 1 minus 1 is x power 0 and we know x power 0 is 1, so it just reduces to integral 0 to infinity e power minus x bx. [00:01:51.268] And we know that integration of e power minus x dx is equal to e power minus x divided by minus 1. [00:02:00.868] Using the fact that e power minus infinity is equal to 0 and e power 0 is equal to 1, we can evaluate this to 1. Thus we have gamma of 1 is equal to 1. [00:02:18.436] We can have an alternate form of gamma function. We can obtain it easily by using definition of gamma function. We know that gamma n is equal to integral 0 to infinity t power minus x x power n minus 1 dx. [00:02:37.092] In this expression, let us substitute x is equal to t square. If x is t square, dx is 2t d. [00:02:47.908]

Therefore, gamma n is equal to integral 0 to infinity e power minus t square t square to the power n minus 1 2 t dt. [00:03:01.860] Observe here, after this substitution, the limits of integration do not change because when x is equal to 0, your t is also 0. When x is infinity, t is also infinite. [00:03:20.004] So it reduces to 2 times integral 0 to infinity t power minus d square t to the power 2n minus 1 dt. This is another form of gamma function. [00:03:34.564] While solving problems on gamma function, we can make use of any of these two forms according to our convenience. Next, we will discuss some properties of gamma function. First of them is reduction formula. [00:03:52.580] It states that gamma of n plus 1 is equal to n gamma n where n is a real number. We can prove this easily by using the definition of gamma function, which is [00:04:07.236] Gamma n is equal to integral 0 to infinity e power minus x x power n minus 1 dx. [00:04:15.812] We want gamma of n plus 1. So let us just replace n by n plus 1 in this expression. So we get gamma of n plus 1 is equal to integral 0 to infinity e power minus x x power n plus 1 minus 1 dx. [00:04:36.004] which is equal to integral 0 to infinity e power minus x x power n dx. We have to evaluate this integral by using integration by parts. We know already that integration by parts [00:04:53.668] is given by integral uv dx is equal to u into integral v dx minus integral of derivative of u or d by dx of u into integral of v dx.

EV0000002 · EV0000003 · EV0000004 · EV0000005

Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.532] When you take x power n is equal to u and e power minus x is equal to v, you will get it as x power n e power minus x divided by minus 1 plus n into integral 0 to infinity. [00:05:27.472] E power minus x power n minus one dx. When you substitute the limits of integration for the first term, [00:05:37.700] E power minus infinity zero and e power zero is one. So first term becomes a zero and observe the second term along with n, you have integral zero to infinity, e power minus x, x power n minus one. [00:05:54.304] dx which is nothing but equal to gamma n. So your gamma of n plus 1 is n into gamma n. So this is the reduction formula which is very very important result. [00:06:10.852] When n is a negative non-integer, the formula used to find gamma of n is gamma n is equal to gamma of n plus 1 divided by n. Remember, gamma n is not defined for n is equal to 0. [00:06:29.060] And it is also not defined when n is a negative integer. But we can find the value of gamma n if n is negative real number which is not a negative integer using this formula. [00:06:47.972] When n is a positive integer, gamma of n plus one is equal to n factory. We can prove this easily by using reduction formula which states gamma of n plus 1 is equal to n gamma n. [00:07:03.428] Let us put n is equal to 1 in this formula, we get gamma 2 is equal to 1, gamma 1. That is 1, because we know gamma 1 is equal to 1. [00:07:15.300] When we put n is equal to 2, we get gamma of 3, that is 2 gamma 2, which is 2 factorial. When you put n is equal to 3, we get gamma 4, which is 3 gamma 3, that is 3 factorial. [00:07:32.740] And when you put n is equal to 4, we get gamma 5, which is 4 gamma 4, that is 4 factor. [00:07:42.148] Just look at the pattern here. Gamma two is one factorial, gamma three is two factorial, gamma four is three factorial, and gamma five is four factorial. So we can write in general [00:07:57.168] Gamma of n plus 1 is equal to n factorial where n is a positive integer. Remember, gamma n is not defined when n is 0 or when n is a negative integer. [00:08:16.100] Gamma half value we can find now. So this is also very very important result. Gamma half is equal to square root of pi. [00:08:25.732] We can prove this by considering alternate form of gamma function, which is gamma n is equal to 2 times integral 0 to infinity e power minus x square x to the power 2n minus 1 dx. [00:08:42.148] Let us put n is equal to half in this expression. So you get gamma half, which is equal to 2 times integral 0 to infinity, a power minus x

square dx. [00:08:55.940] Which can also be written as two times integral 0 to infinity e power minus y square dy. [00:09:04.420] Because since we are dealing with definite integrals, here x is a dummy variable. You can write it as x or y or t or any variable since we are dealing with definite integrals. [00:09:18.980] So gamma half whole square is gamma half into gamma half we can consider this as two times integral zero to infinity e power minus x square dx into two times integral zero to infinity e power minus y square dy. [00:09:36.644] which is equal to four times integral 0 to infinity 0 to infinity e power minus x square plus y square dx d y. [00:09:48.356] Now let us make use of polar coordinates by putting x is equal to r cos theta, y is equal to r sin theta. Then x square plus y square is equal to r square.

EV0000006 · EV0000007 · EV0000008 · EV0000009

Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.556] And dx dy is equal to r dr d theta. And observe the limits for x and y. Both are varying from 0 to infinity. That means both x and y are positive. [00:10:23.364] So our region of integration is the first quadrant. In this first quadrant, r value varies from 0 to infinity and theta value varies from 0 to pi by 2. [00:10:38.980] So we can write gamma half the whole square is equal to 4 times integral 0 to pi by 2, 0 to infinity, e to the power minus r square r dr d theta. [00:10:55.076] Then it can be written as four into integral 0 to pi by 2 b theta into integral 0 to infinity e power minus r square r br [00:11:09.060] When you integrate the first integral 0 to pi by 2 d theta, integration of d theta is theta, substitute the limits of integration, it reduces to 4 into pi by 2. [00:11:23.524] into integral 0 to infinity e power minus r square r dr. That is equal to 2 by into minus 1 by 2 e power minus r square limits vary from 0 to infinity. [00:11:42.180] So here the evaluation of integral is done by taking the substitution r square is equal to t, hence 2 r dr is equal to dt. [00:11:54.404] Thus, we get gamma half the whole square is equal to pi, so we can conclude gamma half is equal to square root of pi, which is a very very important result. Now let us find the graph of the gamma function. [00:12:11.972] Before that, just have a glance at this table where gamma n values are given for various values of n between 1 and 2. This is the graph of gamma function. [00:12:24.260] Observe here, gamma function is continuous for positive values of n and it is not continuous for negative values of n. We can find or you can plot this curve by using the properties gamma of n plus 1 is equal to n gamma n. [00:12:41.028] Gamma n is equal to gamma n plus 1 divided by n. This is for negative real numbers which are not indigenous. And gamma half is equal to square root of pi. And also referring to the table which is given in the previous slide. Just try [00:12:59.012] With these properties and the table, try to get the shape. This is very interesting thing. You can really do it. You will enjoy doing it. With this, we will end the session. Thank you.

EV0000010 · EV0000011 · EV0000012 · EV0000013

Full transcript: 1708 normalized words, 52 timestamped segments, 793.984 seconds duration. Low-confidence segments flagged: 0.

EV0000002 · EV0000003 · EV0000004 · EV0000005

3 Definition of Beta function and its properties

This lesson is indexed from the source video titled Definition of Beta function and its properties. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json.

Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000014 · EV0000015

Graphs and diagrams

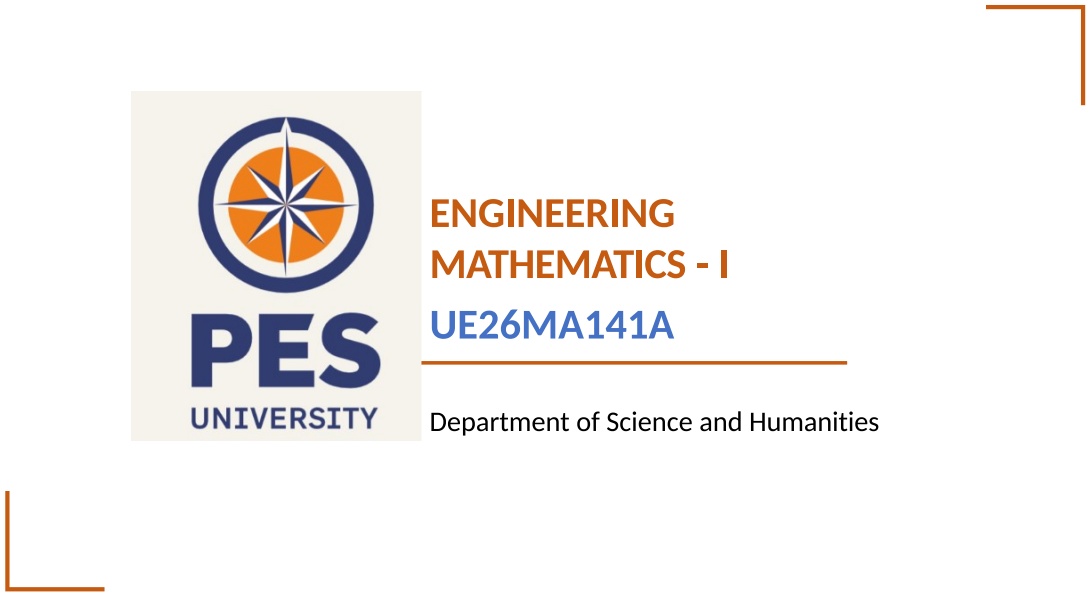


Figure G000017. Beta Function · source slide page 1.

EV0000143

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

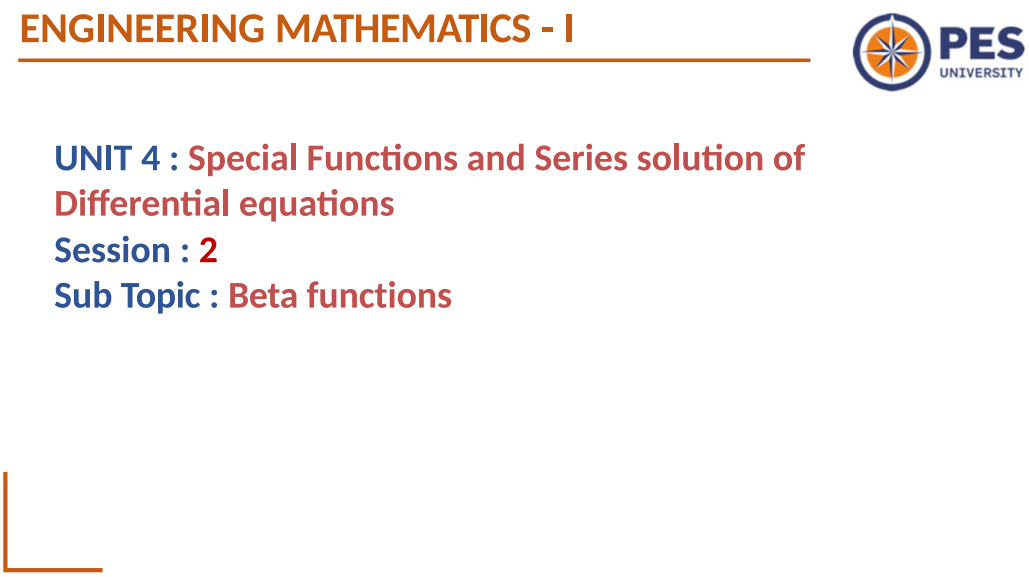


Figure G000018. Beta Function · source slide page 2.

EV0000146

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

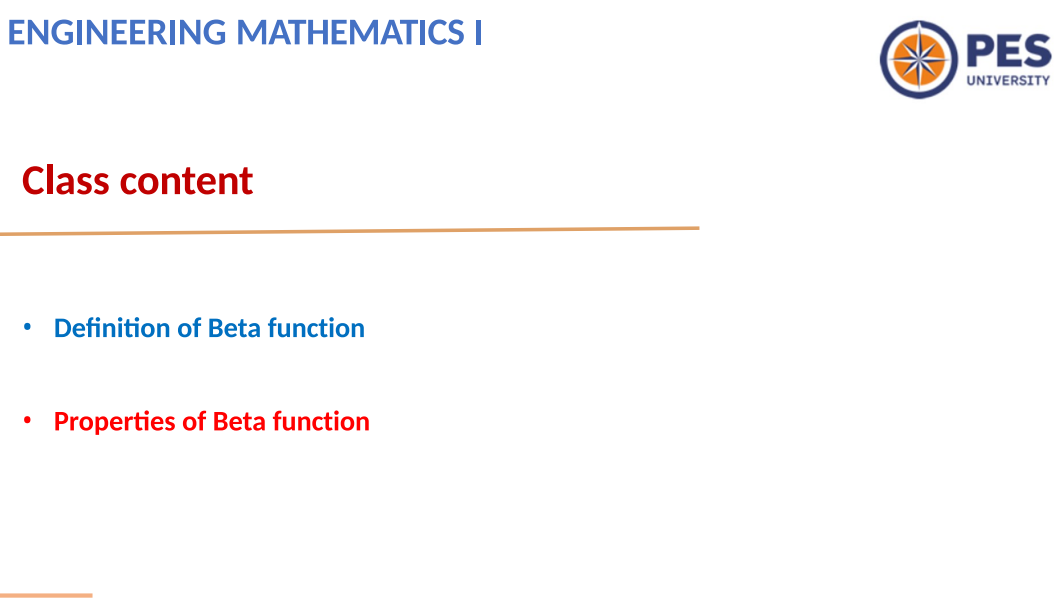


Figure G000019. Beta Function · source slide page 3.

EV0000149

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

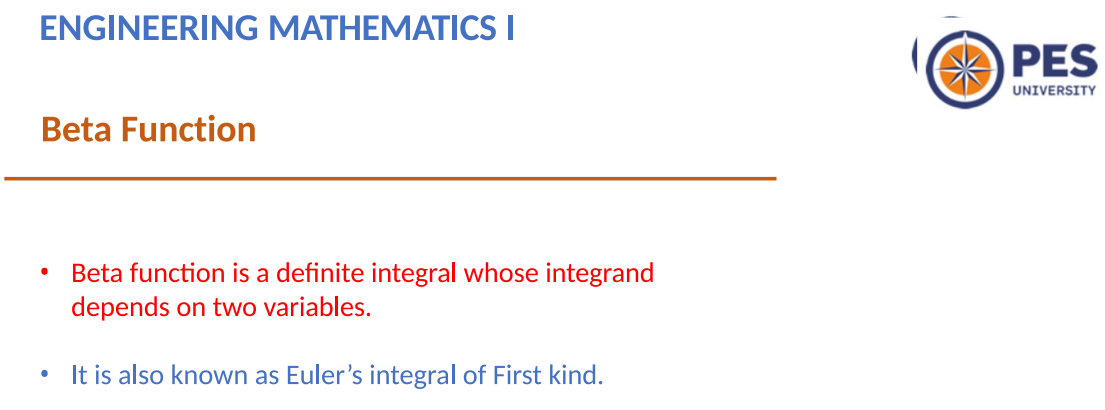


Figure G000020. Beta Function · source slide page 4.

EV0000152

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

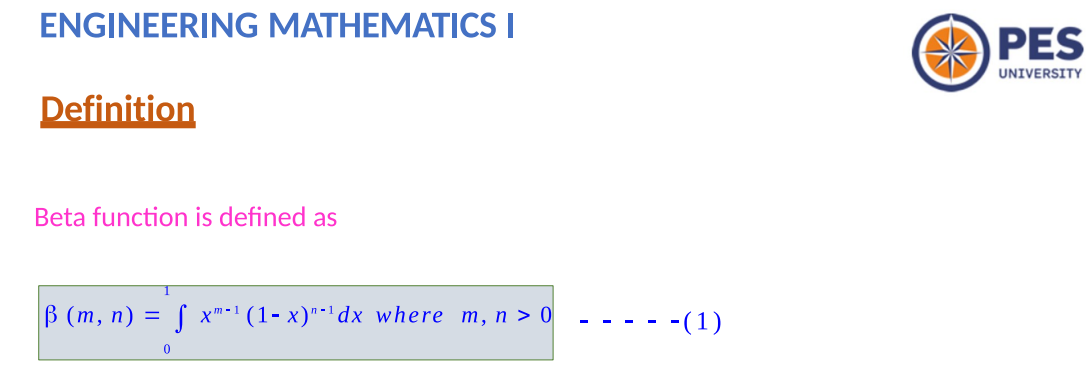


Figure G000021. Beta Function · source slide page 5.

EV0000155

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

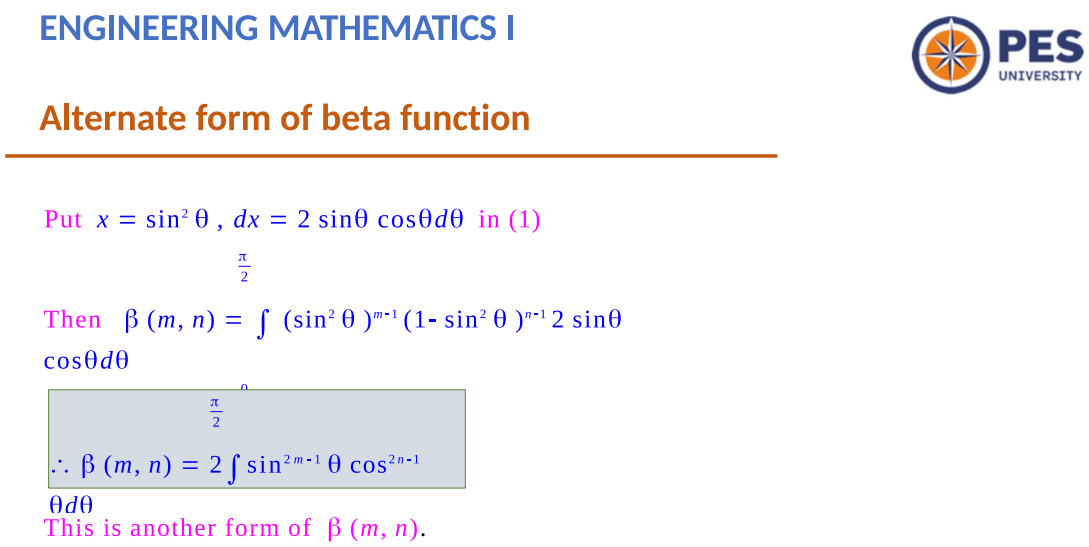


Figure G000022. Beta Function · source slide page 6.

EV0000158

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

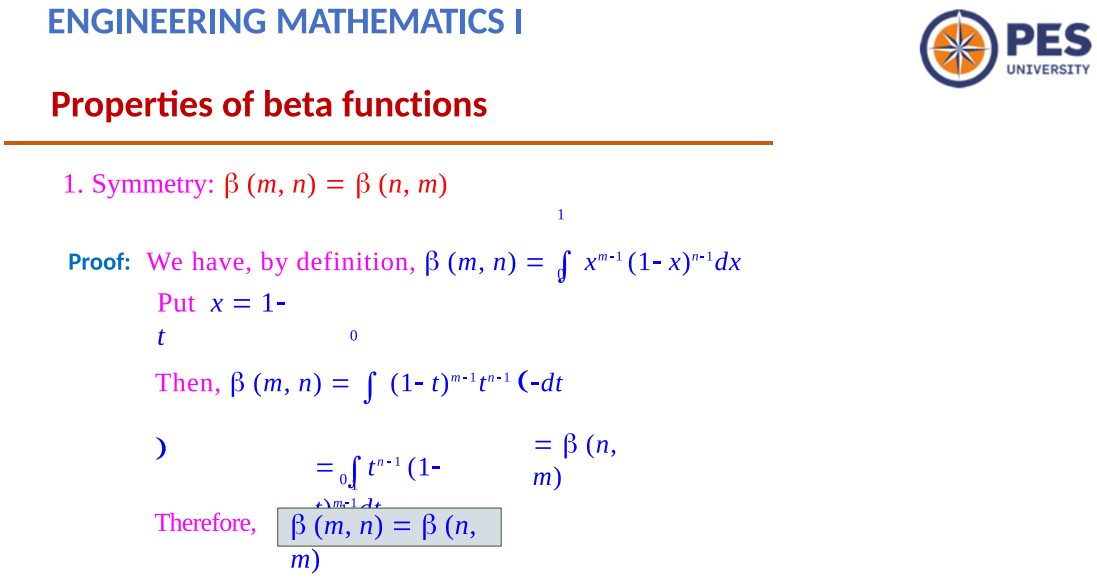


Figure G000023. Beta Function · source slide page 7.

EV0000161

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

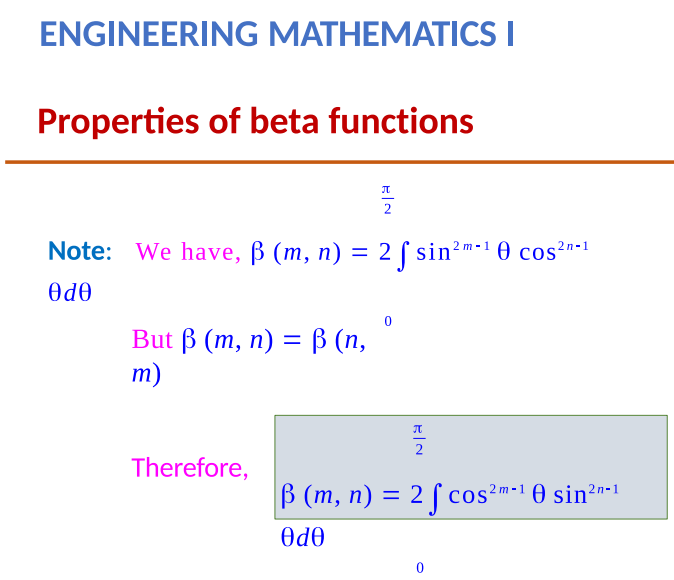


Figure G000024. Beta Function · source slide page 8.

EV0000164

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

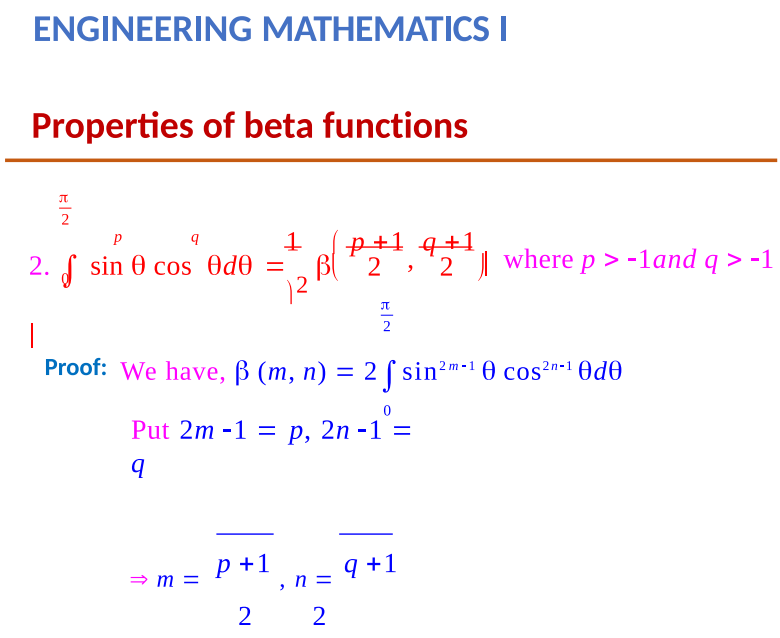


Figure G000025. Beta Function · source slide page 9.

EV0000167

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

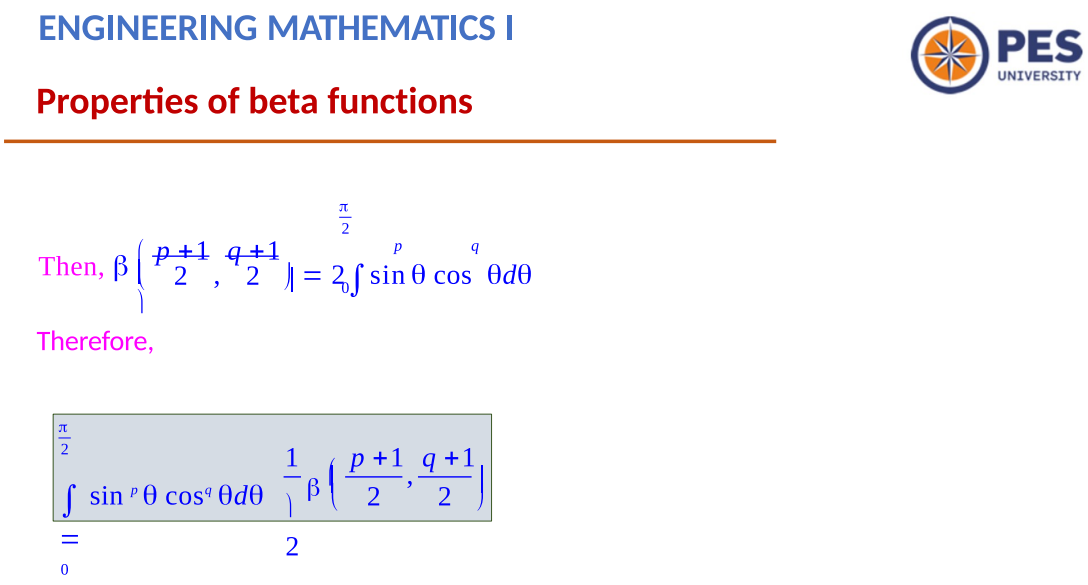


Figure G000026. Beta Function · source slide page 10.

EV0000170

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

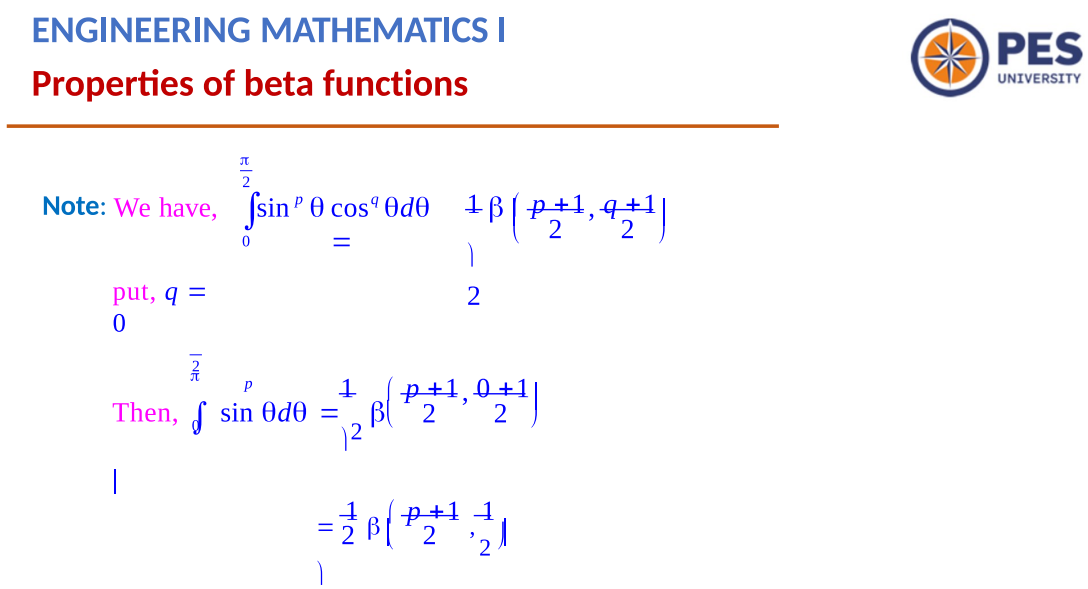


Figure G000027. Beta Function · source slide page 11.

EV0000173

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

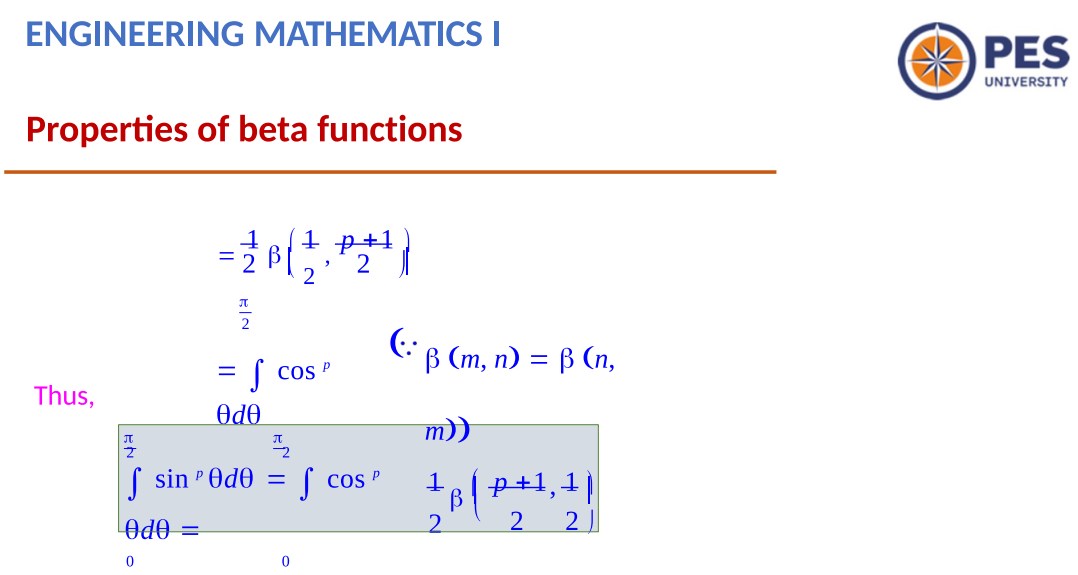


Figure G000028. Beta Function · source slide page 12.

EV0000176

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

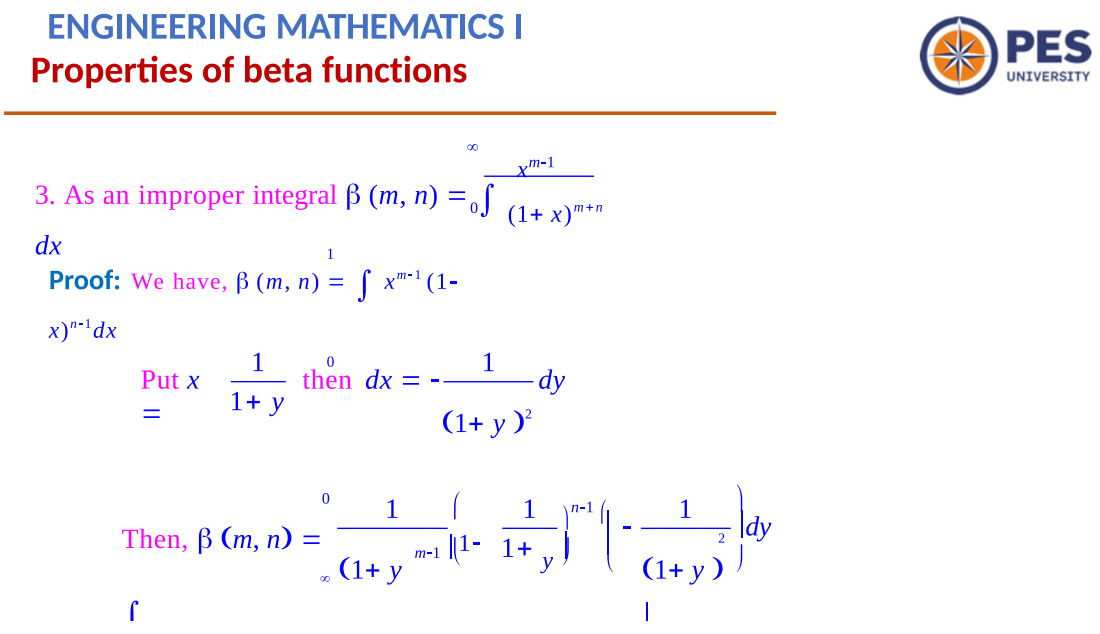


Figure G000029. Beta Function · source slide page 13.

EV0000179

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

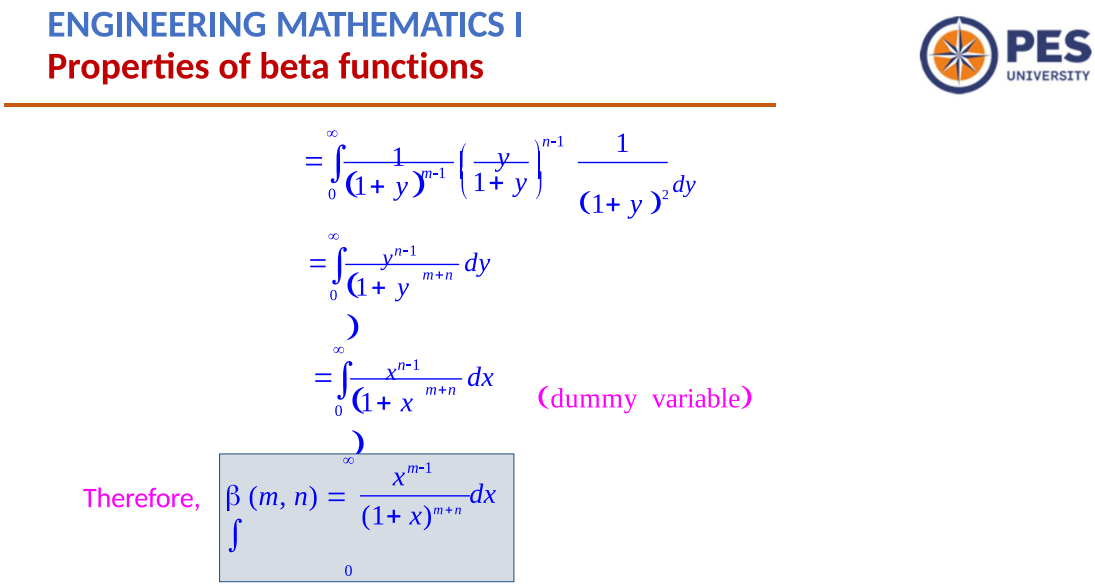


Figure G000030. Beta Function · source slide page 14.

EV0000182

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

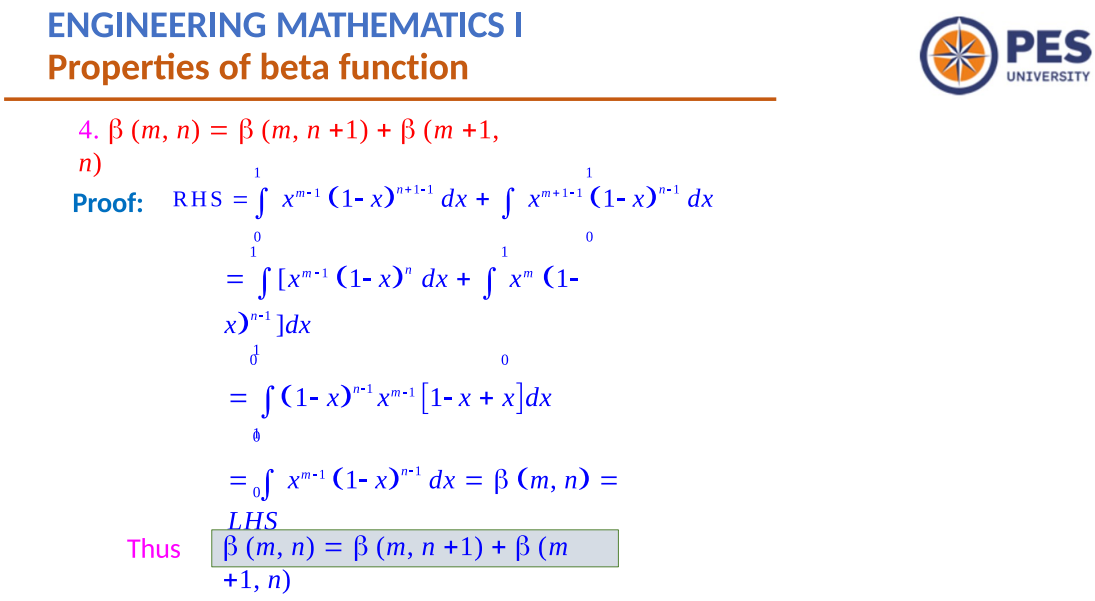


Figure G000031. Beta Function · source slide page 15.

EV0000185

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000032. Beta Function · source slide page 16.

EV0000188

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000142 · EV0000143 · EV0000144
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 4 : Special Functions and Series solution of Differential equations Session : 2 Sub Topic : Beta functions EV0000145 · EV0000146 · EV0000147
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I Class content Definition of Beta function Properties of Beta function EV0000148 · EV0000149 · EV0000150
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I Beta Function Beta function is a definite integral whose integrand depends on two variables. It is also known as Euler's integral of First kind. EV0000151 · EV0000152 · EV0000153
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I Definition Beta function is defined as 1 ☒ (m, n) ☒ ☒ xm☒1 (1☒ x)n☒1dx where m, n ☒ 0 0 ☒ ☒ ☒ ☒ ☒(1) EV0000154 · EV0000155 · EV0000156
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I Alternate form of beta function Put x ☒ sin2 ☒ , dx ☒ 2 sin☒ cos☒d☒ in (1) ☒ Then ☒ (m, n) ☒ ☒ (sin2 ☒)m☒1 (1☒ sin2 ☒)n☒1 2 sin☒ cos☒d☒ 0 2 ☒ 2 ☒ ☒ (m, n) ☒ 2☒sin2m☒1 ☒ cos2n☒1 ☒d☒ 0 This is another form of ☒ (m, n). . EV0000157 · EV0000158 · EV0000159

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I Properties of beta functions 1. Symmetry: ☒ (m, n) ☒ ☒ (n, m) 1 Proof: We have, by definition, ☒ (m, n) ☒ ☒ xm☒1 (1☒ x)n☒1dx 0 Put x ☒ 1☒ t Then, ☒ (m, n) ☒ ☒ (1☒ t)m☒1tn☒1 ☒☒dt ☒ 1 1 0 ☒ ☒tn☒1 (1☒ t)m☒1dt ☒ ☒ (n, m) 0 ☒ (m, n) ☒ ☒ (n, m) Therefore, EV0000160 · EV0000161 · EV0000162
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I Properties of beta functions ☒ 2 Note: We have, ☒ (m, n) ☒ 2☒sin2m☒1 ☒ cos2n☒1 ☒d☒ 0 But ☒ (m, n) ☒ ☒ (n, m) ☒ 2 ☒ (m, n) ☒ 2☒cos2m☒1 ☒ sin2n☒1 ☒d☒ 0 Therefore, EV0000163 · EV0000164 · EV0000165
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I Properties of beta functions ☒ 2 1 ☒ p ☒1 q ☒1 ☒ 2. ☒ sin ☒ cos ☒d☒ ☒ 2 ☒ ☒ p q , ☒ where p ☒ ☒1and q ☒ ☒1 2 2 ☒ ☒ ☒ 0 Proof: We have, ☒ (m, n) ☒ 2☒sin2m☒1 ☒ cos2n☒1 ☒d☒ 0 2 Put 2m ☒1 ☒ p, 2n ☒1 ☒ q ☒ m ☒ p ☒1 , n ☒ q ☒1 2 2 EV0000166 · EV0000167 · EV0000168
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I Properties of beta functions ☒ 2 ☒ p ☒1 q ☒1 ☒ , ☒ ☒ 2☒sin ☒ cos ☒d☒ Then, ☒ ☒ p q 2 2 ☒ ☒ 0 Therefore, ☒ 2 ☒ sin p ☒ cosq ☒d☒ ☒ 0 1 p ☒1 q ☒1 ☒ 2 ☒ ☒ , ☒ ☒ 2 2 ☒ ☒ EV0000169 · EV0000170 · EV0000171
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I Properties of beta functions ☒ 1 ☒ p ☒1 q ☒1 ☒ 2 2 ☒ 0 Note: sin ☒ cos ☒d☒ ☒ ☒ p q We have, , ☒ ☒ 2 2 ☒ ☒ put, q ☒ 0 ☒ 2 1 ☒ p ☒1 0 ☒1 ☒ Then, ☒ sin ☒d☒ ☒ 2 ☒ ☒ p , ☒ 2 2 ☒ ☒ 0 ☒ 1 ☒ ☒ p ☒1 , 1 ☒ ☒ 2 ☒ 2 2 ☒ ☒ EV0000172 · EV0000173 · EV0000174
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I Properties of beta functions ☒ 1 ☒ ☒ 1 , p ☒1 ☒ 2 ☒ ☒ 2 ☒ 2 ☒ ☒ ☒ ☒ cos p ☒d☒ 0 2 ☒ ☒m, n☒ ☒ ☒ ☒n, m☒☒ ☒ Thus, ☒ ☒ 1 p ☒1 1 ☒ 2 2 2 ☒ sin p ☒d☒ ☒ ☒ cos p ☒d☒ ☒ 0 0 ☒ ☒ , ☒ ☒ 2 2 ☒ ☒ EV0000175 · EV0000176 · EV0000177
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I Properties of beta functions xm☒1 ☒ 3. As an improper integral ☒ (m, n) ☒ ☒ (1☒ x)m☒n dx 0 1 Proof: We have, ☒ (m, n) ☒ ☒ xm☒1 (1☒ x)n☒1dx 0 1 ☒1☒ y ☒2 1 Put x ☒ then dx ☒ ☒ dy 1☒ y ☒n☒1 ☒ ☒ 0 ☒ 1 1 1 Then, ☒ ☒m, n☒ ☒ ☒ ☒ ☒ 2 ☒dy y ☒ m☒1 ☒1☒ ☒ ☒1☒ y ☒ ☒1☒ y ☒ ☒ ☒ 1☒ ☒ ☒ ☒ ☒ EV0000178 · EV0000179 · EV0000180
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I Properties of beta functions ☒n☒1 ☒ 1 ☒ y 1 ☒ 0 ☒ ☒ ☒ ☒1☒ y ☒2 dy ☒ ☒ ☒ ☒ m☒1 1☒ y 1☒ y yn☒1 ☒ ☒ 0 ☒ dy ☒ ☒ m☒n 1☒ y xn☒1 ☒ ☒dummy variable☒ ☒ 0 ☒ dx ☒ ☒ m☒n 1☒ x ☒ ☒ (m, n) ☒ ☒ 0 m☒1 x dx Therefore, (1☒ x)m☒n EV0000181 · EV0000182 · EV0000183
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS I Properties of beta function 4. ☒ (m, n) ☒ ☒ (m, n ☒1) ☒ ☒ (m ☒1, n) 1 1 RHS ☒☒ xm☒1 ☒1☒ x☒n☒1☒1 dx ☒ ☒ xm☒1☒1 ☒1☒ x☒n☒1 dx 0 0 Proof: 1 1 ☒ ☒[xm☒1 ☒1☒ x☒n dx ☒ ☒ xm ☒1☒ x☒n☒1]dx 0 0 ☒ ☒☒1☒ x☒n☒1 xm☒1 ☒1☒ x ☒ x☒dx 0 1 1 ☒ ☒ xm☒1 ☒1☒ x☒n☒1 dx ☒ ☒ ☒m, n☒ ☒ LHS 0 ☒ (m, n) ☒ ☒ (m, n ☒1) ☒ ☒ (m ☒1, n) Thus EV0000184 · EV0000185 · EV0000186
16. Source slide page 16 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000187 · EV0000188 · EV0000189

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Hello everyone, welcome back to Engineering Mathematics 1. In last class you have studied definition, properties and graph of gamma function. In this class, we are going to study definition of beta function and properties of beta function. [00:00:19.428] Beta function is a definite integral whose integrant depends on two variables. It is also known as Euler's integral of first time. Now let us define beta function. [00:00:33.796] Beta function is defined as beta of m comma n is equal to integral 0 to 1 x power m minus 1 into 1 minus x power n minus 1 is, where m and n both are positive. [00:00:51.108] Let me call this as equation number one. We can also have an alternate form for beta function. We can get it by putting x is equal to sine square theta in equation number 1. [00:01:06.468] When x is equal to sine square theta, dx is equal to two sine theta cos theta d theta. [00:01:15.076] Also, the limits of integration change with the substitution. When x is equal to 0, that implies sine square theta is equal to 0, that implies theta is equal to 0. [00:01:29.060] When x is equal to 1, that means sine square theta is equal to 1, or theta is equal to pi by 2. With this, we will get beta of m comma n as integral 0 to pi by 2, sin square theta all power m minus 1. [00:01:46.416] One minus sine square theta whole power n minus one into two sine theta cos theta b theta. [00:01:54.244] 2 is a constant we can keep outside the integral sign. So it is 2 times integral 0 to pi by 2. Sin square theta wall power m minus 1 is sin power 2 m minus 2 theta. [00:02:07.108] One minus sin square theta we know it is cos square theta. So cos square power n minus one is cos power two n minus two theta into sin theta cos theta d theta. Right? That is [00:02:23.428] Beta of m comma n is now two times integral 0 to pi by 2 sin power 2 m minus 1 theta cos power 2n minus 1 theta d theta. This is another form of beta function, we can also call it as trigonometric form of beta function. [00:02:41.988] Both the forms are important, so you can remember both the forms here. Both are useful when solving the problems. We can choose any one of these forms according to our convenience. Next, we will study the properties of beta function. [00:03:01.796] First property is symmetry, which states beta of m comma n is equal to beta of L comma n. [00:03:09.284] We can prove this by using the definition of beta function, which states beta of m, n is equal to integral 0 to 1 x power m minus 1 into 1 minus x power n minus 1 dx. Now let me put x is equal to 1 minus t in this. [00:03:28.548] then dx becomes minus of dt. Also the limits of integration change with the substitution [00:03:37.252] x equal to 1 minus t implies t is equal to 1 minus x. Therefore, when x is equal to 0, t is equal to 1. And when x is equal to 1, right, t is equal to 0. [00:03:55.236] Therefore, now we can have beta of m comma n is equal to integral 1 to 0 1 minus t power m minus 1 t power n minus 1 minus t. [00:04:07.844] So now you can take the negative sign outside and also we know minus of integral one to zero becomes integral zero to one. So beta of m comma n is now integral zero to one [00:04:19.888] T power n minus 1 1 minus t to the power m minus 2 t, which is nothing but equal to beta of n comma m. Thus we have proved now beta of m comma n is exactly same as beta of n comma m. [00:04:38.692] Now uh just now you have proved that beta of m comma n can also be written as two times integral zero to pi by two sine power two m minus one theta cos power two n minus one theta e theta. [00:04:52.420] And now we know beta of m n is same as beta of n m, so we can also write beta of m, comma n is equal to 2 times integral 0 to pi by 2 cos power 2 m minus 1 theta into sine power 2n minus 1 theta d.

EV0000014 · EV0000015 · EV0000016 · EV0000017

Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.060] Now second property states integral 0 to pi by 2 sin power p theta cos power q theta d theta is equal to 1 by 2 times beta of p plus 1 by 2 comma q plus 1 by 2 where p is greater than minus 1 and q is also greater

than minus 1. [00:05:32.676] To prove this, consider the diagnometric form of beta function that is beta of m, comma n is equal to 2 times integral 0 to pi by 2 sin power 2m minus 1 theta cos power 2n minus 1 theta d theta. [00:05:50.596] Now let us have a substitution here. Take 2m minus 1 is equal to p and 2n minus 1 is equal to q. Then we get m is equal to p plus 1 by 2 and n is equal to q plus 1 divided by 2. [00:06:09.732] According to the definition of beta function, we should have m in beta of m comma n, m and n both are positive. [00:06:19.780] So, m is greater than 0 implies now p plus 1 divided by 2 is greater than 0, that means p is greater than minus 1. [00:06:31.044] Similarly, when n is greater than 0, you will be having q plus 1 by 2 greater than 0, hence q is also greater than minus 1. This is what the conditions mentioned in property number 2. [00:06:48.676] Now beta of p plus one by two, comma q plus one by two is two times integral zero to pi by two sine power p theta cos power q theta d theta. [00:07:00.452] Therefore, integral 0 to pi by 2 sine power p theta cos power q theta d theta is equal to 1 by 2 times beta of p plus 1 by 2, comma q plus 1 divided by 2. [00:07:16.420] This is also an important property. Whenever we come across the integrals in the form integral 0 to pi by 2, sin power p theta, cos power q theta, always we can express them in terms of beta functions. [00:07:36.932] Already we proved now that integral 0 to pi by 2 sin power p theta cos power q theta d theta is 1 by 2 times beta of p plus 1 by 2 comma q plus 1 by 2. In this let us put q is equal to 0. [00:07:53.764] Then we get integral 0 to pi by 2 sin power p theta is e theta is equal to 1 by 2 times beta of p plus 1 by 2 comma 0 plus 1 divided by 2. [00:08:08.996] Which is equal to 1 by 2 times beta of P plus 1 by 2, comma 1 by 2. And also we know that beta of m comma n is equal to beta of n, comma m. So this can be also written as half into beta of [00:08:27.620] 1 by 2 comma p plus 1 divided by 2, which is equal to integral 0 to pi by 2 cos power p theta d theta. [00:08:38.500] Thus we can conclude integral 0 to pi by 2 sine power p theta d theta is same as integral 0 to pi by 2 cos power p theta d theta and that is equal to half into beta of p plus 1 by 2 comma 1 by 2. [00:08:59.620] Next property says that we can also express beta function in the form of an improper integral 0 to infinity x power m minus 1 divided by 1 plus x power m plus n dx. [00:09:16.484] This property is also very useful because whenever we want to evaluate the integrals in the form integral 0 to infinity x power m minus 1 by 1 plus x power m plus n dx, always we can write them as [00:09:33.712] beta of m comma n and we can find the value of beta of m comma n easily. To prove this property, first let us consider the definition of beta function [00:09:46.756] which says beta of m, n is equal to integral 0 to 1 x power m minus 1 1 minus x to the power n minus 1 bx. Let me call this as equation number 1.

EV0000018 · EV0000019 · EV0000020 · EV0000021

Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.820] So now in this equation let me put x is equal to 1 divided by 1 plus y. That means dx is equal to minus 1 divided by 1 plus y the whole square dy. [00:10:17.924] Now let us see how the limits of integration change. So when x is equal to 1 by 1 plus y, 1 plus y becomes 1 by x. So when x is equal to 0, 1 plus y is equal to infinity. [00:10:35.376] That implies y is equal to infinity. When x is equal to 1, 1 plus y is equal to 1, that means y is equal to 0. So the now limits, new limits of integration are infinity to 0. [00:10:54.084] So if we substitute it in equation number one, we get beta of m comma n is equal to integral infinity to 0, 1 divided by 1 plus 5 whole power m minus 1. [00:11:11.972] into 1 minus 1 by 1 plus y whole to the power n minus 1 into minus dy divided by 1 plus y the whole square, which is equal to minus of integral infinity to 0. [00:11:29.700] 1 divided by 1 plus y whole power m minus 1 into 1 plus y minus 1 divided by 1 plus y whole to the power n minus 1. So into [00:11:47.524] 1 divided by 1 plus y the whole square into d by which is equal to integral 0 to infinity [00:11:58.404]

1 divided by 1 plus y whole to the power m minus 1 into y power n minus 1 divided by 1 plus y power n minus 1 into 1 by 1 plus y the whole square divide. [00:12:13.956] which is equal to integral 0 to infinity y power n minus 1 divided by 1 plus y to the power m minus 1 into 1 plus y to the power n minus 1 into 1 plus y whole square into dy. [00:12:32.612] which is equal to integral 0 to infinity y power n minus 1 divided by 1 plus y to the power m minus 1 plus n minus 1 plus 2. That is m plus n divide. [00:12:49.988] So it can be also written as integral 0 to infinity x power n minus 1 divided by 1 plus x to the power m plus n dx. Because since we are dealing with the definite integrals [00:13:04.736] X and Y both are dummy variables. We can use any variable here. [00:13:10.564] So thus your beta of m comma n is integral 0 to infinity x to the power m minus 1 divided by 1 plus x power m plus n dx. Remember this, this is also an important result. [00:13:27.268] Next consider the fourth property that says beta of m comma n can be written as beta of m comma n plus one plus beta of m plus one comma n. [00:13:39.204] This can be proved very easily by considering the definition of beta function. So we know that beta of m comma n is equal to integral 0 to 1 x power m minus 1 into 1 minus x power n minus 1 dx. [00:13:54.852] Let us use this definition. First let us consider the right hand side. So on right hand side we have two terms, one is beta of m comma n plus 1, another one is beta of m plus 1, n [00:14:08.324] So now apply beta of m comma n formula to both of these terms. So first term becomes now integral 0 to 1 x power m minus 1 1 minus x to the power n plus 1 minus 1 dx. [00:14:22.884] And second terms after applying bit of m per m formula, so we will get integral 0 to 1 x power m plus 1 minus 1 into 1 plus x to the power n minus 1 dx. [00:14:37.444] That is equal to integral 0 to 1 x power m minus 1 1 minus x to the power n dx plus integral 0 to 1 x power m into 1 minus x to the power n minus 1 dx. [00:14:53.924] Now uh in both these integrals, limits are varying from 0 to 1. So let us take the common factors outside. So I can write this as integral 0 to 1. 1 minus x power n minus 1 into x power m minus 1 is the common factor.

EV0000022 · EV0000023 · EV0000024 · EV0000025

Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.648] Take it outside. So inside the bracket we get 1 minus x plus x dx. So minus x plus x gets cancelled. So we get integral 0 to 1 x power m 1 minus x power n minus 1 dx. [00:15:26.564] That is nothing but equal to beta of m comma n. So we have proved left hand side is equal to right hand side. Thus we have beta of m comma n is equal to beta of m comma n plus 1 plus beta of m plus 1 comma n. [00:15:47.044] With this we will end this session. Thank you.

EV0000026 · EV0000027 · EV0000028 · EV0000029

Full transcript: 2259 normalized words, 62 timestamped segments, 951.317333 seconds duration. Low-confidence segments flagged: 0.

EV0000014 · EV0000015 · EV0000016 · EV0000017

4 Relation between Beta and Gamma functions, Duplication formula

This lesson is indexed from the source video titled Relation between Beta and Gamma functions, Duplication formula. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or

rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000030 · EV0000031

Graphs and diagrams

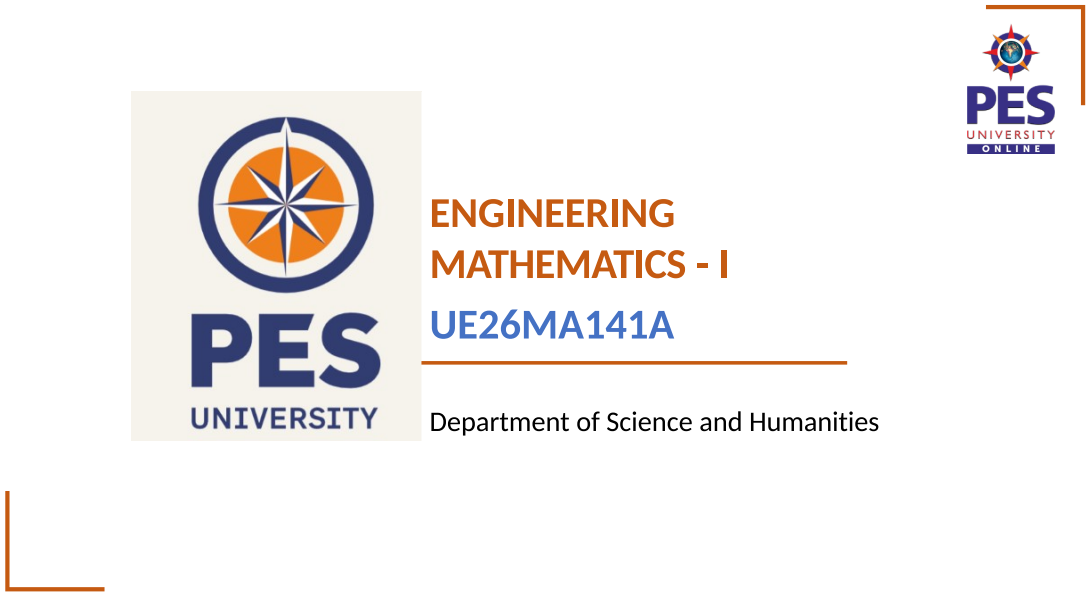


Figure G000033. Relation between beta and gamma functions · source slide page 1. EV0000191

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

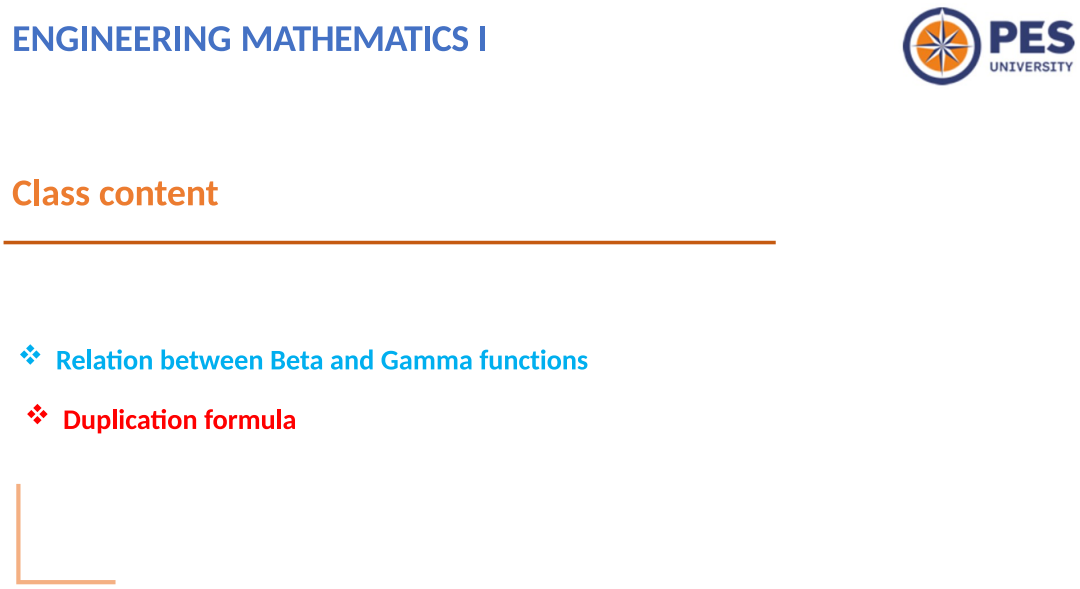


Figure G000034. Relation between beta and gamma functions · source slide page 2.

EV0000194

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

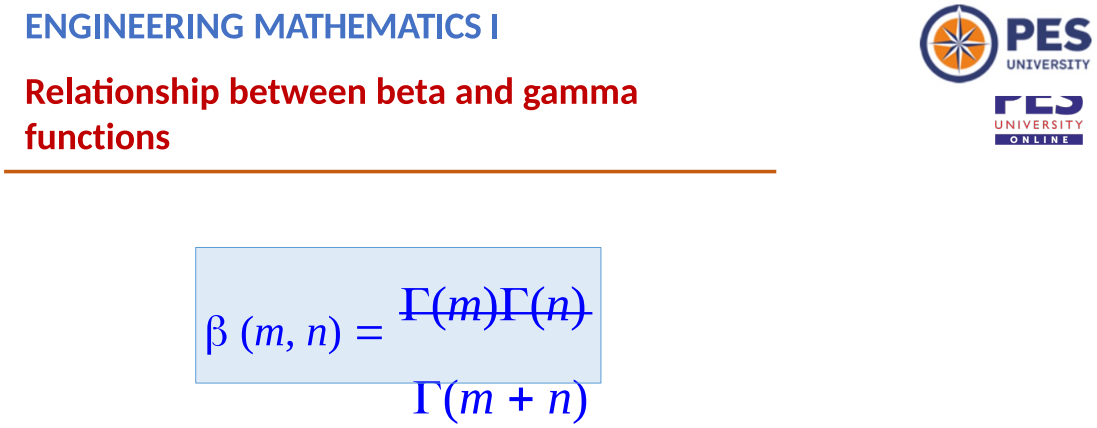


Figure G000035. Relation between beta and gamma functions · source slide page 3.

EV0000197

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

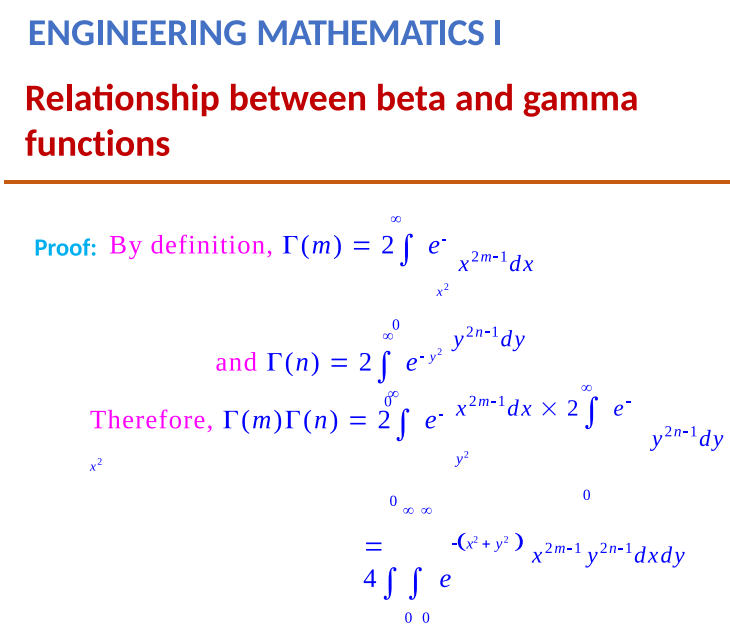


Figure G000036. Relation between beta and gamma functions · source slide page 4.

EV0000200

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

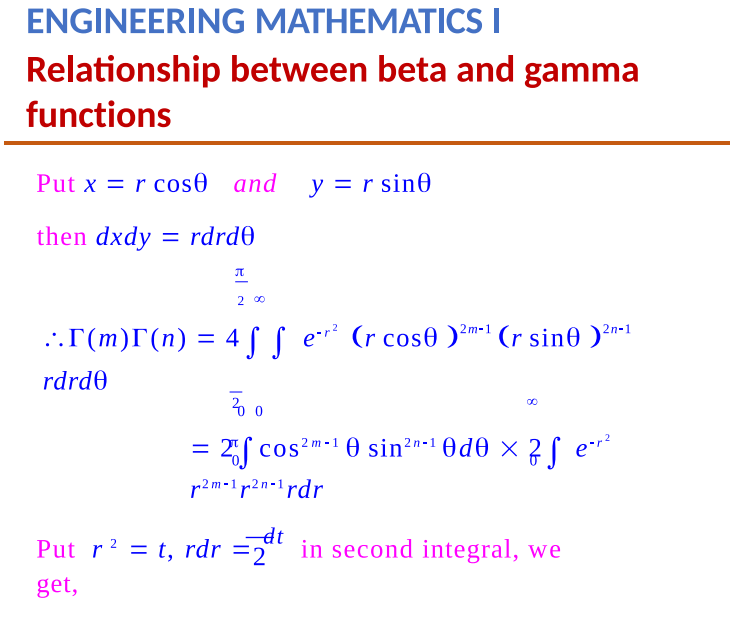


Figure G000037. Relation between beta and gamma functions · source slide page 5.

EV0000203

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

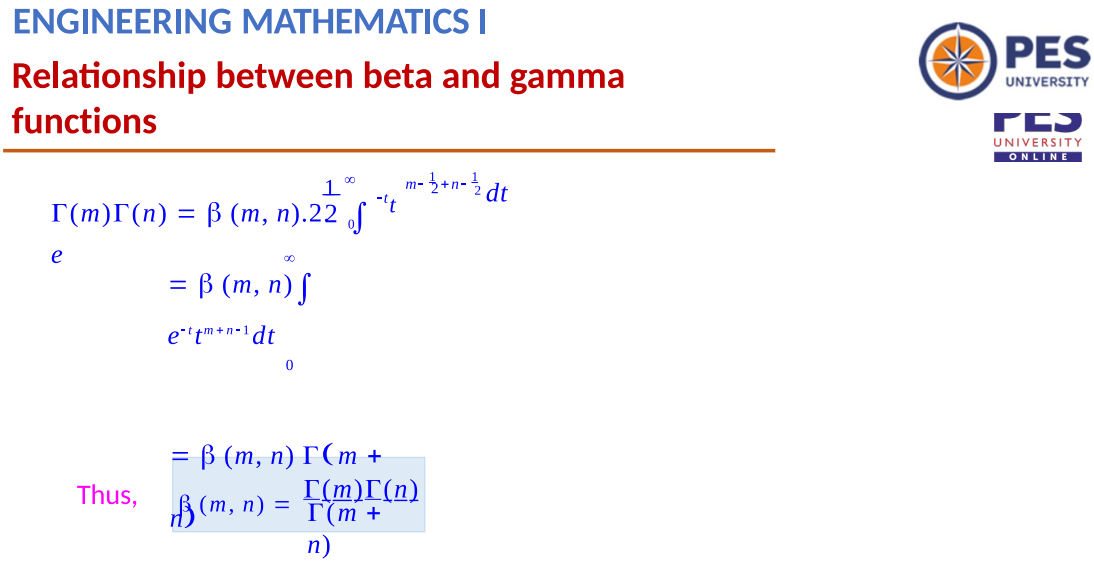


Figure G000038. Relation between beta and gamma functions · source slide page 6.

EV0000206

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

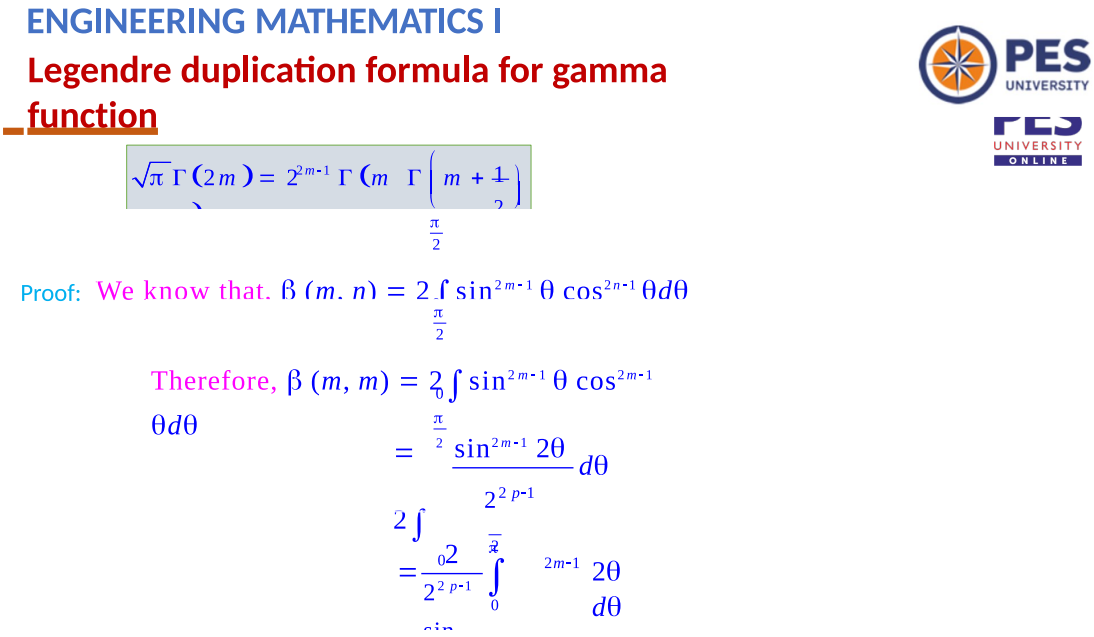


Figure G000039. Relation between beta and gamma functions · source slide page 7.

EV0000209

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

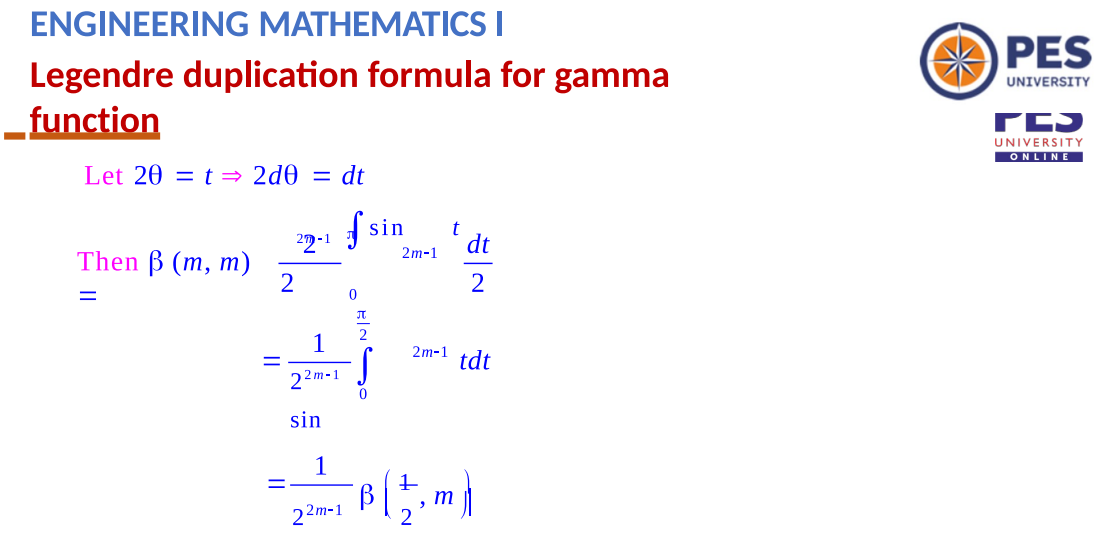


Figure G000040. Relation between beta and gamma functions · source slide page 8.

EV0000212

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

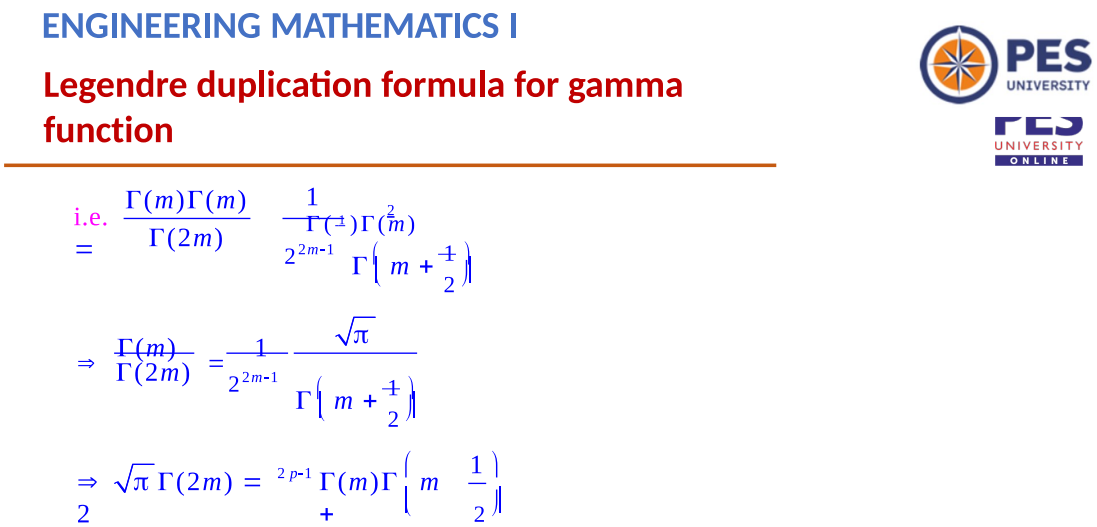


Figure G000041. Relation between beta and gamma functions · source slide page 9.

EV0000215

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

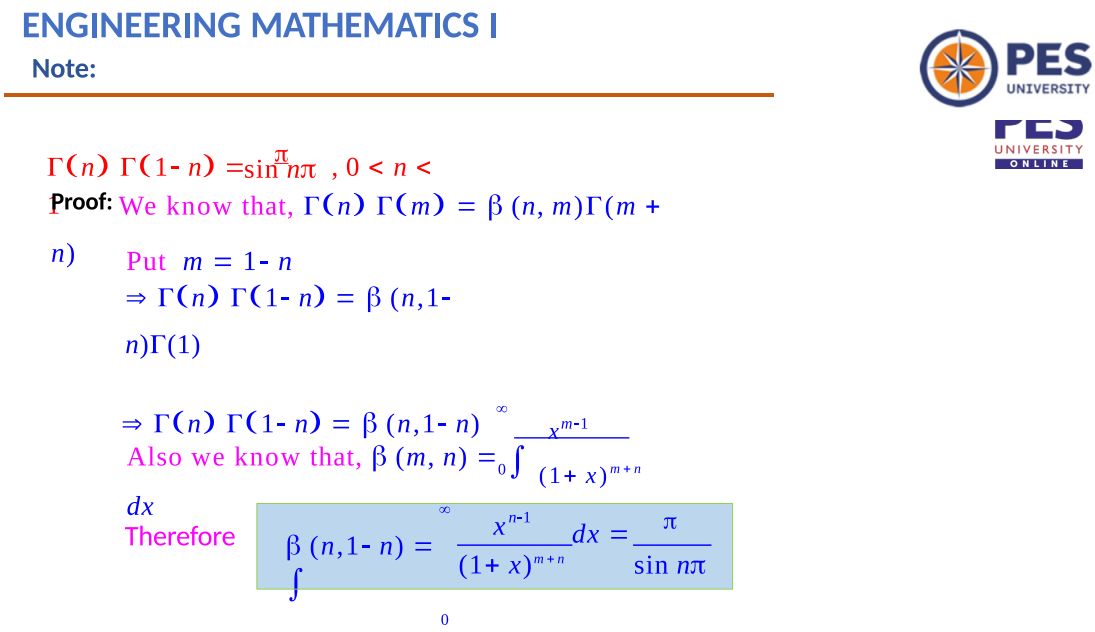


Figure G000042. Relation between beta and gamma functions · source slide page 10.

EV0000218

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000043. Relation between beta and gamma functions · source slide page 11.

EV0000221

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000190 · EV0000191 · EV0000192
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I Class content Relation between Beta and Gamma functions Duplication formula EV0000193 · EV0000194 · EV0000195
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I Relationship between beta and gamma functions ☒ (m, n) ☒ ☒(m)☒(n) ☒(m ☒ n) EV0000196 · EV0000197 · EV0000198
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I Relationship between beta and gamma functions ☒ Proof: By definition, ☒(m) ☒ 2☒ e☒ x2 0 ☒ and ☒(n) ☒ 2☒ e☒ y2 0 x2m☒1dx y2n☒1dy ☒ x2m☒1dx ☒ 2☒ e☒ y2 0 ☒ Therefore, ☒(m)☒(n) ☒ 2☒ e☒ x2 0 y2n☒1dy ☒ ☒ ☒ 4☒☒ e 0 0 ☒☒x2 ☒ y2 ☒ x2m☒1 y2n☒1dxdy EV0000199 · EV0000200 · EV0000201
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I Relationship between beta and gamma functions Put x ☒ r cos☒ and y ☒ r sin☒ then dxdy ☒ rdrd☒ ☒ 2 ☒ ☒☒(m)☒(n) ☒ 4☒☒ e☒r2 ☒r cos☒ ☒2m☒1 ☒r sin☒ ☒2n☒1 rdrd☒ 0 0 ☒ ☒ 2 ☒ 2☒cos2m☒1 ☒ sin2n☒1 ☒d☒ ☒ 2☒ e☒r2 r2m☒1r2n☒1rdr 0 0 Put r 2 ☒ t, rdr ☒ dt in second integral, we get, 2 EV0000202 · EV0000203 · EV0000204
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I Relationship between beta and gamma functions 1 ☒ ☒(m)☒(n) ☒ ☒ (m, n).2. ☒ e ☒tt m☒ 1 ☒n☒ 1 2 dt 2 2 0 ☒ ☒ ☒ (m, n)☒ e☒ttm☒n☒1dt 0 ☒ ☒ (m, n) ☒☒m ☒ n☒ ☒ (m, n) ☒ ☒(m)☒(n) Thus, ☒(m ☒ n) EV0000205 · EV0000206 · EV0000207
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I Legendre duplication formula for gamma function 1 ☒ ☒ ☒ ☒ ☒ ☒ ☒ ☒ 2m 2m☒1 ☒ 2 ☒ m ☒ m ☒ ☒ 2 ☒ ☒ ☒ ☒ 2 Proof: We know that, ☒ (m, n) ☒ 2☒sin2m☒1 ☒ cos2n☒1 ☒d☒ 0 ☒ Therefore, ☒ (m, m) ☒ 2☒sin2m☒1 ☒ cos2m☒1 ☒d☒ 2 0 ☒ ☒ 2☒ 0 2 sin2m☒1 2☒ d☒ 22 p☒1 ☒ 2 2 22 p☒1 ☒ sin 2☒ d☒ 2m☒1 ☒ 0 EV0000208 · EV0000209 · EV0000210
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I Legendre duplication formula for gamma function Let 2☒ ☒ t ☒ 2d☒ ☒ dt ☒ 2 dt 2m☒1 ☒sin t 2m☒1 0 ☒ Then ☒ (m, m) ☒ 2 2 2 1 22m☒1 ☒ sin 2m☒1 ☒ tdt 0 1 ☒ ☒ 1 , m ☒ ☒ ☒ 2 ☒ 22m☒1 ☒ ☒ EV0000211 · EV0000212 · EV0000213
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I Legendre duplication formula for gamma function ☒(m)☒(m) 1 ☒(1)☒(m) 22m☒1 i.e. ☒ 2 ☒(2m) ☒☒ m ☒ 1 ☒ ☒ 2 ☒ ☒ ☒ ☒ ☒ ☒(m) ☒ 1 22m☒1 ☒(2m) ☒☒ m ☒ 1 ☒ ☒ 2 ☒ ☒ ☒ 1 ☒ ☒ ☒ ☒ ☒(2m) ☒ 2 2 p☒1 ☒(m)☒ m ☒ ☒ 2 ☒ ☒ ☒ EV0000214 · EV0000215 · EV0000216
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I Note: ☒☒n☒ ☒☒1☒ n☒ ☒ ☒ , 0 ☒ n ☒ 1 sin n☒ Proof: We know that, ☒☒n☒ ☒☒m☒ ☒ ☒ (n, m)☒(m ☒ n) Put m ☒ 1☒ n ☒ ☒☒n☒ ☒☒1☒ n☒ ☒ ☒ (n,1☒ n)☒(1) ☒ ☒☒n☒ ☒☒1☒ n☒ ☒ ☒ (n,1☒ n) xm☒1 ☒ Also we know that, ☒ (m, n) ☒ ☒ (1☒ x)m☒n dx 0 ☒ ☒ (n,1☒ n) ☒ ☒ 0 n☒1 x Therefore dx ☒ ☒ sin n☒ (1☒ x)m☒n EV0000217 · EV0000218 · EV0000219
11. Source slide page 11 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000220 · EV0000221 · EV0000222

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.260] Hello everyone, welcome to Engineering Mathematics One. In last class you have learnt definition and properties of beta function. [00:00:09.732] In this class, you are going to learn relationship between beta gamma function and duplication formula. First, let us see what is relationship between beta gamma functions. [00:00:21.008] It is a very important relation which is very useful and it states that beta of m comma n is equal to gamma of n into gamma of n divided by gamma of m plus n. Now let us see how to prove this. [00:00:36.624] First, we will consider the definition of gamma function. So we know gamma n is equal to 2 times integral 0 to infinity e power minus x square x power 2m minus 1 dx. Then consider gamma n. So gamma n is equal to 2 times integral. [00:00:54.256] integral 0 to infinity e power minus y square y power to n minus 1 dy. So here in gamma m I have used x and in gamma n I have used y doesn't matter because we are dealing with definite integrals. Next [00:01:10.688] Consider the product gamma m into gamma n is equal to two times integral zero to infinity e power minus x square into x power two m minus one dx into [00:01:23.088] 2 times integral 0 to infinity e power minus y square y power 2n minus 1 d y. That is equal to 4 times integral 0 to infinity 0 to infinity e power minus x square into e power minus y square which is e power minus [00:01:40.720] minus of x square plus y square into x power two m minus one into y power two n minus one dx dy. Now let us make use of polar coordinates [00:01:52.512] So let us take x is equal to r cos theta and y is equal to r sin theta. Therefore, your x square plus y square is now r square and dx dy is r into dr into d theta. [00:02:08.992] Now, let us see how the limits of integration change after using the polar coordinates. [00:02:17.156] In Cartesian form, we know that x varies from 0 to infinity and y also varies from 0 to infinity. That means x and y both are positive. So, region of integration is the first quadrant. [00:02:35.300] In this first quadrant, now let us see how does R and theta vary. So in this region, r varies from 0 to infinity and theta varies from 0 to pi by 2. [00:02:49.248] Thus, we can now consider gamma m into gamma n is equal to 4 times integral 0 to pi by 2, integral 0 to infinity, e power minus r square [00:03:03.600] R cos theta whole power two m minus one into r sin theta whole power two n minus one into r into dr into d theta. Now here 0 to pi by 2 are limits for theta and 0 to infinity are limits for r. [00:03:22.276] So I can write this as 4 times integral 0 to pi by 2, integral 0 to infinity. So first let me write all the terms containing r together and then write all the terms containing theta together. So I get it as [00:03:38.032] E power minus r square r power two m minus one r power two n minus one r into cos power two m minus one theta sin power two n minus one theta into d r d theta. [00:03:54.212] Here you can observe, so e power minus r square into r power 2n minus 1 into r power 2n minus 1 into r is purely a function of r and cos power 2n minus 1 theta into sin power 2n minus 1 theta is purely a function of theta. [00:04:14.148] So I can write this as 2 times integral 0 to pi by 2 cos power 2n minus 1 theta sin power 2n minus 1 theta d theta into [00:04:26.272] Two times integral zero to infinity e power minus r square r power two m minus one r power two n minus one r into dr. You can observe here the first integral is nothing but equal to beta m comma n. [00:04:43.716] Now we need to evaluate the second integral. In the second integral, let us put r square is equal to t. So r dr is equal to dt divided by 2. [00:04:57.124] So if we substitute this we get gamma m into gamma n is equal to beta of m comma n into 2 into 1 by 2 into integral 0 to infinity.

EV0000030 · EV0000031 · EV0000032 · EV0000033

Transcript block 2 (source interval 000300-000599 seconds): [00:05:11.728] e to the power minus t because we are putting r square is equal to t, so it is e power minus t into t to the power m minus

one by two plus n minus one by two d t. [00:05:26.048] So, when we simplify this, we get it as beta of m comma n into integral 0 to infinity e power minus t t to the power m plus n minus 1 dt. [00:05:40.164] We know that gamma n is equal to integral 0 to infinity e power minus t t power n minus 1 dt. So here you can observe [00:05:50.832] The expression whatever we got for gamma m into gamma n, which is beta of m, n into gamma m plus n. [00:06:01.700] Thus we can conclude that beta of m comma n is equal to gamma m into gamma n divided by gamma of m plus n. So this is a very very useful result. Remember this. [00:06:17.572] Next let us go for duplication formula for gamma function. [00:06:22.528] It states that square root of pi into gamma of 2m is equal to 2 to the power 2m minus 1 into gamma of m into gamma of m plus 1 by 2. To prove this, we first consider beta of m, n. [00:06:40.420] which is equal to two times integral 0 to pi by 2 sine power 2 n minus 1 theta cos power 2n minus 1 theta d theta. Now [00:06:52.580] Uh we consider beta of m comma m. That means in the place of n I am putting m. So why we are doing this is you can observe in the duplication formula there is only m, there are there are no [00:07:08.708] Two variables there is only one variable. So let me first consider beta of m comma m which is equal to just just replace n by m in the above expression. So I get two times integral 0 to pi by 2 sine power 2 m minus 1 theta cos power 2. [00:07:26.480] 2 m minus 1 theta d theta. So we know that sine 2 theta is equal to 2 sine theta cos theta. So now in the integ integrant you have sine theta cos theta whole power two m minus 1. So we can write it as [00:07:43.232] two times integral zero to pi by two sin two theta divided by two whole power two m minus one into d theta. So it is equal to [00:07:54.224] 2 divided by 2 to the power 2 m minus 1 into integral 0 to pi by 2 sine power 2 m minus 1 theta 2 theta d theta. Now let us put 2 theta is equal to t. That means 2 d theta is equal to dt. So now [00:08:13.828] When theta is equal to 0, t is also equal to 0. When theta is equal to [00:08:22.468] Pi by two, your t is equal to pi. So now you get beta of m comma m is equal to two divided by two to the power two m minus one into integral zero to pi sine power two m minus one t. [00:08:39.296] into dt by 2. So we can cancel this to here. And also you can use the result here 0 to pi sine power 2 m minus 1 t dt is equal to 2 times integral 0 to pi by 2 sin power 2 m minus 1 t. [00:08:56.336] dt. Thus my beta of m comma m now becomes two divided by two to the power two m minus one into integral zero to pi by two sine power two m minus one t dt, which is equal to one by two to the power two m minus one [00:09:14.960] into beta of m comma half. So how do we got this result? That is by using the formula integral 0 to pi by 2 sine power p theta d theta is 1 by 2 into beta of p plus 1 by [00:09:32.592] 2,1 by 2. So this formula you have learnt while studying the properties of beta function. We know that beta of m comma n is equal to gamma m into gamma n divided by gamma of m plus n. [00:09:50.468] Apply this formula to the left hand side which is beta of m comma m so we get left hand side as gamma m into gamma m divided by gamma of m plus m which is gamma of two m.

EV0000034 · EV0000035 · EV0000036 · EV0000037

Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.164] Apply the same formula to the right hand side also, where 1 by 2 to the power 2m minus 1 is a constant and you have beta of m, half, so it is gamma half into gamma m divided by gamma of m plus half. [00:10:22.372] So we can cancel gamma m on both sides of the equation. We get gamma m divided by gamma 2m is equal to 1 by 2 to the power 2m minus 1 into square root of pi because we know gamma half is equal to square root of pi divided by gamma of m plus half. [00:10:42.340] Thus we have square root of pi into gamma of 2m is equal to 2 to the power 2m minus 1 into gamma of m into gamma of m plus half. This is the required result. [00:10:55.108] Next, we consider one important result

that is gamma n into gamma of 1 minus n is equal to pi divided by sin n pi, where n lies between 0 and 1. [00:11:07.972] To prove this, we consider the relationship between beta gamma function. So we can have gamma n into gamma m is equal to beta of m comma n divided by gamma m plus n. Since we need gamma n into gamma of 1 minus n, [00:11:25.264] Let us put M is equal to one minus N. [00:11:29.028] So we have gamma n into gamma of 1 minus n is equal to beta of 1 minus n, comma n, or you can also write it as n comma 1 minus n into gamma of 1 minus n plus n which is gamma 1. [00:11:45.668] And we know gamma 1 is equal to 1, so we have gamma n into gamma of 1 minus n is equal to beta of n, 1 minus n. [00:11:56.676] And also we know beta of m comma n is equal to integral 0 to infinity x power n minus 1 divided by 1 plus x power m plus n dx. This is another form that is of a beta function. You have studied it. [00:12:12.912] while studying the properties of beta function. Beta function as an improper integral. Now therefore we can have beta of n comma 1 minus n as [00:12:24.720] Integral zero to infinity x power n minus one divided by one plus x dx which is equal to pi divided by sin n pi. Why this is because [00:12:36.912] of the fact that integral 0 to infinity x power n minus 1 divided by 1 plus x dx is equal to pi divided by sin n pi. The proof of this you will learn while studying the complex analysis [00:12:56.196] Now you need not have to bother about the proof of this, just concentrate on the result. Thus, in this class you have studied relationship between beta gamma functions, duplication formula, [00:13:11.984] And also this result, which says gamma n into gamma of 1 minus n is equal to pi divided by sine n. With this we end this session. Thank you.

EV0000038 · EV0000039 · EV0000040 · EV0000041

Full transcript: 1990 normalized words, 55 timestamped segments, 803.818667 seconds duration. Low-confidence segments flagged: 0.

EV0000030 · EV0000031 · EV0000032 · EV0000033

5 Problems on Beta an Gamma functions

This lesson is indexed from the source video titled Problems on Beta an Gamma functions. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000042 · EV0000043

Graphs and diagrams

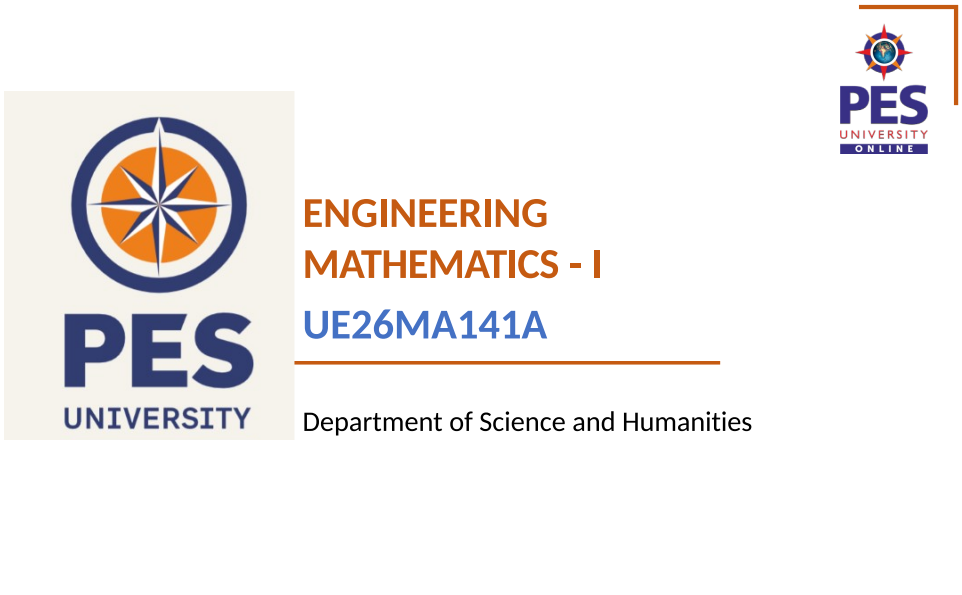


Figure G000044. Problems on Beta and Gamma Functions · source slide page 1.

EV0000224

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

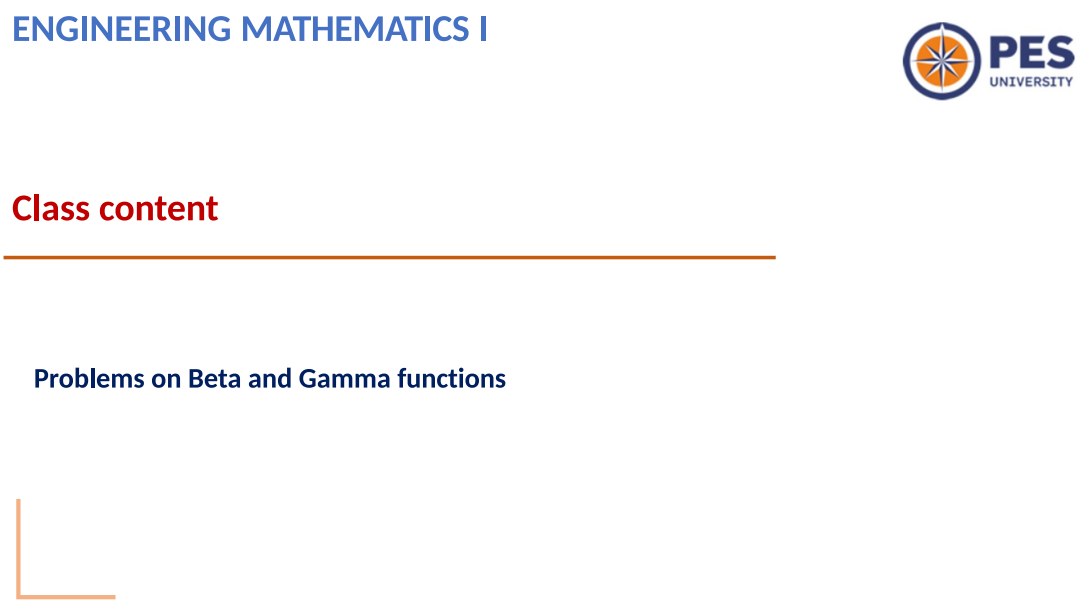


Figure G000045. Problems on Beta and Gamma Functions · source slide page 2.

EV0000227

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

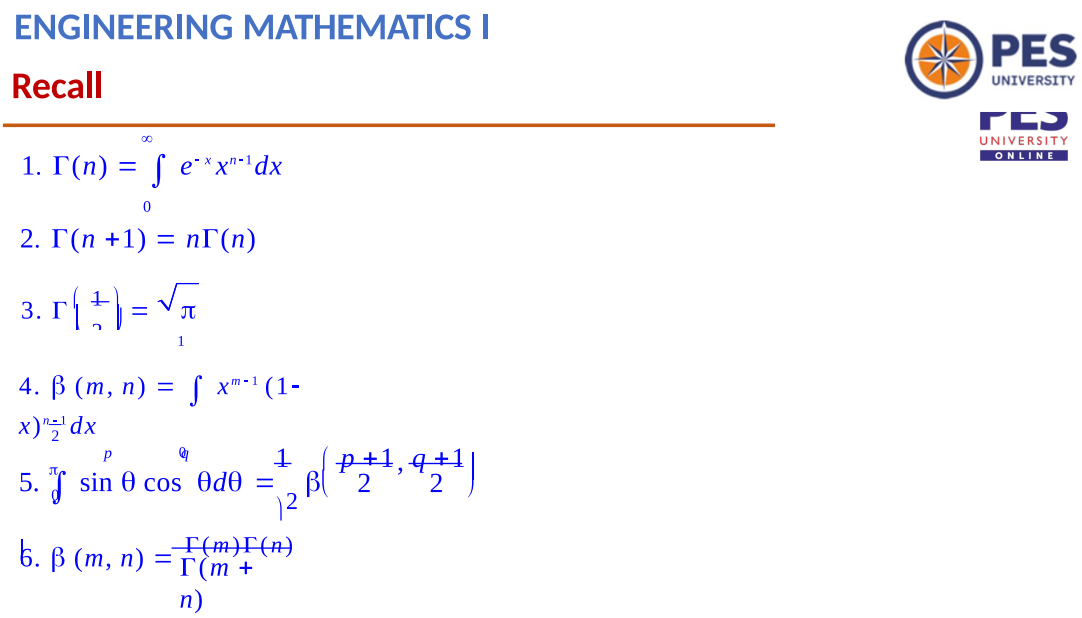


Figure G000046. Problems on Beta and Gamma Functions · source slide page 3.

EV0000230

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

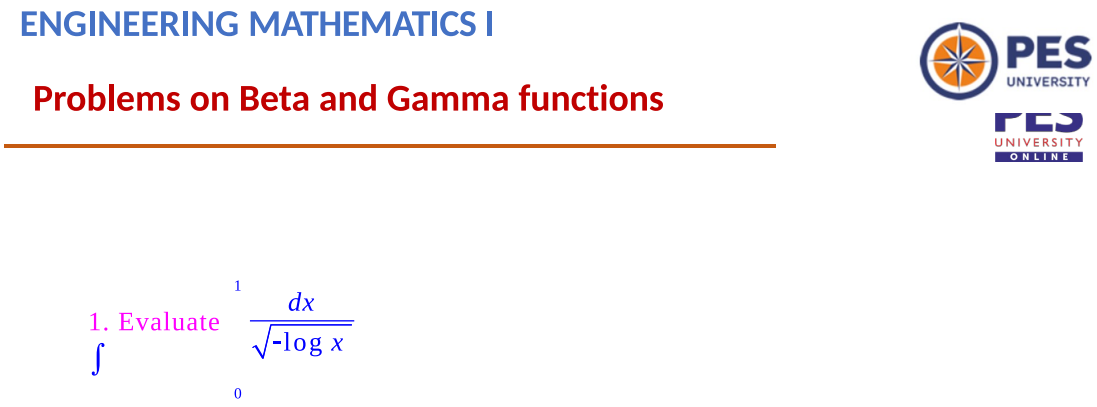


Figure G000047. Problems on Beta and Gamma Functions · source slide page 4.

EV0000233

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

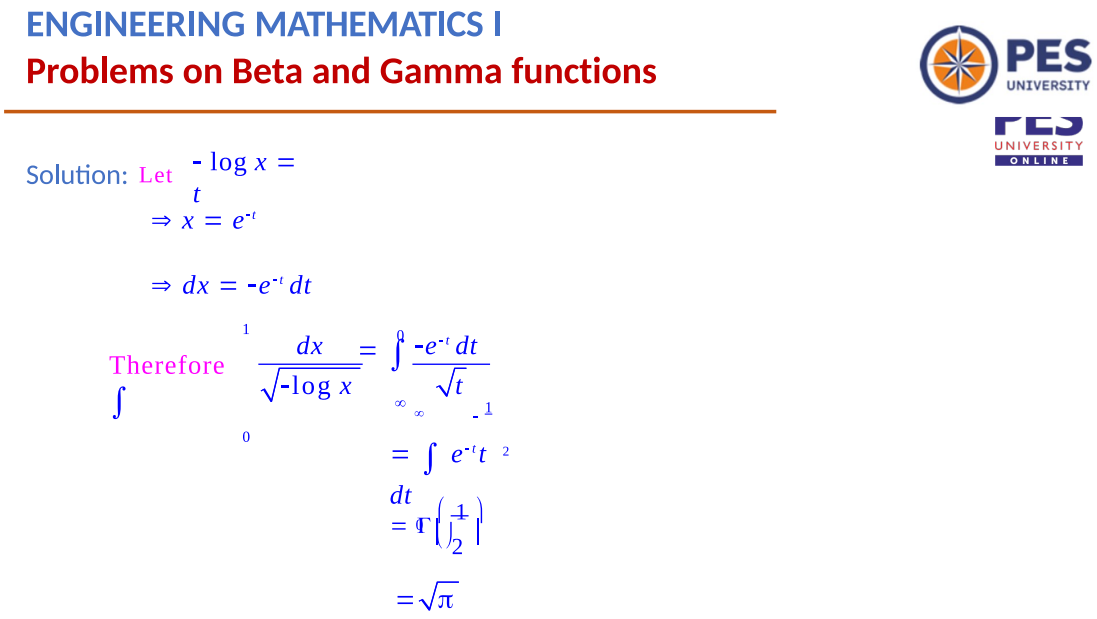


Figure G000048. Problems on Beta and Gamma Functions · source slide page 5.

EV0000236

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000049. Problems on Beta and Gamma Functions · source slide page 6.

EV0000239

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

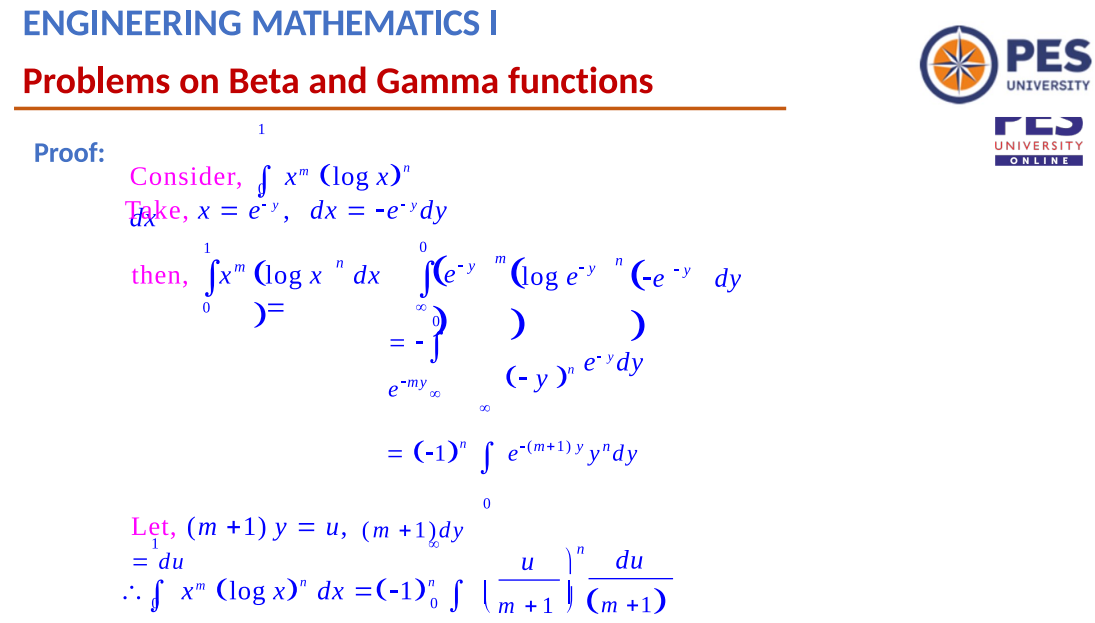


Figure G000050. Problems on Beta and Gamma Functions · source slide page 7.

EV0000242

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

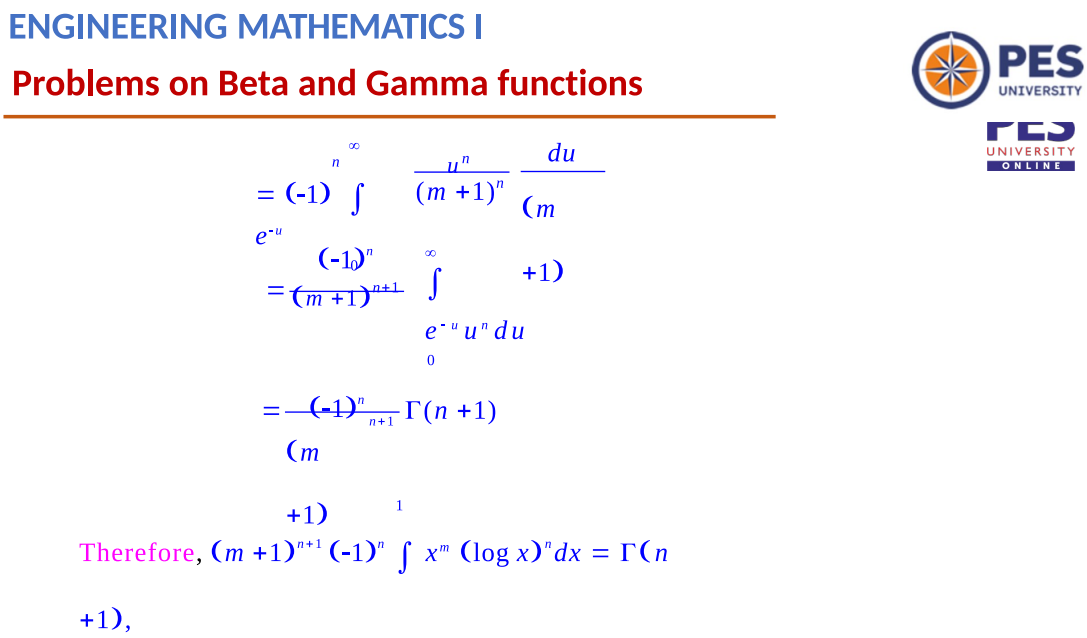


Figure G000051. Problems on Beta and Gamma Functions · source slide page 8.

EV0000245

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

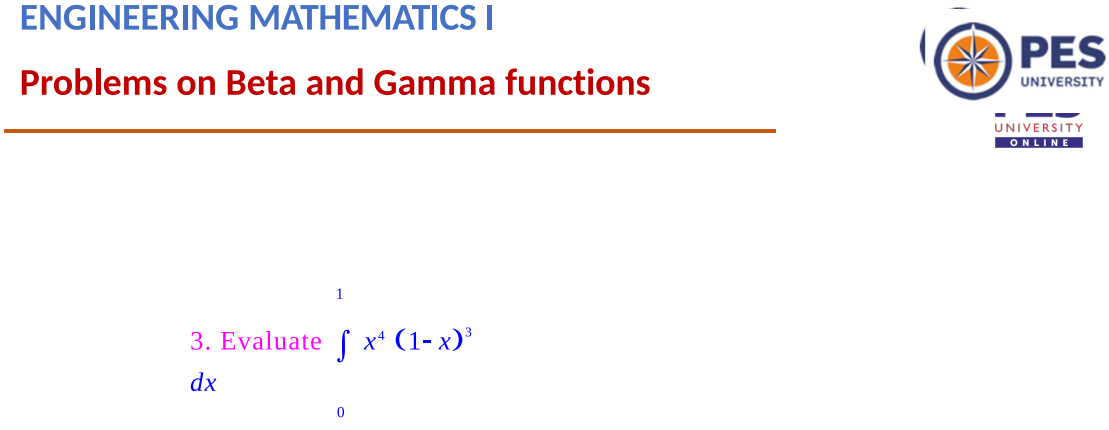


Figure G000052. Problems on Beta and Gamma Functions · source slide page 9.

EV0000248

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

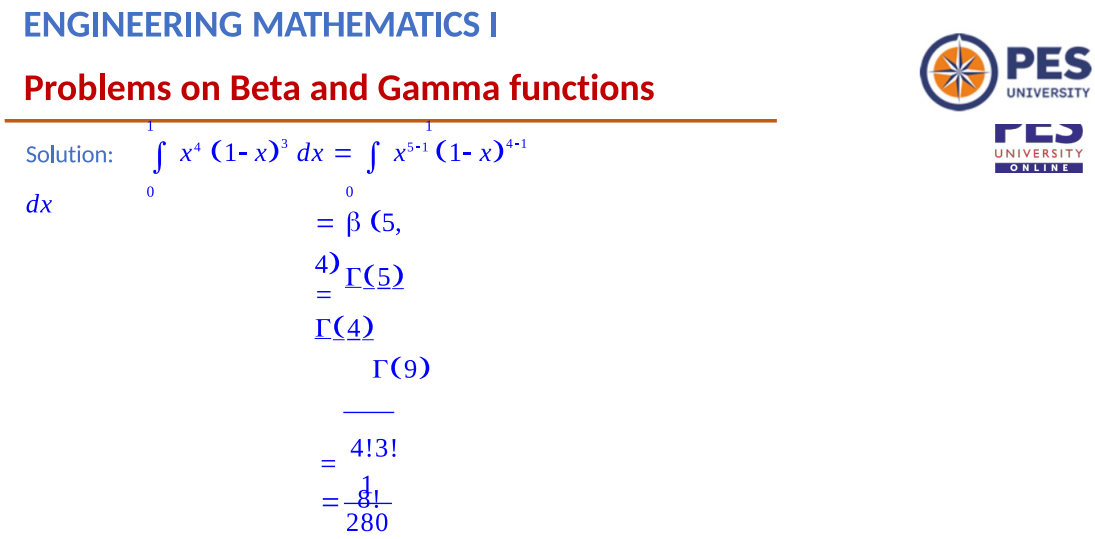


Figure G000053. Problems on Beta and Gamma Functions · source slide page 10.

EV0000251

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

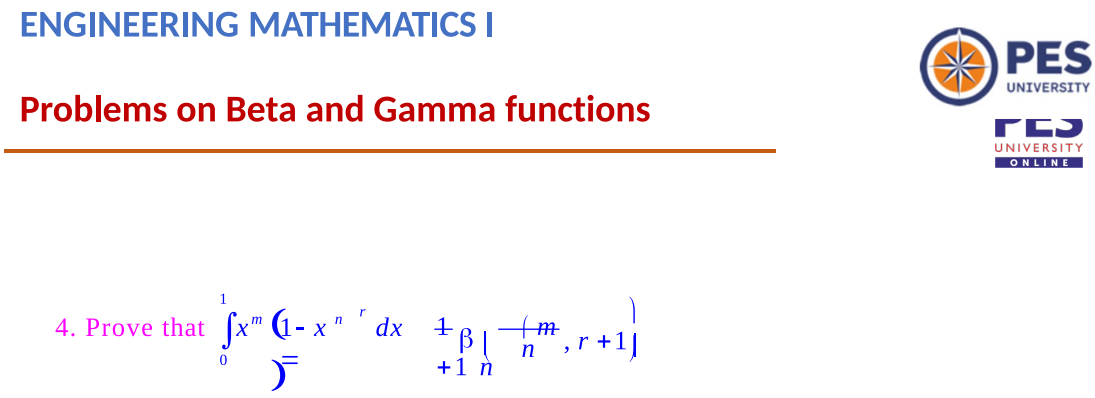


Figure G000054. Problems on Beta and Gamma Functions · source slide page 11.

EV0000254

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

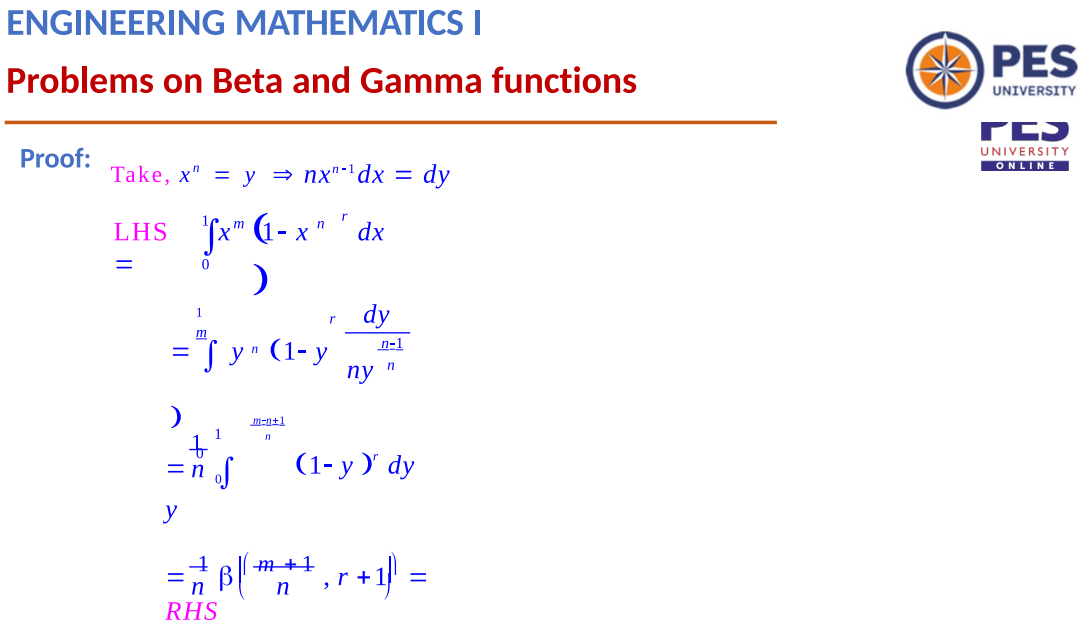


Figure G000055. Problems on Beta and Gamma Functions · source slide page 12.

EV0000257

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

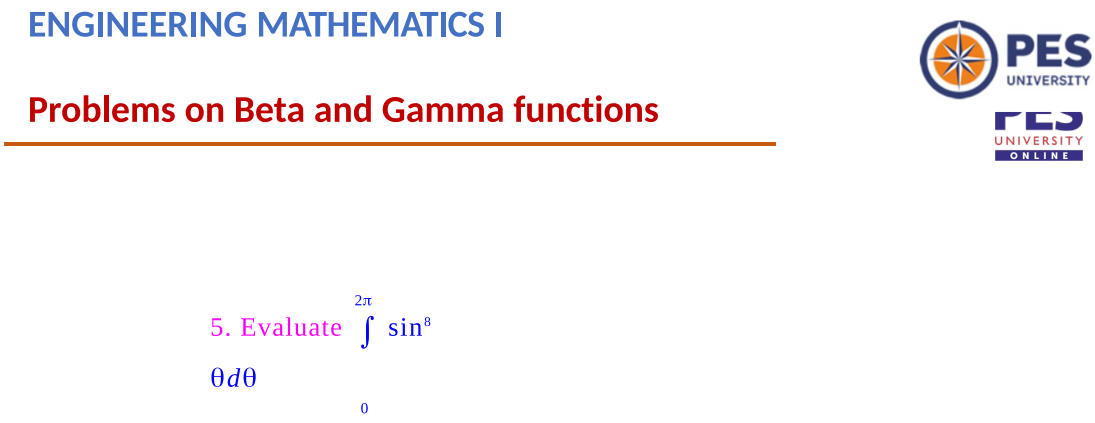


Figure G000056. Problems on Beta and Gamma Functions · source slide page 13.

EV0000260

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

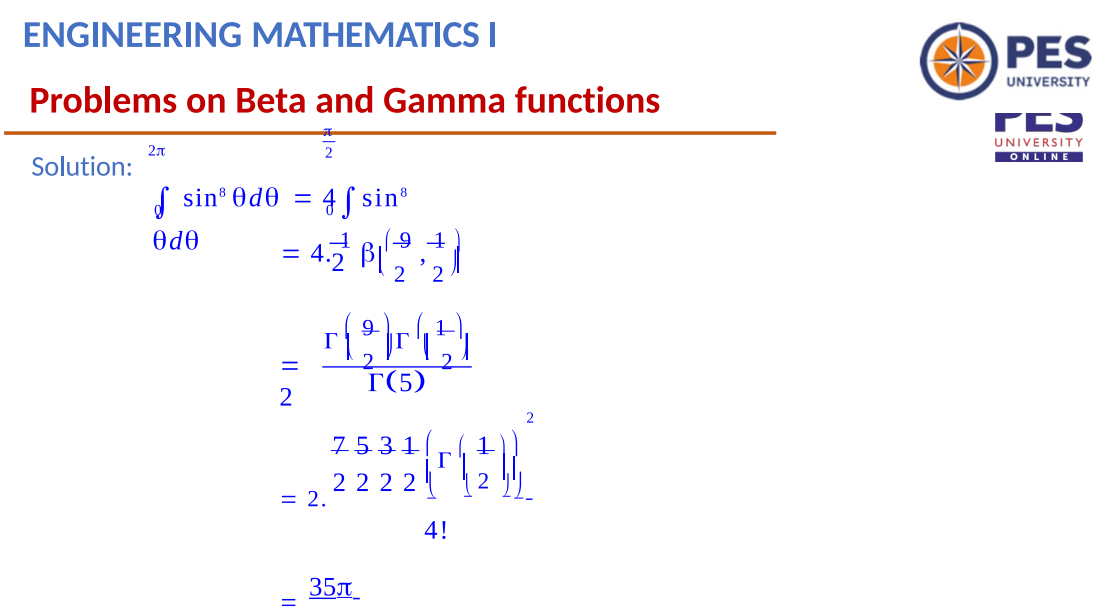


Figure G000057. Problems on Beta and Gamma Functions · source slide page 14.

EV0000263

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

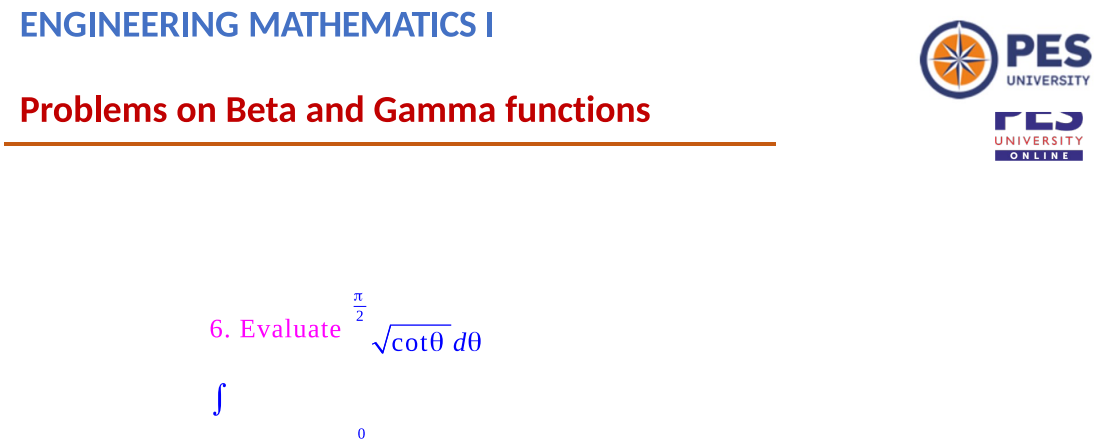


Figure G000058. Problems on Beta and Gamma Functions · source slide page 15.

EV0000266

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

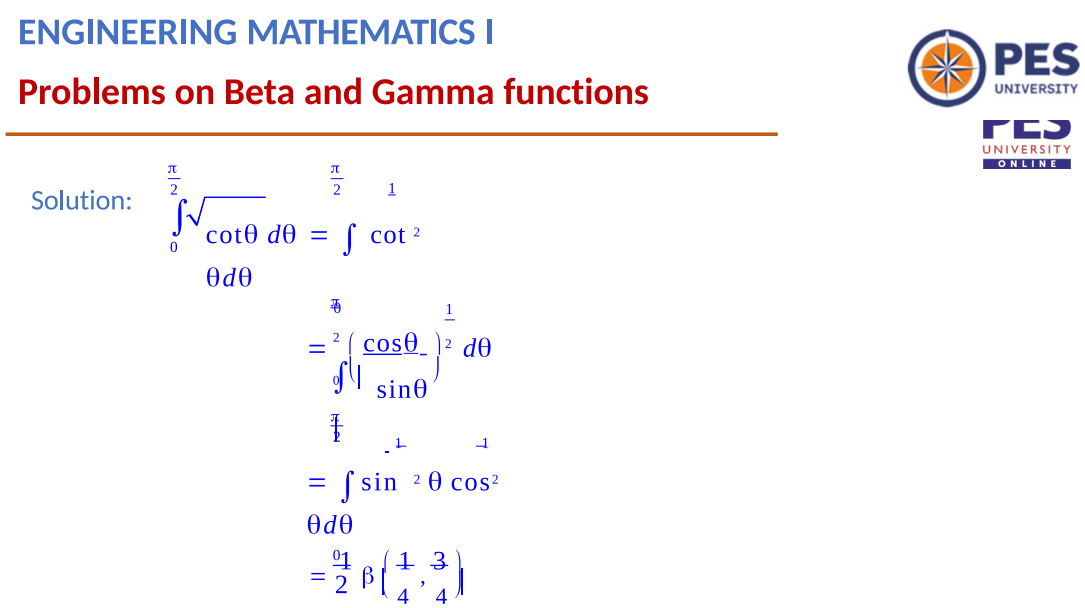


Figure G000059. Problems on Beta and Gamma Functions · source slide page 16.

EV0000269

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

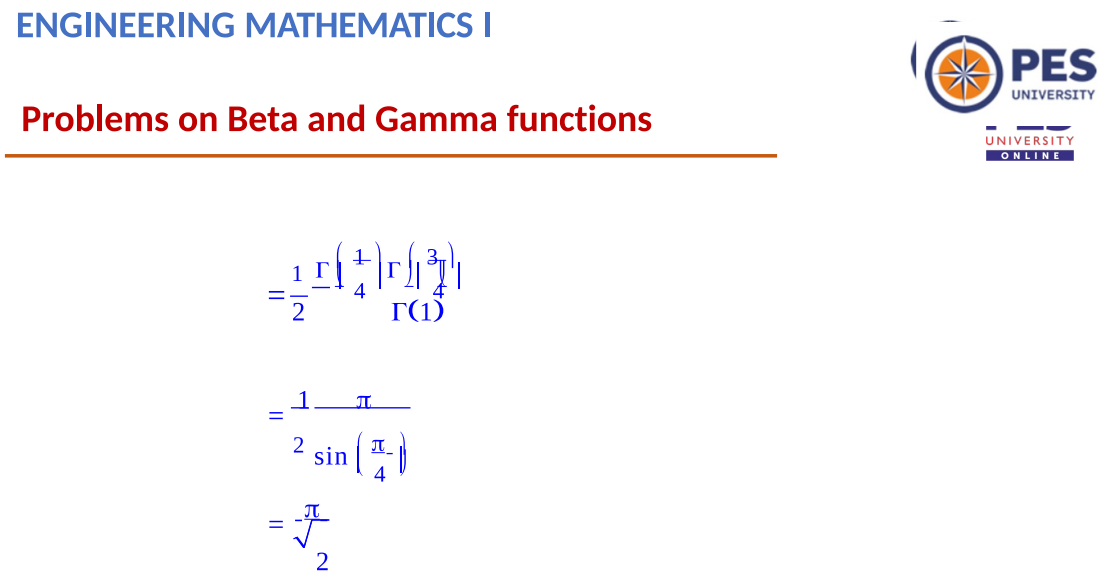


Figure G000060. Problems on Beta and Gamma Functions · source slide page 17.

EV0000272

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000061. Problems on Beta and Gamma Functions · source slide page 18.

EV0000275

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

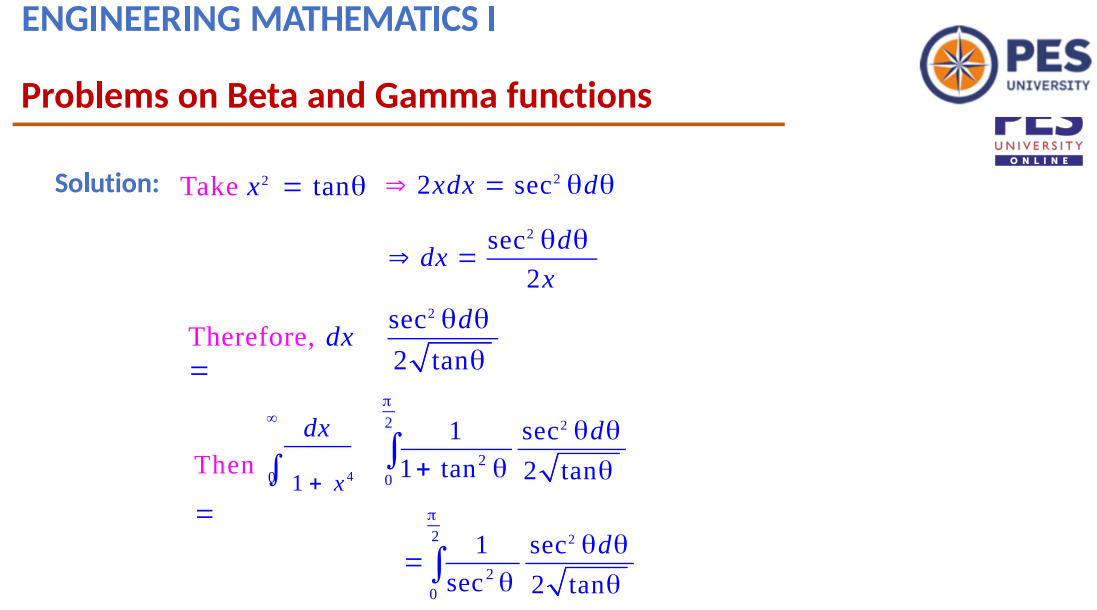


Figure G000062. Problems on Beta and Gamma Functions · source slide page 19.

EV0000278

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

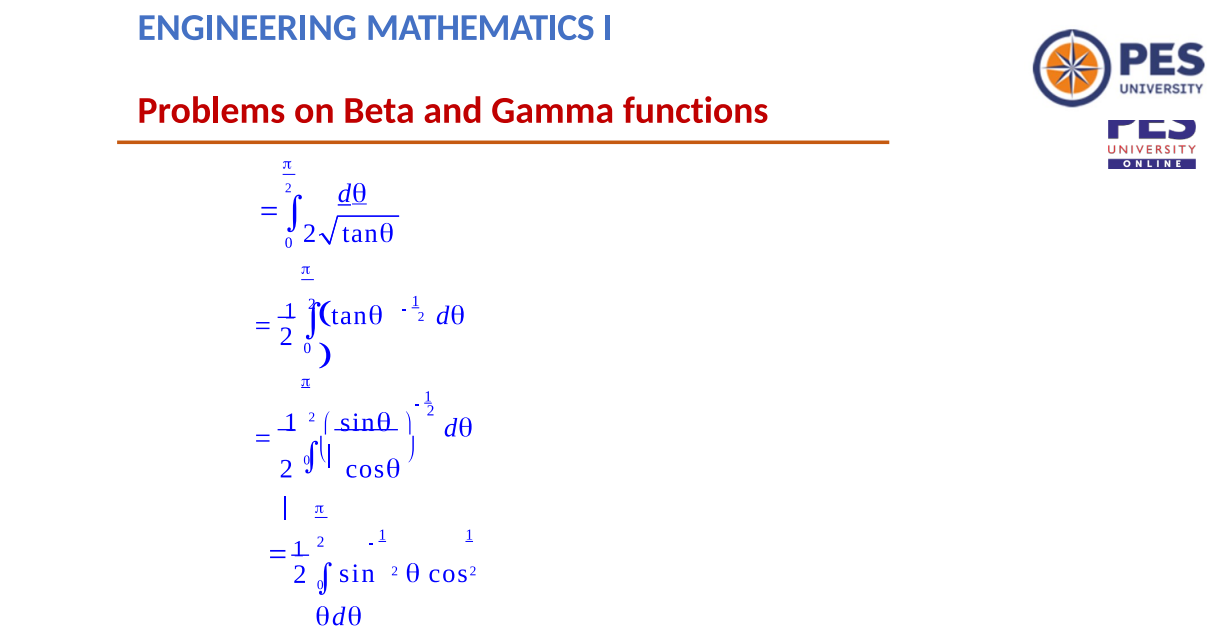


Figure G000063. Problems on Beta and Gamma Functions · source slide page 20.

EV0000281

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

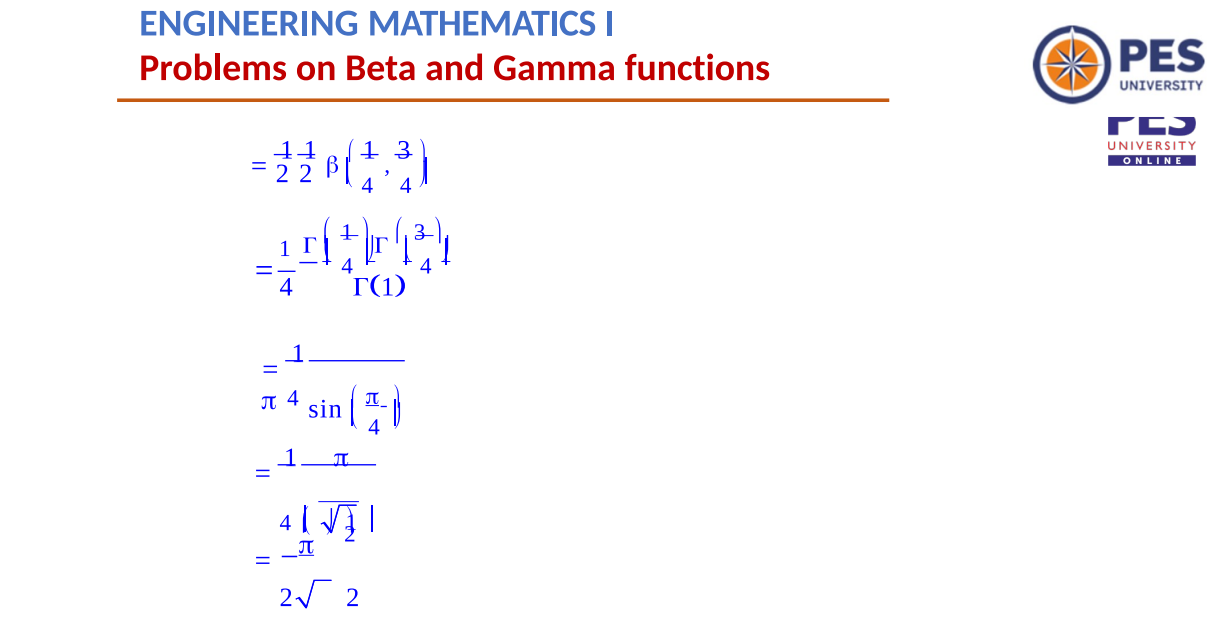


Figure G000064. Problems on Beta and Gamma Functions · source slide page 21.

EV0000284

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000065. Problems on Beta and Gamma Functions · source slide page 22.

EV0000287

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000223 · EV0000224 · EV0000225
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I Class content Problems on Beta and Gamma functions EV0000226 · EV0000227 · EV0000228
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I Recall ☒ ☒(n) ☒ ☒ e☒ x xn☒1dx 0 ☒(n ☒1) ☒ n☒(n) 3. ☒☒ 1 ☒ ☒ ☒ ☒ 2 ☒ ☒ ☒ 4. ☒ (m, n) ☒ ☒ xm☒1 (1☒ x)n☒1dx 0 ☒ 1 2 1 ☒ p ☒1 q ☒1 ☒ 5. ☒ sin ☒ cos ☒d☒ ☒ 2 ☒ ☒ p q , ☒ 2 2 ☒ ☒ 0 6. ☒ (m, n) ☒ ☒(m)☒(n) ☒(m ☒ n) EV0000229 · EV0000230 · EV0000231
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions 1 1. Evaluate ☒ 0 dx ☒log x EV0000232 · EV0000233 · EV0000234
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions ☒ log x ☒ t Solution: Let ☒ x ☒ e☒t ☒ dx ☒ ☒e☒t dt 1 Therefore ☒ 0 0 dx ☒e☒t dt ☒ ☒ ☒log x t ☒ ☒ ☒ ☒ e☒tt 2 dt 0 1 ☒ ☒ ☒☒ 1 ☒ ☒ 2 ☒ ☒ ☒ ☒ ☒ EV0000235 · EV0000236 · EV0000237
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions 2. Prove that ☒m ☒1☒n☒1 ☒☒1☒n ☒ xm ☒log x☒ndx ☒ ☒☒n ☒1☒, 0 where n is positive integer and m ☒ ☒1 1 EV0000238 · EV0000239 · EV0000240

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions 1 Consider, ☒ xm ☒log x☒n dx Proof: 0 dx ☒ ☒e☒ ydy 0 Take, x ☒ e☒ y , 1 ☒ ☒ ☒ ☒ m ☒ ☒ n ☒ ☒ ☒ n ☒ 0 ☒ y e m then, x log x dx ☒ ☒ y log e ☒ y ☒e dy ☒ 0 ☒ ☒☒ e☒my ☒☒ y ☒n e☒ ydy ☒ ☒ ☒☒1☒n ☒ e☒(m☒1) y yndy 0 Let, (m ☒1) y ☒ u, (m ☒1)dy ☒ du ☒ 1 ☒ n ☒☒ xm ☒log x☒n dx ☒☒☒1☒n ☒ e☒u ☒ du u ☒ ☒ m ☒1 ☒ ☒m ☒1☒ ☒ ☒ 0 0 EV0000241 · EV0000242 · EV0000243
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions un ☒ ☒☒1☒ ☒ e☒u 0 du ☒ n ☒m ☒1☒ n (m ☒1) ☒☒1☒n ☒m ☒1☒n☒1 ☒ ☒ e☒uundu 0 ☒ ☒☒1☒n ☒ n☒1 ☒(n ☒1) ☒m ☒1☒ Therefore, ☒m ☒1☒n☒1 ☒☒1☒n ☒ xm ☒log x☒ndx ☒ ☒☒n ☒1☒, 0 1 EV0000244 · EV0000245 · EV0000246
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions 3. Evaluate ☒ x4 ☒1☒ x☒3 dx 0 1 EV0000247 · EV0000248 · EV0000249
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions 1 1 Solution: ☒ x4 ☒1☒ x☒3 dx ☒ ☒ x5☒1 ☒1☒ x☒4☒1 dx 0 0 ☒ ☒ ☒5, 4☒ ☒ ☒☒5☒ ☒☒4☒ ☒☒9☒ ☒ 4!3! 8! 1 ☒ 280 EV0000250 · EV0000251 · EV0000252
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions 1 ☒ m ☒1 n 1 ☒ 0 ☒ ☒ ☒ r ☒ ☒ m n 4. Prove that x 1☒ x dx ☒ , r ☒1☒ n ☒ ☒ EV0000253 · EV0000254 · EV0000255
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions Proof: Take, xn ☒ y ☒ nxn☒1dx ☒ dy 1 ☒ ☒ r ☒ m n LHS ☒ x 1☒ x dx 0 ☒ ☒ y n ☒1☒ y ☒ 0 dy n☒1 1 m r ny n 1 1 m☒n☒1 n ☒ ☒ y ☒1☒ y ☒r dy n 0 ☒ 1 ☒ ☒ m ☒1 , r ☒1☒ ☒ RHS ☒ ☒ n n ☒ ☒ EV0000256 · EV0000257 · EV0000258
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions 2☒ 5. Evaluate ☒ sin8 ☒d☒ 0 EV0000259 · EV0000260 · EV0000261
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions ☒ 2☒ 2 ☒ sin8 ☒d☒ ☒ 4☒sin8 ☒d☒ Solution: 0 0 ☒ 4. 1 ☒ ☒ 9 , 1 ☒ ☒ 2 2 ☒ 2 ☒ ☒ ☒☒ 9 ☒☒☒ 1 ☒ ☒ 2 ☒ ☒ 2 ☒ ☒ ☒ ☒ ☒ ☒☒5☒ ☒ 2 2 7 5 3 1 ☒☒☒ 1 ☒☒ ☒ 2 ☒☒ 2 2 2 2 ☒ ☒ 2. ☒ ☒ ☒☒ 4! ☒ 35☒ 2 EV0000262 · EV0000263 · EV0000264
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions ☒ 2 cot☒ d☒ 6. Evaluate ☒ 0 EV0000265 · EV0000266 · EV0000267
16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions ☒ ☒ cot☒ d☒ ☒ ☒ cot 2 ☒d☒ 0 1 2 2 Solution: ☒ 0 ☒ 1 2 ☒ cos☒ ☒2 d☒ ☒ ☒☒ sin☒ ☒ 0 ☒ ☒ ☒ ☒ ☒sin 2 ☒ cos2 ☒d☒ 0 ☒ 1 1 2 ☒ 1 ☒ ☒ 1 , 3 ☒ ☒ 4 4 ☒ 2 ☒ ☒ EV0000268 · EV0000269 · EV0000270
17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions ☒☒ 1 ☒☒☒ 3 ☒ ☒ 4 ☒ ☒ 4 ☒ 1 ☒ ☒ ☒ ☒ 2 ☒☒1☒ ☒ 1 ☒ ☒ 2 sin ☒ ☒ ☒ ☒ 4 ☒ ☒ ☒ ☒ ☒ 2 EV0000271 · EV0000272 · EV0000273
18. Source slide page 18 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions ☒ dx 7. Evaluate ☒1☒ x4 0 EV0000274 · EV0000275 · EV0000276

19. Source slide page 19 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions Solution: ☒ 2xdx ☒ sec2 ☒d☒ Take x2 ☒ tan☒ sec2 ☒d☒ ☒ dx ☒ 2x sec2 ☒d☒ Therefore, dx ☒ 2 tan☒ ☒ ☒ dx 1 sec2 ☒d☒ 2 Then ☒1☒ x4 ☒ ☒ 0 1☒ tan ☒ 2 2 tan☒ 0 ☒ 1 sec2 ☒d☒ 2 ☒ ☒ 0 sec ☒ 2 2 tan☒ EV0000277 · EV0000278 · EV0000279

20. Source slide page 20 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions ☒ 2 d☒ ☒ 0 ☒ 2 tan☒ ☒ ☒ 1 2 ☒ 1 ☒ ☒ ☒ tan☒ 2 d☒ 2 0 ☒ ☒ 1 ☒ 1 2 ☒ sin☒ ☒ 2 d☒ 2 ☒☒ cos☒ ☒ 0 ☒ ☒ ☒ 1 2 ☒ 1 1 ☒sin 2 ☒ cos2 ☒d☒ ☒ 2 0 EV0000280 · EV0000281 · EV0000282

21. Source slide page 21 (extracted text): ENGINEERING MATHEMATICS I Problems on Beta and Gamma functions ☒ 1 1 ☒ ☒ 1 , 3 ☒ ☒ 4 4 ☒ 2 2 ☒ ☒ ☒☒ 1 ☒☒☒ 3 ☒ ☒ 4 ☒ ☒ 4 ☒ 1 ☒ ☒ ☒ ☒ 4 ☒☒1☒ ☒ ☒ 1 ☒ 4 sin ☒ ☒ ☒ ☒ 4 ☒ ☒ ☒ ☒ 1 ☒ 4 ☒ 1 ☒ ☒ 2 ☒ ☒ ☒ ☒ ☒ 2 2 EV0000283 · EV0000284 · EV0000285

22. Source slide page 22 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000286 · EV0000287 · EV0000288

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.164] Hello students, welcome back to Engineering Mathematics 1. In previous few classes, you have studied definition and properties of beta gamma functions, relationship between beta gamma function and duplication formula. [00:00:15.460] In today's class, we are going to solve some problems on beta gamma function. Before that, we will recall some of the formulas you have studied earlier, which are useful in solving problems for today's class. [00:00:32.452] First one is gamma n is equal to integral 0 to infinity e to the power minus x x power n minus 1 dx. This is definition of gamma function. Second one is gamma of n plus 1 is equal to n gamma n. [00:00:47.012] You can also consider the result gamma of n plus one is equal to n factorial when n is a positive integer which is derived from this result. Third one is gamma half is equal to square root of pi. [00:01:00.548] Porth one is beta of m comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 1 dx. [00:01:12.260] Next one is integral zero to pi by two sine power p theta cos power q theta t theta is equal to one by two times beta of p plus one by two comma q plus one by two, which is derived from the previous result itself. [00:01:28.612] And the last one, but very important one, that is beta of m, comma n is equal to gamma m into gamma n divided by gamma of m plus n. That is the relationship between beta gamma function. [00:01:43.396] So let us remember these formulas before going to the problems in this class. Lear these formulas, all these formulas are very very important. Consider the first problem evaluate integral 0 to 1 dx divided by square root of minus log x. [00:02:03.012] Now in the problem there is minus log x to the power half in the denominator. So let me take a substitution for minus log x. So in the solution, let me take minus log x is equal to p. [00:02:17.796] If minus log x is equal to t, x will be e to the power minus t, and dx is minus a power minus t dt. Now let us see how the limits of integration change with the substitution. So we know minus log x is equal to [00:02:35.332] log x to the power minus 1, which is equal to log of 1 divided by x, and we have taken it as t. Therefore, when x is equal to 0, we have t is equal to log of 1 by 0 [00:02:51.344] Which is log infinity, which you can consider it as infinite. Thus you have when x is 0, t is equal to infinity, and when x is equal to 1, we have t is equal to log of 1 divided by 1, that is log 1 and log 1 is equal to 0. [00:03:10.788] So new limits of integration are infinity to zero. [00:03:15.172] Therefore, integral 0 to 1 dx divided by square root of minus log

x is equal to integral infinity to 0 minus a power minus d dt divided by square root of t. [00:03:31.396] which is equal to integral 0 to infinity e to the power minus t. So why the limits have become 0 to infinity here is you have a negative sign, when you take the negative sign outside, it becomes minus integral infinity to 0 [00:03:46.496] And you know minus of integral a to b is integral b to a so it is integral zero to infinity a power minus t t to the power minus half dt. [00:03:58.180] So now you can compare this with the definition of gamma function. So you have gamma n is equal to integral 0 to infinity e to the power minus t t to the power n minus 1 dt. So you can express this as [00:04:15.344] Integral zero to infinity e to the power minus t t to the power minus one by two d we can write it as integral zero to infinity e power minus t t to the power one by two minus one dt. [00:04:31.968] So when you compare these two, so you will get to know that n is equal to 1 divided by 2. So I can call this as gamma half 1 by 2. And we have already studied the result that gamma half is equal to square root of pi. So we have evaluated [00:04:51.716] the given integral with the answer square root of 5. [00:04:56.836] Now consider the second problem. True that m plus 1 whole to the power n plus 1 into minus 1 power n into integral 0 to 1 x power m log x to the power n dx is equal to gamma of n plus 1.

EV0000042 · EV0000043 · EV0000044 · EV0000045

Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.796] where n is a positive integer and m is greater than minus. To prove this result, first let me consider the integral 0 to 1 x power m log x to the power n dx. [00:05:29.860] In this integral, let me make some substitution so that I will get gamma function. In gamma function, we need an exponential term. So let me take the substitution x is equal to e to the power minus 1. [00:05:45.572] When x is equal to e to the power minus y, dx is equal to minus of e to the power minus y, dy. Now let us see how the limits of integration change with this substitution. x is equal to e to the power minus y implies. [00:06:04.068] log x is equal to minus y or we can say minus of log x is equal to y or we have log of 1 divided by x is equal to y which is uh [00:06:19.716] Similar to the previous problem, where also we got the same limits of integration. So here also when x is [00:06:28.484] equal to 0 that means y is equal to log of 1 by 0 or log infinity which is infinity. When x is equal to 1, we have y is equal to log of 1 by 1 and log of 1 by 1 is log 1 [00:06:44.688] and log one is equal to zero. So the new limits of integration are infinity to zero. Thus you have integral 0 to 1 x power m log x power n dx is equal to [00:07:01.264] integral infinity to 0 e to the power minus y whole to the power m into log of e power minus phi whole to the power n into in the place of dx now I am substituting minus [00:07:18.448] e to the power minus y d y which is equal to give the minus n outside so it is minus integral 0 to infinity e to the power minus y whole to the power m. So this term e to the power minus y whole to the power m is nothing but equal to e to the power [00:07:38.240] minus m y and log of e to the power minus y whole to the power n. You know log of e to the power minus y is [00:07:48.356] Minus y so it is minus y to the power n and in the last term we have already taken the negative sign outside so it is e to the power minus y d y. [00:08:03.140] And now minus of integral infinity to 0, we can write it as integral 0 to infinity. And from this term, minus y whole to the power n, I have taken [00:08:16.356] Minus 1 to the power n outside. So it is minus 1 whole to the power n into integral 0 to infinity e to the power minus m plus 1 y. How we got this is [00:08:30.980] From the previous step, here you have e to the power minus m y, here you have e to the power minus y. So you will get e to the power minus of m plus 1 into y. Which is [00:08:47.300] Now let us substitute m plus 1 is into y is equal to u. Then m plus 1 into dy is equal to du and also you can consider here y is equal to u divided by m plus 1. [00:09:07.268] Therefore, integral 0

to 1 x power m log x all power n dx is equal to minus 1 power n integral 0 to infinity e to the power minus u because we have taken m plus 1 y is equal to u into [00:09:26.848] Y power n y is u by m plus one. So it is u by m plus one whole to the power n into dy dy is du divided by m plus one. Now consider the third problem. Evaluate integral 0 to 1 x to the power 4 1 minus x whole q dx. [00:09:46.628] Now let us see how to solve this. Consider the left hand side that is integral 0 to 1 x power 4 1 minus x whole cube dx that can be written as x to the power 5 minus 1 into 1 minus x whole to the power 4 minus 1 dx. So we have d

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 3 (source interval 000600-000899 seconds): [00:10:06.340] Like this, just to compare this with the beta function definition, which is beta of m, comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 1 dx. [00:10:23.748] So we can write it as beta of 5 comma 4. And now you can use the relationship between beta and gamma function which is beta of m comma n. If you want to write it in general, you can write it for beta of [00:10:39.440] M comma n which is equal to gamma of m into gamma of n divided by gamma of m plus n. So now beta five comma four can be written as gamma five into gamma four divided by gamma of five plus four which is gamma of nine. [00:10:58.628] Now we can consider this result gamma of n plus 1 is equal to n factorial where n is a positive integer. So gamma of 5 is equal to 4 factorial. [00:11:11.920] Gamma four is three factorial and gamma nine is eight factorial. So we can write it as four factorial three factorial divided by eight factorial which is equal to one hundred two hundred and eighty. [00:11:26.756] Now let us consider the fourth problem that is prove that integral 0 to 1 x to the power m 1 minus x power n whole power r dx is equal to 1 by n into beta of m plus 1 divided by n comma r plus 1. [00:11:46.628] Now let us consider the left hand side. Where you have 1 minus x power n whole to the power r. But we want to write the integral on the left hand side in terms of beta function. [00:12:01.380] And beta function definition says that beta of m comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 20x. [00:12:17.316] But in this problem we have x to the power n here. So let us make a substitution for this. So to prove this result, first take x power n is equal to y. Then n into x power n minus 1 dx is equal to dy. [00:12:35.780] With this substitution, the limits of integration will not change because when x is equal to 0, y is 0. When x is equal to 1, y is also equal to 1. [00:12:49.956] So, when you consider the left hand side integral 0 to 1 x power m 1 minus x power n whole to the power r dx, we get this as integral 0 to 1 y to the power m by n. [00:13:04.868] So why we got this y to the power f by n is you have x power m and it is equal to x O to the power m and by the substitution x is equal to y to the power 1 by n O to the power m so it is [00:13:22.160] y to the power m divided n into 1 minus x power n is nothing but equal to y. So it is 1 minus y or to the power r into dx. What is dx? dx is equal to [00:13:39.108] From the substitution it is dy divided by n into x power n minus 1, which is nothing but n into y power n minus 1 divided by n [00:13:52.548] That is equal to 1 by n, we can keep it outside the integral sign, integral 0 to 1. So here you have y to the power m by n. [00:14:07.300] and into 1 divided by y to the power n minus 1 divided by n. [00:14:15.492] which is equal to y to the power m by n divided by y to the power 1 minus 1 by n, which is equal to y to the power m by n minus 1 plus 1 by n. [00:14:32.324] So I can write this as y to the power m plus 1 minus n divided by n. [00:14:44.740] So that is what I got here y to the power m minus n plus 1 divided by n into 1 minus y 1 to the power r into dy. [00:14:55.428] So it is nothing but equal to 1 by n into beta of m plus 1 by n comma r plus 1. So because now uh this integral, integral 0 to 1

EV0000050 · EV0000051 · EV0000052 · EV0000053

Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.552] Y to the power m minus n plus one divided by n you can write it as m plus one divided by n minus one into one minus phi whole to the power r. R you can write it as r plus one minus one. [00:15:29.348] into dy. So compare it with the definition of beta of m, comma n, which says integral 0 to 1 y to the power m minus 1 into 1 minus y to the power n minus 1 dy which is nothing but equal to what? Beta of m comma n. [00:15:48.484] So you can conclude this integral is nothing but equal to beta of m plus 1 by n, r plus 1, which is exactly equal to right hand side.

EV0000054 · EV0000055 · EV0000056 · EV0000057

Full transcript: 2377 normalized words, 62 timestamped segments, 962.88 seconds duration. Low-confidence segments flagged: 0.

EV0000042 · EV0000043 · EV0000044 · EV0000045

6 Problems on Beta and Gamma Function_continued

This lesson is indexed from the source video titled Problems on Beta and Gamma Function_continued. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000058 · EV0000059

Graphs and diagrams



Figure G000066. Problems on Beta and Gamma Functions Continued · source slide page 1.

EV0000290

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000067. Problems on Beta and Gamma Functions Continued · source slide page 2.

EV0000293

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

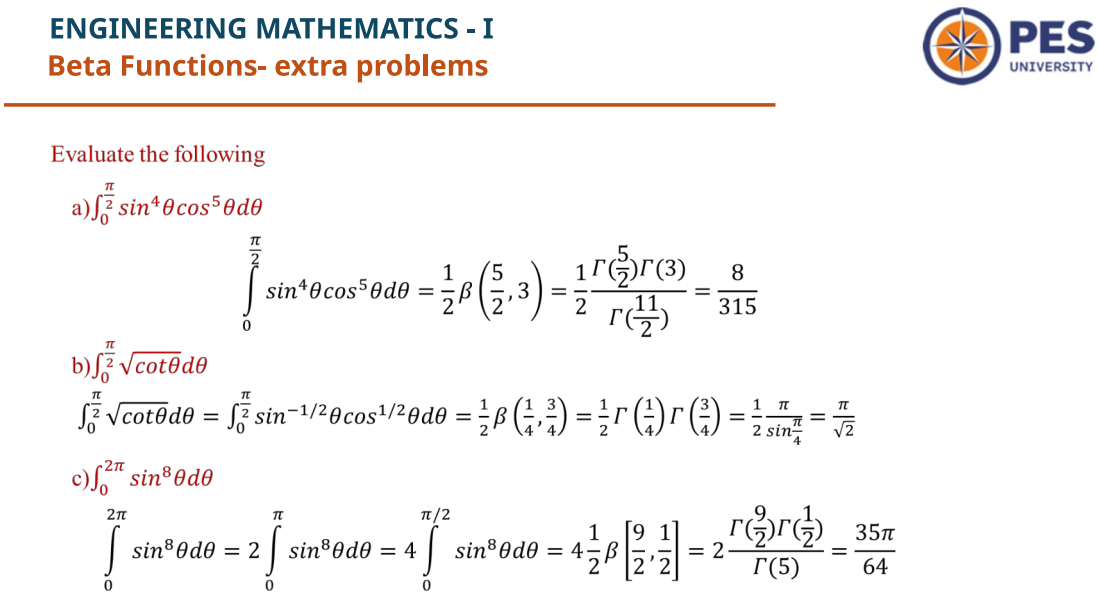


Figure G000068. Problems on Beta and Gamma Functions Continued · source slide page 3.

EV0000296

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

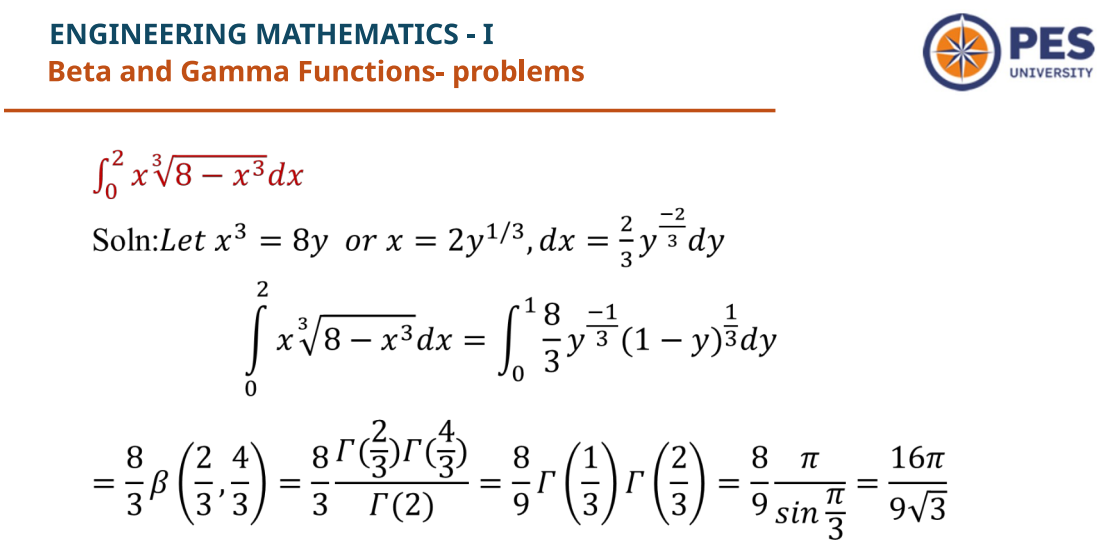


Figure G000069. Problems on Beta and Gamma Functions Continued · source slide page 4.

EV0000299

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

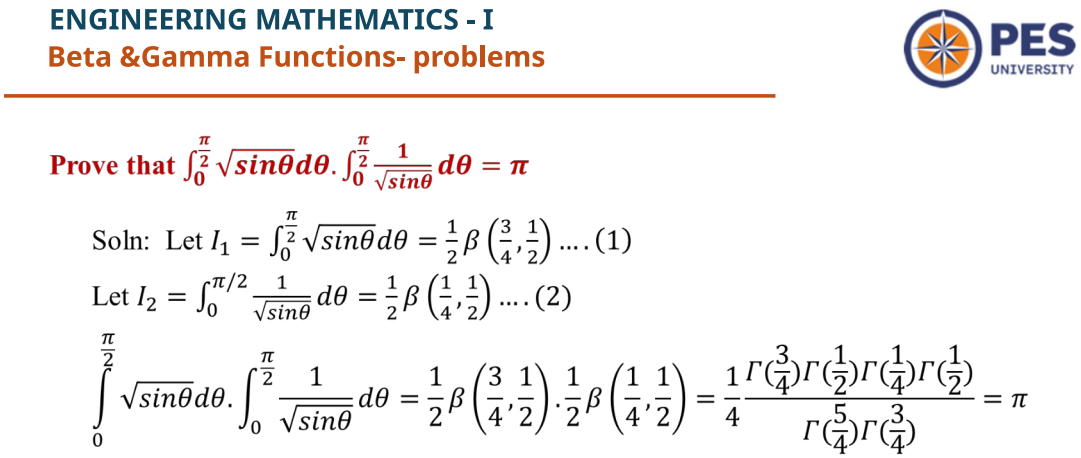


Figure G000070. Problems on Beta and Gamma Functions Continued · source slide page 5.

EV0000302

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000071. Problems on Beta and Gamma Functions Continued · source slide page 6.

EV0000305

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000289 · EV0000290 · EV0000291
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 4: SPECIAL FUNCTIONS AND SERIES SOLUTION OF DE SESSION: 5 SUB-TOPIC: Problems on beta and gamma function Department of Science and Humanities EV0000292 · EV0000293 · EV0000294
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Beta Functions- extra problems EV0000295 · EV0000296 · EV0000297
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Beta and Gamma Functions- problems EV0000298 · EV0000299 · EV0000300
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Beta &Gamma Functions- problems EV0000301 · EV0000302 · EV0000303
6. Source slide page 6 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000304 · EV0000305 · EV0000306

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.868] Consider the fifth problem evaluate integral zero to two pi sine power eight theta d theta. Consider integral zero to

two pi sine power eight theta d theta, which is equal to four times integral zero to pi by two sine power eight theta d theta. [00:00:19.172] This I got by using one of the properties of the definite integral, which is integral 0 to 2a f of x dx is equal to 2 times integral 0 to a f of 2a minus x dx. [00:00:40.100] If you use this property twice, you will get this. And after this, to simplify, I use one of the properties of beta function we have studied earlier, which is integral 0 to pi by 2, sine power p theta, cos power q theta. [00:00:57.776] d theta is equal to one by two times beta of p plus one divided by two, comma q plus one divided by two. [00:01:09.796] In this case, you have P is equal to 8 and Q is equal to 0. So we can write this as 4 times 1 by 2 into beta of P plus 1 by 2. That is 8 plus 1 by 2, which is 9 by 2. [00:01:26.340] Comma q plus one by two that is zero plus one divided by two that is one by two. [00:01:33.668] Now you can apply relationship between beta gamma function and write it as 2 times gamma of 9 by 2 into gamma of 1 by 2 divided by gamma of 9 by 2 plus 1 by 2 which is gamma of n by 2 that is gamma 5. [00:01:50.884] And now use the property of gamma function that is gamma of n plus 1 is equal to n into gamma n. [00:02:00.100] So gamma of 9 by 2 we can write it as gamma of 7 by 2 plus 1. So it is equal to 7 by 2 into gamma of 7 by 2. [00:02:13.380] And gamma 7 by 2 can be written as gamma of 5 by 2 plus 1. So it is equal to 7 by 2 into gamma 5 by 2 is 5 by 2 into gamma of 5 by 2. Again gamma 5 by 2 can be written as [00:02:32.356] 3 by 2 plus 1 and gamma 3 by 2 can be written as gamma of 1 by 2 plus 1. So we can write this by continuing like this 7 by 2 into 5 by 2 into 3 by 2 into 1 by 2 into gamma of 1 by 2. [00:02:50.692] And we know gamma half. Gamma of 1 by 2 is equal to square root of pi. So thus we can write this as gamma half whole square in the numerator because there is one more gamma half and denominator gamma 5. Gamma 5 is equal to 4 factory. [00:03:11.108] So when you simplify this, you will get the answer as thirty five by divided by two. [00:03:21.124] Next one is evaluate integral 0 to pi by 2 square root of cot theta d theta. So since it is cot theta, we can write it as cos by sine. So it is equal to integral 0 to pi by 2. [00:03:39.620] cos theta divided by sine theta whole to the power one by two d theta, which is equal to integral zero to five by two sine power minus half theta cos power half theta d theta. [00:03:55.716] Again it is in the form now integral 0 to pi by 2 sine power p theta cos power q theta d theta, which is equal to 1 by 2 times beta of p plus 1 divided by 2. [00:04:13.264] comma q plus 1 divided by 2. In this particular problem we have p is equal to minus 1 by 2 and q is equal to 1 by 2. So when we apply this formula we get 1 by 2 into beta of 1 by 4 comma 3 by 4 [00:04:33.124] Because minus half plus 1 divided by 2 is 1 by 4, 1 by 2 plus 1 divided by 2 is 3 by 4. Now you use the relationship between beta gamma function, which is beta of m comma n is equal to gamma m into gamma n divided by gamma m plus n. [00:04:52.260] So we get 1 by 2 into gamma of 1 by 4 into gamma of 3 by 4 divided by gamma of 1. But how do we simplify gamma of 1 by 4 into gamma of t by 4?

EV0000058 · EV0000059 · EV0000060 · EV0000061

Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.636] You have studied one result which says gamma of n into gamma of one minus n is equal to pi divided by sin n pi. So now you if you take n is equal to one by four. [00:05:20.944] You will get gamma of one by four into gamma of one minus one by four is equal to pi divided by sine of n into pi here n is equal to one by four, so it is sine of pi by four. [00:05:37.732] So this is equal to 1 by 2 into psi a pi by sine of pi by 4 and sine pi by 4 we know it is equal to 1 by root 2. So your answer is pi divided by 2 into square root of 2. [00:05:56.612] Consider the seventh problem evaluate integral zero to infinity dx divided by one plus x to the power four. Consider the [00:06:08.964] Term x square let us take it as tan theta. When x square is tan theta, two x dx is equal to second square theta d theta. [00:06:18.500]

So you can write dx is equal to secant square theta d theta divided by 2x and we know x square is tan theta, so x is equal to square root of tan theta. So it is second square theta d theta divided by 2 into square root of tan theta. [00:06:38.052] Therefore, integral zero to infinity dx divided by one plus x power four is equal to integral zero to pi by two one divided by one plus tan square because x square is tan. [00:06:51.264] x power four is tan square theta into dx dx is second square theta d theta divided by two into square root of tan theta. [00:07:00.804] Observe here the limits of integration change to 0 to pi by 2. How we got this is you have x square is equal to tan theta. So when x is equal to 0, that means tan theta is equal to 0. [00:07:16.676] Tan theta is equal to zero implies theta is equal to zero. And when x is equal to infinity, that means tan theta is equal to infinity and we know tan theta is infinity when theta is equal to 5 bit. [00:07:32.868] So that is how we got the limits from 0 to pi by 2 after substituting x square is equal to n. [00:07:40.740] So it is equal to now integral 0 to pi by 2. We know 1 plus tan square theta. 1 plus 10 square theta is equal to second square theta. So I write 1 plus tan square theta is equal to second square theta here. Second to second square theta divided by [00:07:58.144] two times ten theta. We can observe here we can cancel second square theta. So it is equal to [00:08:08.548] Integral 0 to pi by 2 d theta divided by 2 into square root of tan theta. That is 1 by 2 into integral 0 to pi by 2. 1 by root tan theta we can write it as tan theta to the power minus half. [00:08:27.620] into d theta. That is equal to 1 by 2 into integral 0 to pi by 2 sin theta divided by cos theta all to the power minus h into d theta. [00:08:43.236] Which is equal to 1 by 2 into integral 0 to pi by 2, we get it as sin power minus 1 by 2 theta into cos power plus 1 by 2 theta. Cos power plus 1 by 2 theta d theta. [00:08:59.812] So again it is equal to half into half into beta of three by four, comma, one by four because we know integral 0 to pi by 2 sin power p theta cos power q theta d theta. [00:09:17.232] is equal to 1 by 2 times beta of p plus 1 divided by 2, comma q plus 1 divided by 2. We had p is equal to minus 1 by 2, q is equal to 1 by 2. So we it is p plus 1 by 2 is equal to [00:09:37.252] 1 minus half is half, that is 1 by 4, and q plus 1 divided by 2 is equal to 3 divided by 4. [00:09:46.724] So anyway I have written it as beta of three by four comma one by four. It can be also written as beta of one by four comma three by four because we know that beta of m comma n is same as

EV0000062 · EV0000063 · EV0000064 · EV0000065

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.328] beta of n comma n so we have already proved this property while studying the properties of beta function. So now apply [00:10:13.220] Relationship between beta gamma function so it is one by four into gamma of three by four into gamma of one by four divided by gamma of one. [00:10:23.140] So we know gamma one is equal to one and gamma three by four into gamma one by four just in the previous problem we have evaluated it is pi divided by sin three pi divided by four, which is equal to one by four into [00:10:39.680] Pi divided by 1 by square root of 2 which is equal to pi divided by root 2. Thus we have solved some problems on beta and gamma functions in this class. I hope you got an idea how to solve problems in this topic. [00:10:56.740] Anyways, it is better to work on some more problems in this topic to get a better understanding. With this, we will end this session. Thank you.

EV0000066 · EV0000067 · EV0000068 · EV0000069

Full transcript: 1650 normalized words, 44 timestamped segments, 667.669333 seconds duration. Low-confidence segments flagged: 0.

EV0000058 · EV0000059 · EV0000060 · EV0000061

7 Series solution of Besselâ??s differential equation, Bessel functions

This lesson is indexed from the source video titled Series solution of Besselâ??s differential equation, Bessel functions. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000070 · EV0000071

Graphs and diagrams



Figure G000115. Series solution of Bessel differential equation · source slide page 1.

EV0000437

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000116. Series solution of Bessel differential equation · source slide page 2.

EV0000440

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

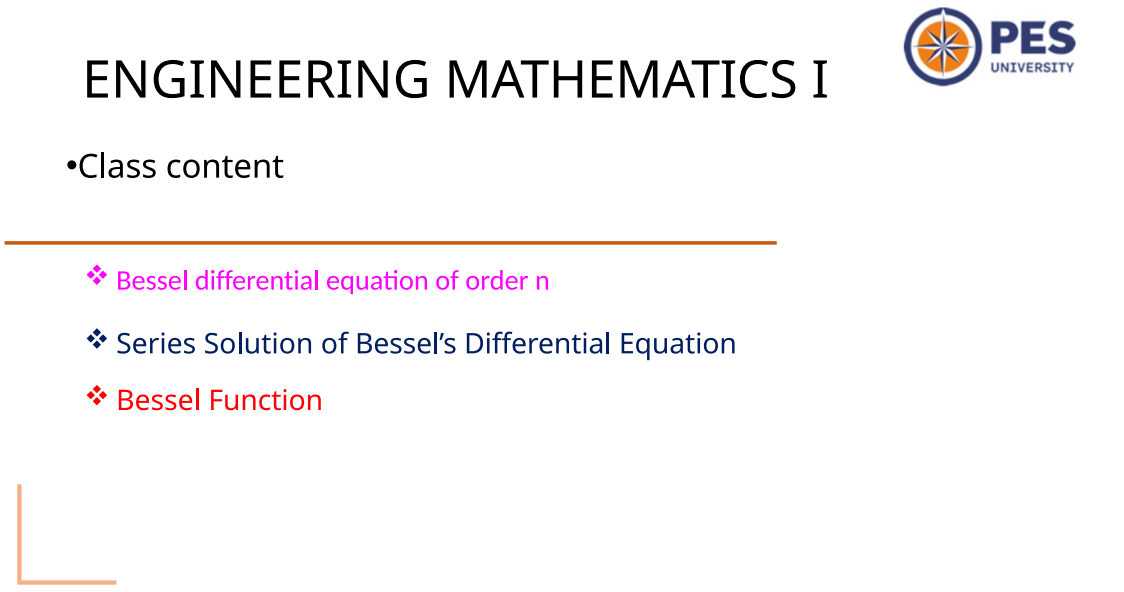


Figure G000117. Series solution of Bessel differential equation · source slide page 3.

EV0000443

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

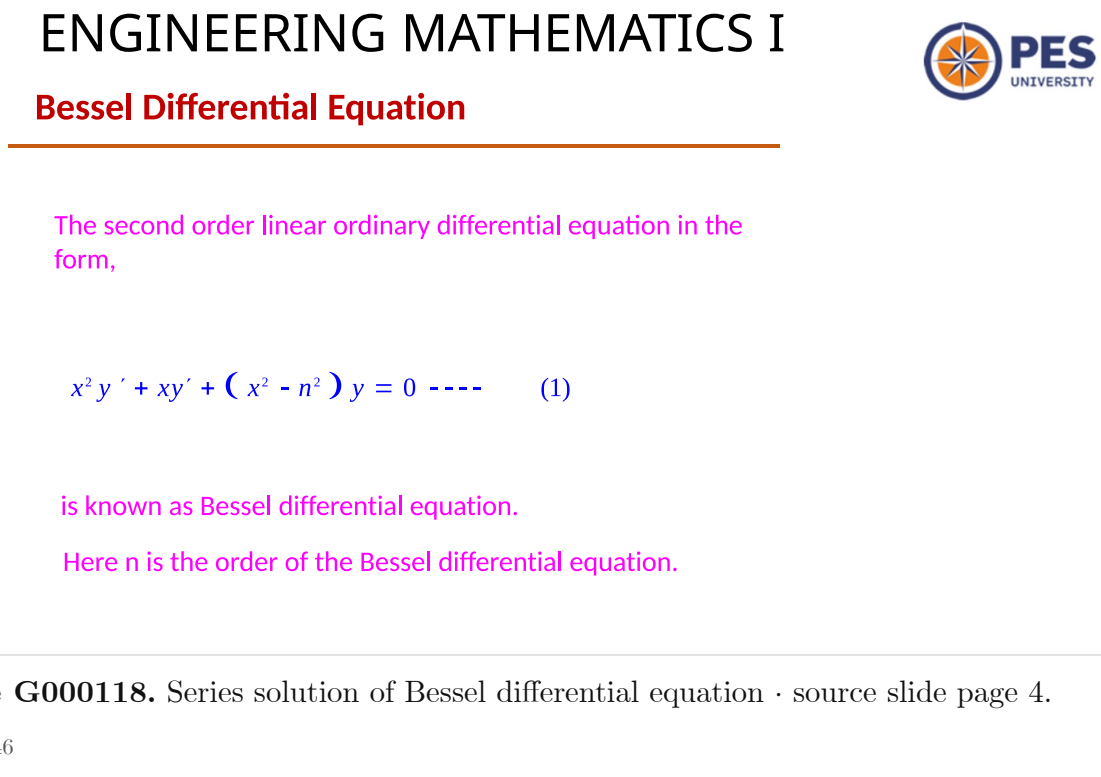


Figure G000118. Series solution of Bessel differential equation · source slide page 4.

EV0000446

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

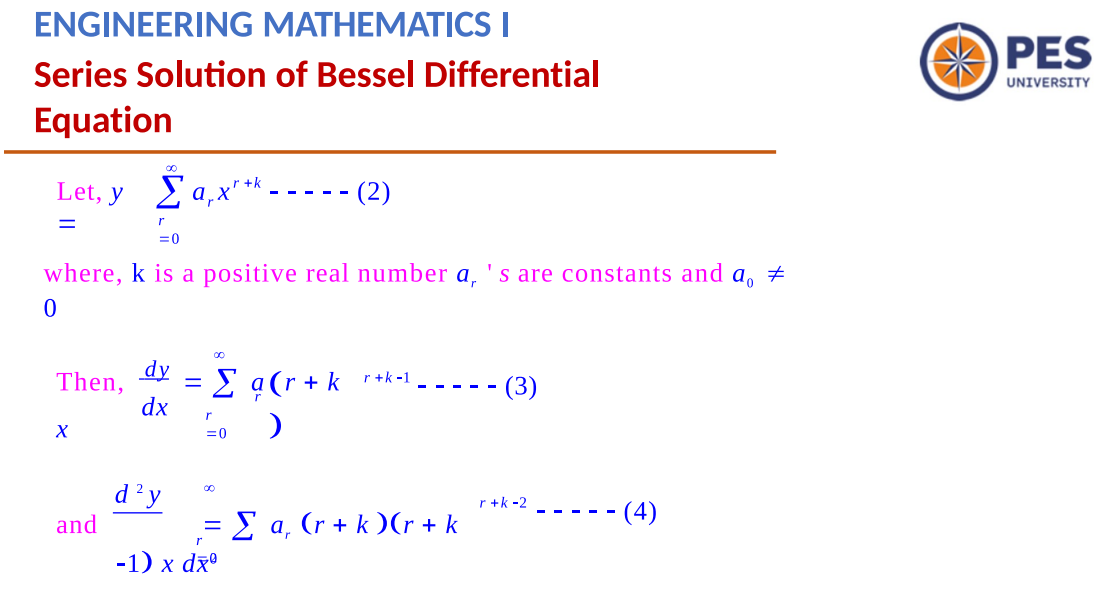


Figure G000119. Series solution of Bessel differential equation · source slide page 5.

EV0000449

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

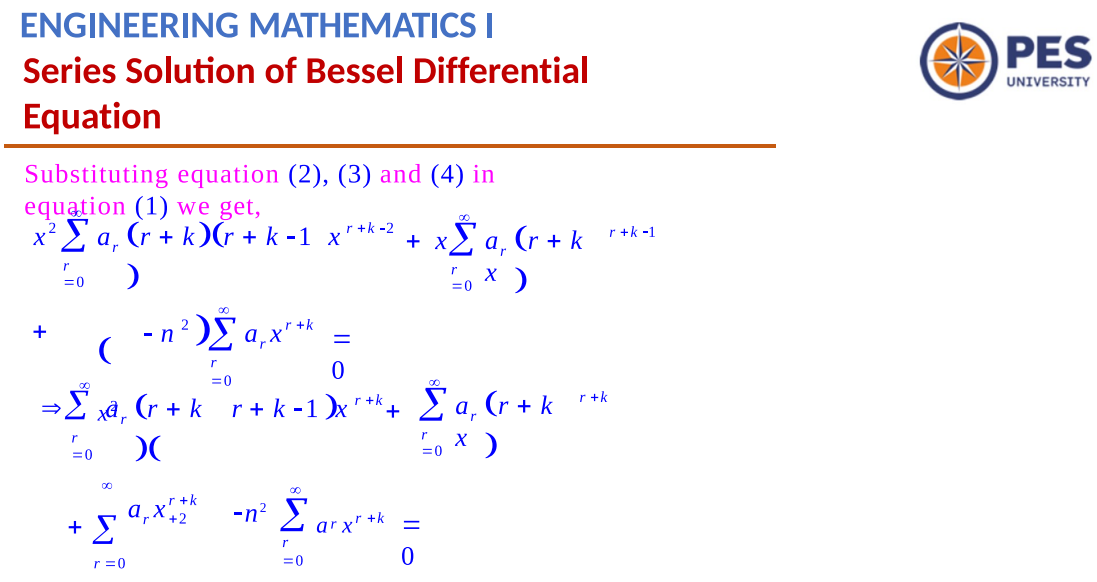


Figure G000120. Series solution of Bessel differential equation · source slide page 6.

EV0000452

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

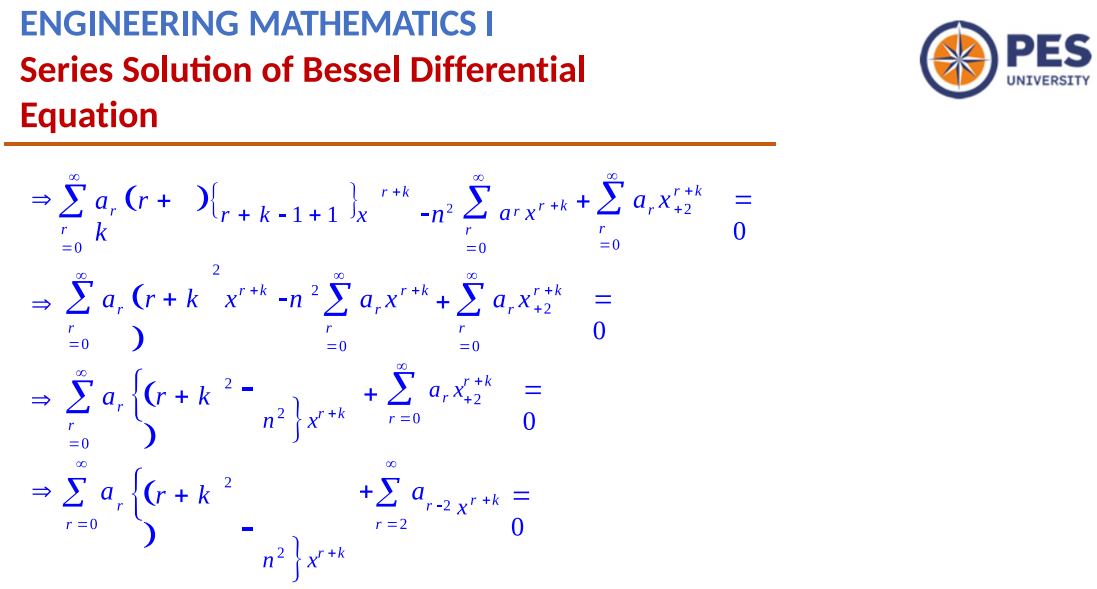


Figure G000121. Series solution of Bessel differential equation · source slide page 7.

EV0000455

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

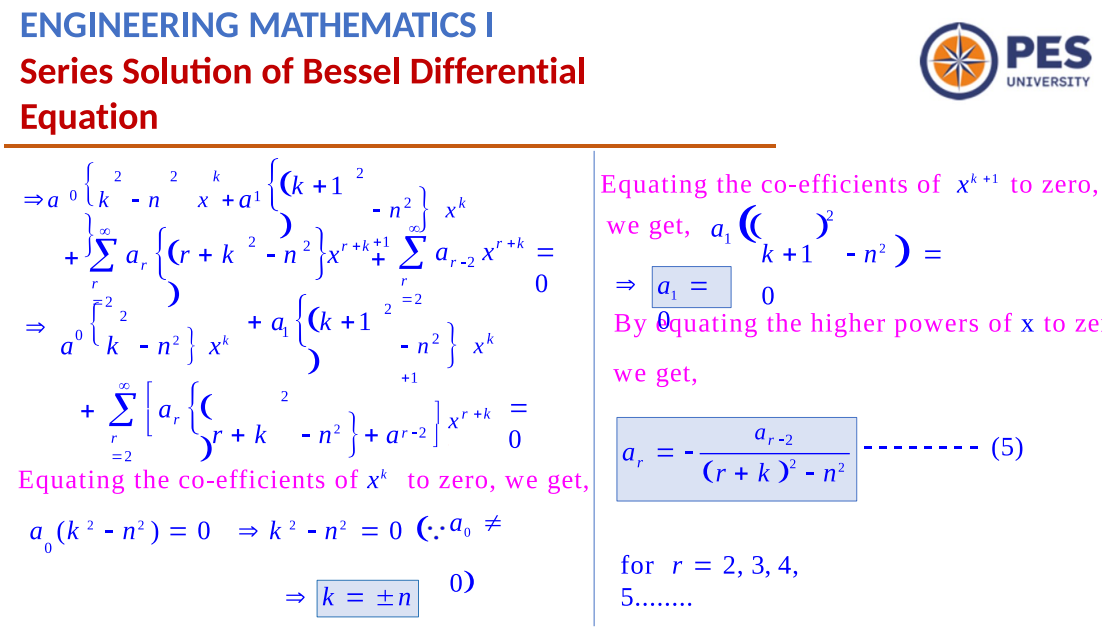


Figure G000122. Series solution of Bessel differential equation · source slide page 8.

EV0000458

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

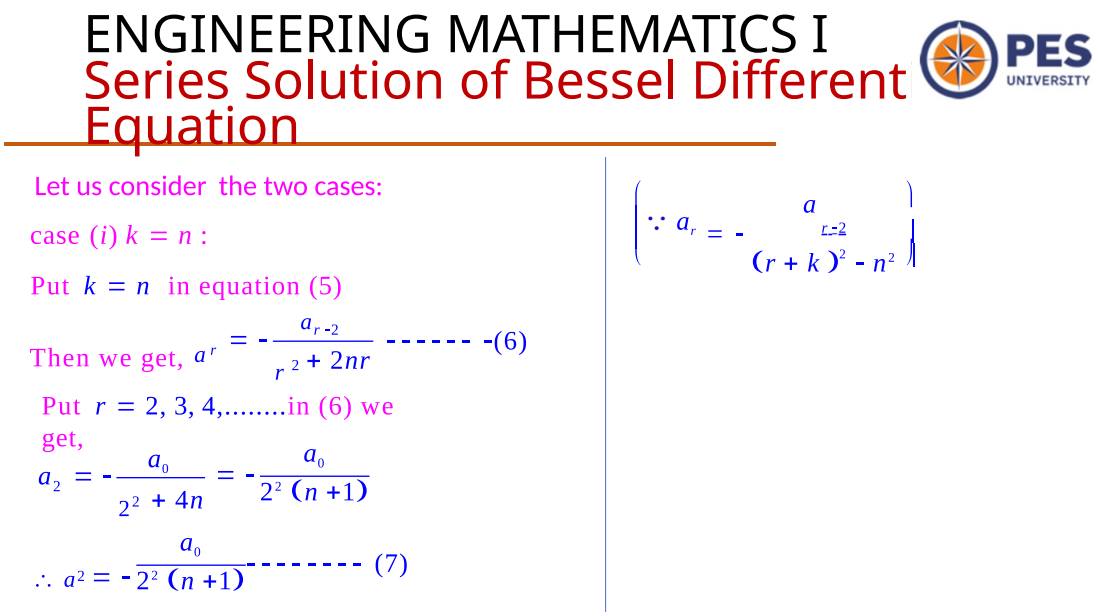


Figure G000123. Series solution of Bessel differential equation · source slide page 9.

EV0000461

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

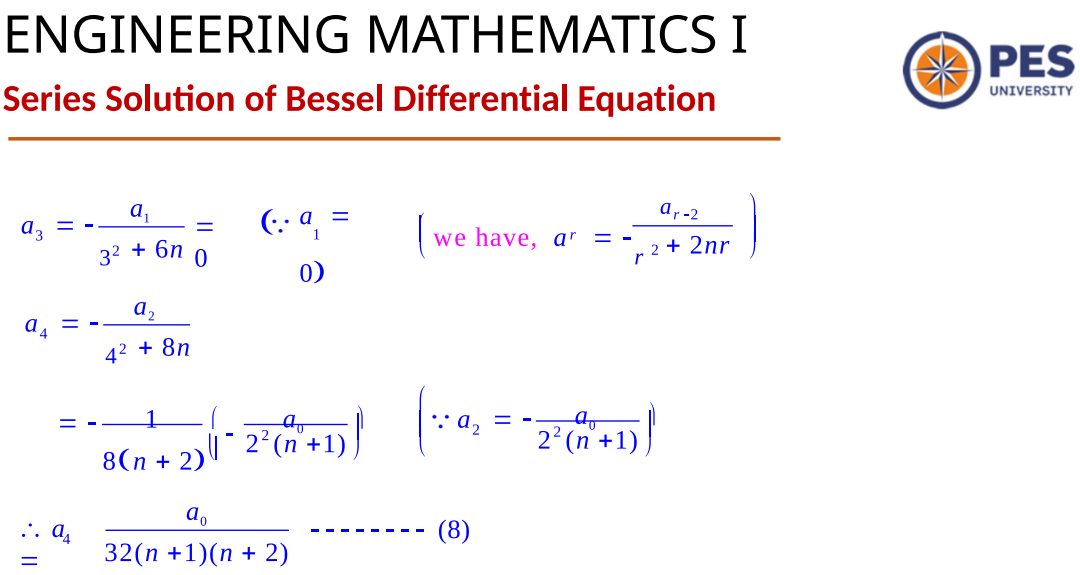


Figure G000124. Series solution of Bessel differential equation · source slide page 10.

EV0000464

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

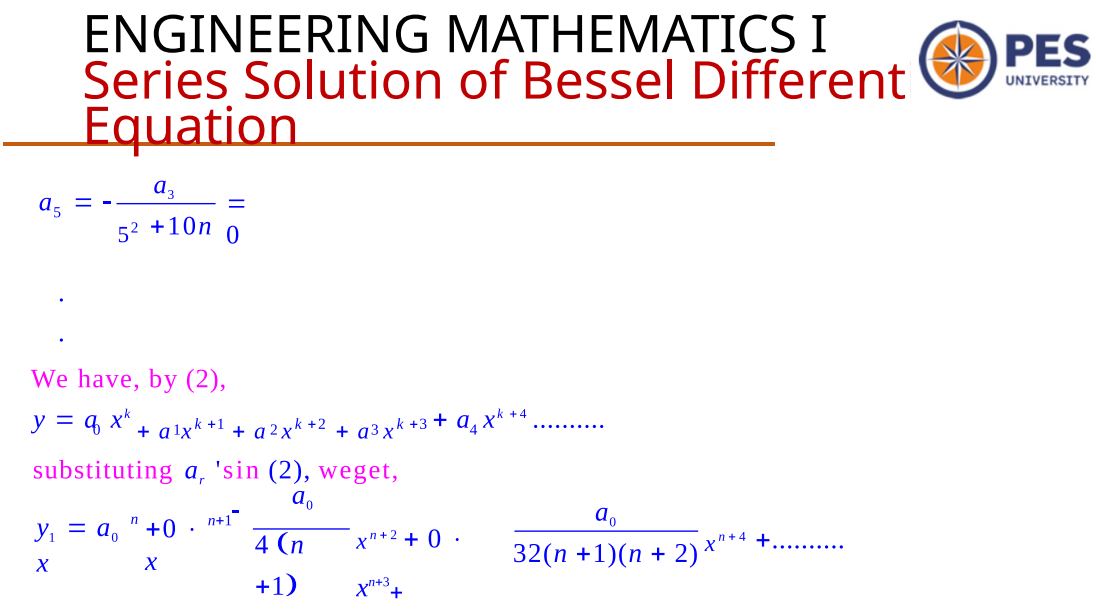


Figure G000125. Series solution of Bessel differential equation · source slide page 11.

EV0000467

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

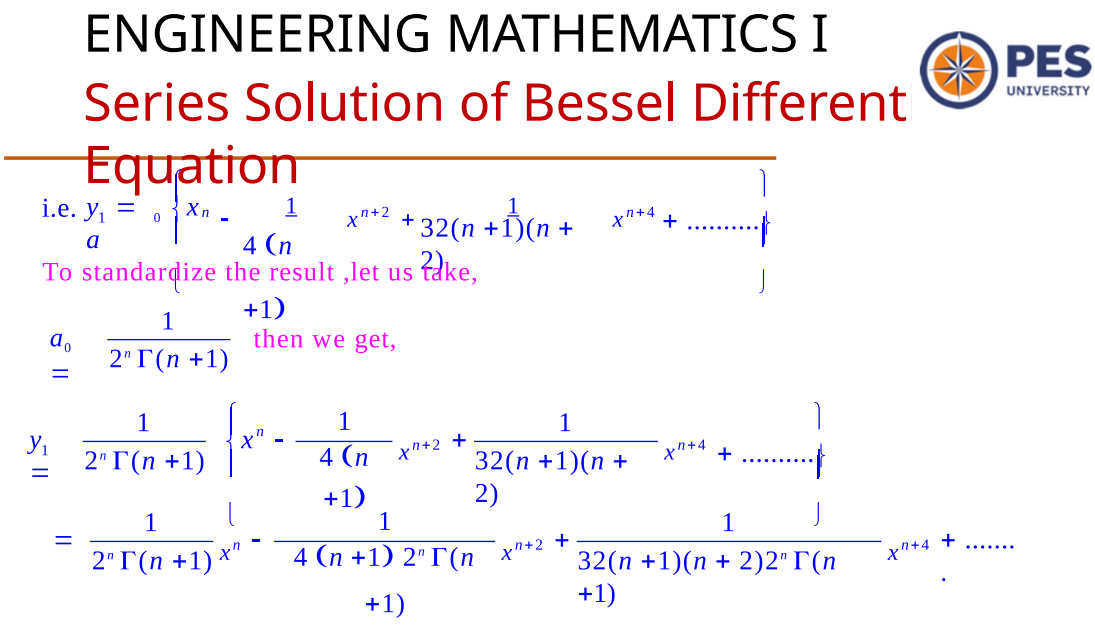


Figure G000126. Series solution of Bessel differential equation · source slide page 12.

EV0000470

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

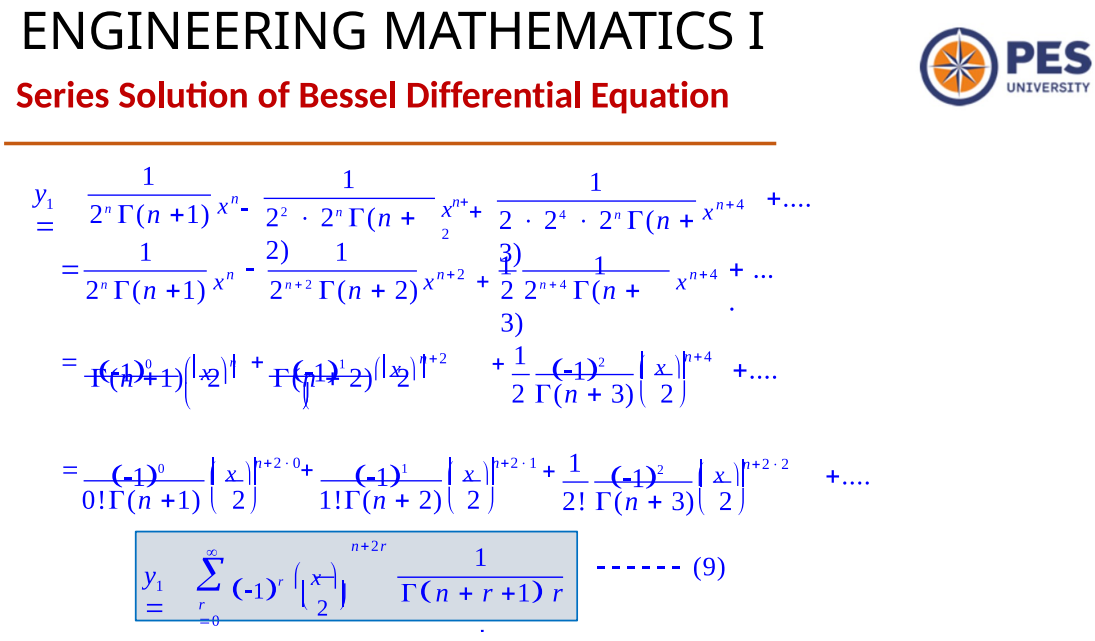


Figure G000127. Series solution of Bessel differential equation · source slide page 13.

EV0000473

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

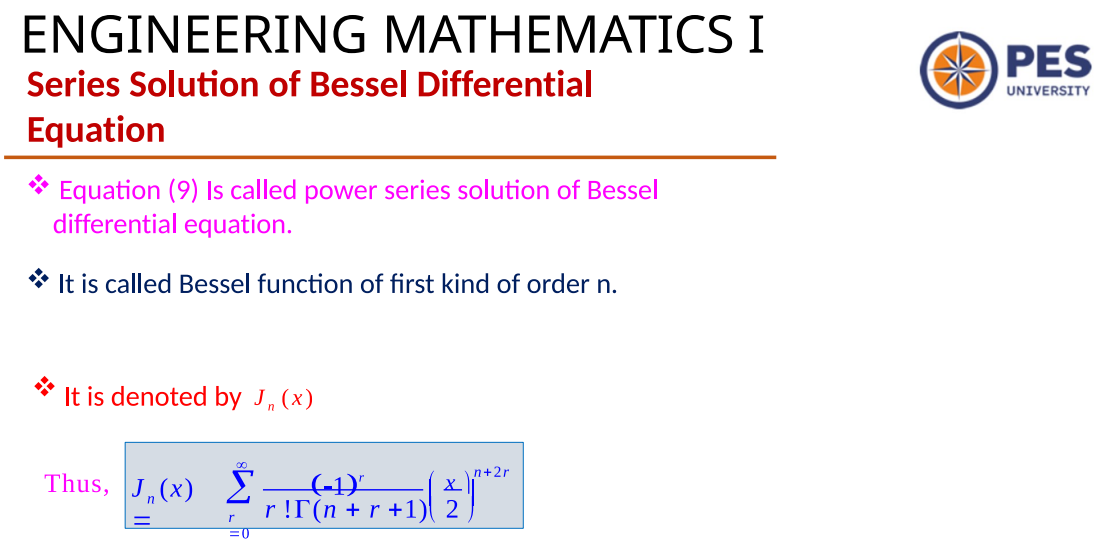


Figure G000128. Series solution of Bessel differential equation · source slide page 14.

EV0000476

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

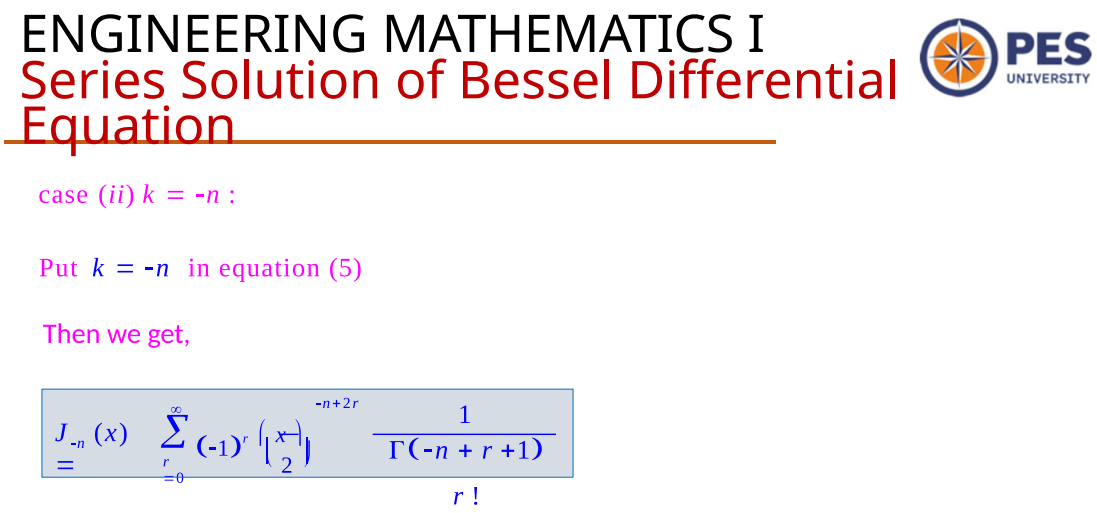


Figure G000129. Series solution of Bessel differential equation · source slide page 15.

EV0000479

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS I

Series Solution of Bessel Differential Equation

Thus the complete solution of Bessel differential equation is given by

$$y = C_1 J_n(x) + C_2 J_{-n}(x)$$ (C_1 and C_2 are arbitrary constants)

provided n is a non integer.

Figure G000130. Series solution of Bessel differential equation · source slide page 16.

EV0000482

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PES UNIVERSITY

Thank you

Department of Science and Humanities

Figure G000131. Series solution of Bessel differential equation · source slide page 17.

EV0000485

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000436 · EV0000437 · EV0000438
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 4: SPECIAL FUNCTIONS AND SERIES SOLUTION OF DE SESSION: 1o SUB-TOPIC: BESSEL FUNCTIONS Department of Science and Humanities EV0000439 · EV0000440 · EV0000441
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I Class content Bessel differential equation of order n Series Solution of Bessel's Differential Equation Bessel Function EV0000442 · EV0000443 · EV0000444
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I Bessel Differential Equation The second order linear ordinary differential equation in the form, x2 y ☒ ☒ xy☒ ☒ ☒ x2 ☒ n2 ☒ y ☒ 0 ☒☒☒☒ (1) is known as Bessel differential equation. Here n is the order of the Bessel differential equation. EV0000445 · EV0000446 · EV0000447
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation ☒ ☒ r ☒k Let, y ☒ a x ☒ ☒ ☒ ☒ ☒ (2) r r ☒0 where, k is a positive real number ar ' s are constants and a0 ☒ 0 ☒ Then, dy ☒ ☒ a r ☒ k x ☒ ☒ r ☒k ☒1 ☒ ☒ ☒ ☒ ☒ (3) r dx r ☒0 d 2 y ☒ and ☒ ☒ ar ☒r ☒ k ☒☒r ☒ k ☒1☒ x dx2 r ☒k ☒2 ☒ ☒ ☒ ☒ ☒ (4) r ☒0 EV0000448 · EV0000449 · EV0000450
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation Substituting equation (2), (3) and (4) in equation (1) we get, ☒ ☒ ☒ r ☒0 ☒ ☒☒ ☒ ☒ r ☒0 ☒ ☒ 2 r ☒k ☒2 x a r ☒ k r ☒ k ☒1 x r ☒k ☒1 a r ☒ k x ☒ x r r ☒ x2 ☒ ☒ ☒ ☒ 2 r ☒k ☒ n a x ☒ 0 r r ☒0 ☒ ☒ ☒ r ☒0 ☒ ☒ ☒☒ r ☒0 ☒ ☒☒ ☒ ☒ r ☒k a r ☒ k x r ☒k a r ☒ k r ☒ k ☒1 x r r ☒ ☒ r ☒0 ☒ ☒ a xr ☒k ☒ r r ☒k ☒2 a x ☒n2 ☒ 0 r r ☒0 EV0000451 · EV0000452 · EV0000453
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation ☒ ☒ ☒ a xr ☒k ☒ r ☒0 ☒ ☒ ☒☒ ☒ ☒ r r ☒k ☒2 a x ☒ 0 ☒ r ☒k a r ☒ k r ☒ k ☒1☒1 x ☒n2 ☒ r r r ☒0 r ☒0 2 ☒ ☒ ☒ ☒ ☒ ☒ r ☒0 ☒ ☒ 2 r ☒k r ☒k ☒2 r ☒k a x a x ☒ 0 ☒ k x ☒n a r ☒ ☒ r r r r ☒0 r ☒0 n2 ☒xr ☒k n2 ☒xr ☒k ☒ ☒ ☒ ☒ ☒ ar x r ☒0 r ☒k ☒2 ☒ ☒ 0 ☒ ☒ 2 a r ☒ k ☒ r r ☒0 ☒ ☒ ☒ ☒ a r ☒0 ☒ ☒☒ a r ☒2 xr ☒k ☒ ☒ 2 r ☒ k ☒ 0 r r ☒2 EV0000454 · EV0000455 · EV0000456
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation ☒ n2 ☒ xk ☒1 ☒ ☒ ☒ ☒ ☒ 2 2 2 k Equating the co-efficients of xk ☒1 to zero, x ☒a ☒a k ☒ n k ☒1 0 k ☒1 ☒ n2 ☒ ☒ 0 1 ☒ ☒ ☒ 2 we get, a ☒ ☒ ☒ ☒ ☒ 1 ☒ ☒ ☒ 2 r ☒k a x ☒ 0 2 r ☒k a r ☒ k ☒ n x ☒ ☒ r ☒2 r ☒ ☒ n2 ☒ xk ☒1 a1 ☒ 0 r ☒2 r ☒2 ☒ By equating the higher powers of x to zero, we get, a k ☒ n2 ☒ xk ☒ ☒ ☒ 2 ☒ a k ☒1 2 ☒ 1 0 r ☒ k ☒ n2 ☒☒ a ☒ ☒ ☒ xr ☒k ☒ ☒ ☒ ☒ ☒ 2 ☒ 0 a ☒ r ☒2 ☒ r ar ☒2 ☒r ☒ k ☒2 ☒ n2 r ☒2 ☒☒☒☒☒☒☒☒ (5) a ☒ ☒ r Equating the co-efficients of xk to zero, we get, a0 ☒ 0☒ ☒ a (k 2 ☒ n2) ☒ 0 ☒ k 2 ☒ n2 ☒ 0 0 for r ☒ 2, 3, 4, 5..... ☒ k ☒ ☒n EV0000457 · EV0000458 · EV0000459
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation Let us consider the two cases: case (i) k ☒ n : Put k ☒ n in equation (5) ☒ ☒ a ☒ ☒ r ☒2 ☒ ☒ ar ☒r ☒ k ☒2 ☒ n2 ☒ ☒ ☒ ☒ ar ☒2 ☒ 2nr ☒ ☒ ☒☒☒☒☒☒ ☒(6) Then we get, a r 2 r Put r ☒ 2, 3, 4,.....in (6) we get, a0 ☒ 4n a0 22 ☒n ☒1☒ ☒ ☒ a ☒ ☒ 22 2 a0 22 ☒n ☒1☒ ☒ a ☒ ☒ ☒☒☒☒☒☒☒☒ (7) 2 EV0000460 · EV0000461 · EV0000462

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation a1 ☒ 6n ☒ ar ☒2 ☒ 2nr a ☒ 0☒ ☒ ☒ we have, a ☒ ☒ a ☒ ☒ ☒ 0 ☒ ☒ r 2 32 1 r 3 ☒ ☒ a2 ☒ 8n a ☒ ☒ 42 4 ☒ a0 ☒ a0 ☒ 1 ☒ ☒ a ☒ ☒ ☒ ☒ ☒ ☒ ☒ 8☒n ☒ 2☒ ☒ 2 2 2 (n ☒1) 2 2 (n ☒1) ☒ ☒ ☒ ☒ a0 32(n ☒1)(n ☒ 2) ☒ a ☒ ☒☒☒☒☒☒☒ (8) 4 EV0000463 · EV0000464 · EV0000465

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation a3 ☒10n a ☒ ☒ ☒ 0 52 5 . . We have, by (2), ☒ a xk ☒1 ☒ a xk ☒2 ☒ a xk ☒3 y ☒ a xk ☒ a xk ☒4 ..... 0 1 2 3 4 substituting ar 'sin (2), weget, a0 32(n ☒1)(n ☒ 2) ☒ a0 4 ☒n ☒1☒ xn☒2 ☒ 0 ☒ xn☒3☒ xn☒4 ☒..... n y1 ☒ a0 x n☒1 ☒0 ☒ x EV0000466 · EV0000467 · EV0000468

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation ☒ ☒☒ ☒ n ☒ 1 xn☒2 ☒ 1 xn☒4 y ☒ a x ☒ .....☒ i.e. 0 ☒ 4 ☒n ☒1☒ 1 32(n ☒1)(n ☒ 2) ☒☒ ☒☒ To standardize the result ,let us take, 1 a0 ☒ then we get, 2n ☒(n ☒1) ☒ ☒☒ 1 1 4 ☒n ☒1☒ 1 ☒ xn☒2 xn☒4 n y ☒ x ☒ ☒ ☒ .....☒ ☒ 1 2n ☒(n ☒1) 32(n ☒1)(n ☒ 2) ☒☒ ☒☒ 1 1 4 ☒n ☒1☒ 2n ☒(n ☒1) 1 xn xn☒2 xn☒4 ☒ ☒ ☒ ☒ ..... 2n ☒(n ☒1) 32(n ☒1)(n ☒ 2)2n ☒(n ☒1) EV0000469 · EV0000470 · EV0000471

13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation 1 1 xn☒ 1 xn☒☒2 xn☒4 y ☒ ☒.... 2n ☒(n ☒1) 1 1 22 ☒ 2n ☒(n ☒ 2) 2 ☒ 24 ☒ 2n ☒(n ☒ 3) 1 ☒ 1 1 xn xn☒2 xn☒4 ☒ ☒ ☒ .... 2n ☒(n ☒1) 2n☒2 ☒(n ☒ 2) 2 2n☒4 ☒(n ☒ 3) ☒ x ☒n☒4 ☒ x ☒n☒2 ☒☒1☒2 ☒☒1☒0 ☒☒1☒1 1 ☒ x ☒n ☒.... ☒ ☒ ☒ 2 ☒(n ☒ 3) 2 ☒ ☒(n ☒1) ☒ 2 ☒ ☒ ☒(n ☒ 2) ☒ 2 ☒ ☒ ☒ ☒ ☒ ☒ ☒ ☒ x ☒n☒2☒0 ☒ x ☒n☒2☒1 ☒ x ☒n☒2☒2 ☒☒1☒0 ☒☒1☒1 ☒☒1☒2 1 ☒.... ☒ ☒ ☒ 0!☒(n ☒1) 2 ☒ ☒ ☒ 1!☒(n ☒ 2) 2 ☒ ☒ ☒ 2! ☒(n ☒ 3) 2 ☒ ☒ ☒ ☒ ☒ ☒ ☒☒1☒r ☒ x ☒ n☒2r 1 ☒☒n ☒ r ☒1☒ r ! ☒ ☒☒☒☒☒☒ (9) ☒ r ☒0 y ☒ ☒ 2 ☒ 1 ☒ ☒ EV0000472 · EV0000473 · EV0000474

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation Equation (9) Is called power series solution of Bessel differential equation. It is called Bessel function of first kind of order n. It is denoted by Jn (x) ☒ x ☒n☒2r ☒☒1☒r ☒ ☒ r ☒0 Thus, J (x) ☒ ☒ ☒ n r !☒(n ☒ r ☒1) 2 ☒ ☒ EV0000475 · EV0000476 · EV0000477

15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation case (ii) k ☒ ☒n : Put k ☒ ☒n in equation (5) Then we get, ☒☒1☒r ☒ x ☒ ☒n☒2r 1 ☒☒☒n ☒ r ☒1☒ r ! ☒ ☒ r ☒0 J (x) ☒ ☒ 2 ☒ ☒n ☒ ☒ EV0000478 · EV0000479 · EV0000480

16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation Thus the complete solution of Bessel differential equation is given by ☒C1 and C2 are arbitraryconstants☒ y ☒ C1Jn (x) ☒ C2 J☒n (x) provided n is a non integer. EV0000481 · EV0000482 · EV0000483

17. Source slide page 17 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000484 · EV0000485 · EV0000486

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.292] Hello everyone, welcome back to Engineering Mathematics 1. We are discussing special functions. In today's class, you are going to learn Basel differential equation of order n, series solution of Basel differential equation, and Basel function. [00:00:19.140] A linear second order ordinary differential equation in the form x square into y double dash plus x into y dash plus x square minus n square into y is equal to zero

[00:00:35.616] Is known as Basel differential equation. Here, n is the order of the basal differential equation. [00:00:44.740] Let me call this equal equation number 1. Now, let us see how to find the solution of this differential equation, which is in the form of an infinite series. [00:00:58.948] That is why we call it as series solution of the Basel differential equation. [00:01:05.604] Now, to find the series solution, let us consider y is equal to summation r from 0 to infinity ar x to the power r plus k, which is an infinite series. Let us call it as equation number 2, where k is a positive real number. [00:01:25.700] And article a0, a1, a2, etc. are constants and a0 is not equal to 0. [00:01:35.332] Now let us find what is y dash. So y dash is nothing but dy by dx which is equal to summation r from 0 to infinity ar into r plus k into x to the power r plus k minus 1. [00:01:50.724] So because we all know that d by dx of x to the power n is equal to n into x to the power n minus 1. Therefore, differentiation of x to the power r plus k is r plus k into x to the power r plus k minus 1. [00:02:09.764] Similarly let us find a second derivative of y also, that is d square y by dx square is equal to summation r from 0 to infinity ar into r plus k into r plus k minus 1. [00:02:26.240] into x to the power r plus k minus 2. Here we have differentiated x to the power r plus k minus 1, which is r plus k minus 1 into x to the power r plus k minus 2. Let us call it as equation number four. [00:02:44.228] Now let us substitute equation number 2, 3, 4 in equation number 1. We get x square into summation r from 0 to infinity, ar into r plus k into r plus k minus 1. [00:02:59.972] into x to the power r plus k minus 2 that is nothing but equal to y double dash this is y double dash then plus x into summation ar into [00:03:13.668] R plus k into x to the power r plus k minus 1. This is nothing but equal to y dash, then plus x square minus n square into summation ar into x to the power r plus k. This is nothing but equal to y. [00:03:30.576] is equal to 0. Thus we have substituted the expressions for y dash and y double dash in the x equation x square y double dash plus x y dash plus [00:03:43.616] x square minus n square into y is equal to z, which is a basal differential equation of order n. [00:03:52.420] Now let us simplify this. So this implies now let me take this x square inside the summation. So I have x square into x to the power r plus k minus 2. So I'll get it as summation r from 0 to infinity ar into r plus k [00:04:09.616] into r plus k minus 1 into x to the power r plus k plus. In the second term also I will take this x inside the summation. So I have x into x to the power r plus k minus 1. [00:04:23.236] So we get summation r from 0 to infinity ar into r plus k into x to the power r plus k plus x square minus n square into summation ar into x power r plus k here. So let us split it into two. [00:04:40.420] Thus, 1 is x square into summation ar x power r plus k. That is nothing but equal to summation r from 0 to infinity, ar into x to the power r plus k plus 2, then minus [00:04:54.464] n square into summation r from 0 to infinity ar into x to the power r plus k is equal to 0.

EV0000070 · EV0000071 · EV0000072 · EV0000073

Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.428] That implies summation r from 0 to infinity ar into r plus k into r plus k minus 1 plus 1 into x to the power r plus k. We have considered the first two terms together here. [00:05:18.084] Then minus n square into summation r from zero to infinity ar plus k plus summation r from zero to infinity ar x to the power r plus k plus two is equal to zero. Here we can cancel these two terms. [00:05:35.776] So we will get this as summation r from 0 to infinity, ar into r plus k closes square into x power r plus k minus n square into summation r from 0 to infinity, ar into x to the power r plus k plus. [00:05:52.480] Summation R from zero to infinity er into x to the power r plus k plus two is equal to zero. That means [00:06:02.532] Summation R from 0 to infinity R plus K whole square minus N square into AR into X to the power R plus K. So that is got by considering these two terms

together. Summation R from 0 to infinity is common. So [00:06:19.396] We consider it as a common summation. A R is common factor, x power r plus k is also common, so we can write it as r plus k whole square minus n square. [00:06:31.620] Plus summation r from 0 to infinity a r into x to the power r plus k plus 2 is equal to 0. [00:06:41.700] That implies summation r from 0 to infinity ar into x r plus k whole square minus n square into x to the power r plus k plus. [00:06:55.140] Summation R is equal to 2 to infinity AR minus 2 into x to the power r plus k is equal to 0. See we obtain this term by replacing r by r minus 2 in the last term here. [00:07:13.828] Okay, so we are replacing r by r minus 2 so that this becomes a r minus 2 and this becomes x to the power r plus k. And here the summation starts from r minus 2 is equal to 0 to infinity, that is nothing but r is equal to 2 to infinity. [00:07:32.068] Now consider these two summations here. In one summation R starts from 0 and in another R starts from 2. [00:07:40.516] So, in the first summation, I will write two terms separately, that is, when r is equal to 0 and when r is equal to 1. That implies a0 into k square minus n square into x power k plus. [00:07:55.920] A1 into K plus one whole square minus N square into X power K plus one plus summation R from two to infinity A R into R plus K whole square minus N square into X power R plus K plus [00:08:12.080] Summation R from 2 to infinity AR minus 2 into x power r plus k is equal to 0. Now let us combine these two summations R from 2 to infinity [00:08:27.076] ar R plus k whole square minus n square into x power r plus k and summation r from 2 to infinity ar minus 2 x power r plus k. [00:08:39.204] So this we get a0 into k square minus n square into x power k plus a one into k plus one whole square minus n square into x power k plus one. After combining these two summations. [00:08:53.376] We get it as summation r from two to infinity AR into R plus K whole square minus N square plus AR minus two whole into x power r plus k is equal to zero. [00:09:08.484] Now, equating the coefficient of x to the power k to 0. Coefficient of x power k is a0 into k square minus n square. So you get a0 into k square minus n square is equal to 0. [00:09:22.800] That implies k square minus n square is equal to 0 because we already know that k0 is not equal to 0. So we have k square is equal to n square or k is equal to plus or minus n. [00:09:38.628] Now, equating the coefficient of x to the power k plus 1 to 0, we get a1 into k plus 1 whole square minus n square is equal to 0. This implies a1 is equal to 0. [00:09:54.596] So why we got this is because if a1 is not equal to 0, we get k plus 1 whole square minus n square is equal to 0. That means k plus 1 whole square is equal to n square.

EV0000074 · EV0000075 · EV0000076 · EV0000077

Transcript block 3 (source interval 000600-000899 seconds): [00:10:11.360] That means k plus 1 is equal to plus or minus n, which is not correct because already we have proved now that k is equal to plus or minus n. Therefore, this term cannot be equal to 0. So we must have a1 is equal to 0. [00:10:32.132] By equating the higher power of x to 0, we get ar is equal to some minus ar minus 2 divided by r plus k whole square minus n square. [00:10:48.260] Call it as equation number five. Now this is valid for r is equal to 2, 3, comma 4, comma 5, etc. So let us consider the two cases. First case is when k is equal to n. Put k is equal to n in equation number 5, we get [00:11:07.492] ar is equal to minus ar minus 2 divided by r square plus 2 n r. [00:11:15.044] Because we know that AR is equal to minus A R minus 2 divided by R plus K whole square minus n square by equation number 5. When you put k is equal to n in the denominator you get r plus n the whole square minus n square. [00:11:31.812] And R plus N the whole square is R square plus 2 N R plus N square minus N square. So cancel plus N square minus N square, you get R square plus 2 N R in the denominator. Call it as equation number 6. [00:11:50.884] Let us substitute r is equal to 2, 3, 4, etc. in equation number 6, we get when you put r is equal to 2, we get a2, a2 is equal to minus.

[00:12:04.068] A0 divided by 2 square plus 4n. Because when r is equal to 2, a r minus 2 becomes a 2 minus 2, which is a0. And in the denominator when you put r is equal to 2, r square becomes 2 square, 2n r becomes 4n. [00:12:20.932] Which is equal to minus of a0 divided by 2 square into n plus 1. Therefore, your a2 is equal to minus a0 divided by 2 square into n plus 1. Call it as equation number 7. [00:12:41.348] Now a3 is equal to minus a 1 divided by t square plus 6n, which is equal to 0 because a1 is equal to 0. This is got by ar is equal to minus ar minus 2 divided by r square plus 2 n r by putting r is equal to 3. [00:12:59.236] In the numerator when R is 3, you get A3 minus 2, which is A1, which is there in the numerator here, and in the denominator 3 square you get and plus 2 Nr becomes a 6n. [00:13:13.796] Now a4 is equal to minus. Now you have to put r is equal to 4, so a 4 minus 2, which is nothing but equal to a2 here divided by r square. R is 4, so it becomes 4 square plus 2n r is equal to a n. [00:13:32.580] Which is equal to minus 1 divided by 8 into m plus 2 into a2. We are substituting for a2 here because we know a2 is equal to minus a0 divided by 2 square into m plus 1. [00:13:47.172] Thus we have a4 is equal to a0 divided by 32 into n plus 1 into n plus 2. Now a5 is equal to minus a3 divided by 5 square plus 10n. [00:14:00.676] which is equal to zero because we know that a3 is equal to zero. We can continue like this. [00:14:11.396] Now we have by equation number two. Y is equal to a0 x power k plus a1 x power k plus one plus a2 x power k plus two plus a three x power k plus three plus a4 x power k plus four and so on. [00:14:27.712] because we have assumed a y as a series solution that is y is equal to summation r from 0 to infinity ar into x to the power k plus r. Now substituting a r in equation number two we get [00:14:46.884] Y1. We call it as Y1 because we are just discussing case 1 where your K is equal to plus N. We had two cases K is plus or minus N. So for the first case K is equal to N we call the solution as Y1.

EV0000078 · EV0000079 · EV0000080 · EV0000081

Transcript block 4 (source interval 000900-001199 seconds): [00:15:02.500] So y1 is equal to a0 into x power n plus 0 into x power n plus 1 because we know that a1 is equal to 0 and k is equal to n here. [00:15:17.028] Then plus a2 x power k plus 2. So a2 is replaced by minus a0 into 4 in a0 divided by 4 into n plus 1 into x to the power n plus 2. [00:15:30.468] Next term is a3x power k plus 3. A3 is equal to 0. So it is 0 into x to the power n plus 3. [00:15:39.844] Then a4 x to the power k plus 4. Substitute for a4, a4 is a0 divided by 32 into n plus 1 into n plus 2 into x to the power n plus 4 plus other terms. [00:15:55.748] That is y1 is equal to a naught into x power n minus 1 divided by 4 into n plus 1 into x power n plus 2 plus 1 divided by 32 into n plus 1 into m plus 2 into x to the power n plus 4 plus other terms. [00:16:13.956] To standardize the result, let us take a0 is equal to 1 divided by 2 power n into gamma of n plus 1. So let us put a0 is equal to 1 divided by 2 power n into gamma of n plus 1. [00:16:29.124] So we get y1 is equal to just replace a naught by 1 divided by 2 to the power n into gamma of n plus 1 here. Remaining terms I write as it is. [00:16:41.860] So we have y1 is equal to let me multiply this 1 by 2 power n into gamma of n plus 1 inside. So you get 1 divided by 2 power n into gamma of n plus 1 into x power n. [00:16:55.460] Minus 1 divided by 4 into n plus 1 into 2 power n into gamma of n plus 1 into x power n plus 2 plus. [00:17:06.596] 1 by 32 into n plus 1 into n plus 2 into 2 power n into gamma of n plus 1 into x to the power n plus 4 plus other terms. [00:17:19.524] Now consider the second term. So where you have n plus 1 into gamma of n plus 1. So we have already studied a property of gamma function that gamma of n plus 1 is equal to n into gamma of n. [00:17:37.764] So what happens to n plus 1 into gamma of n plus 1? So can we call it as gamma of n plus 2? Right? So here you just replace n by n plus 1, so you get n plus 1 into gamma of n plus 1 is equal to gamma

of n plus 1 plus 1, which is gamma of n plus 2. [00:17:57.316] So, this here we can combine these two and call it as gamma of n plus 2. Right? Similarly, consider this n plus 1 into gamma of n plus 1. [00:18:11.204] Which is nothing but equal to gamma of n plus 2. And then you have n plus 2 into gamma of n plus 2. What is n plus 2 into gamma of n plus 2? So again use the same property. So n plus 2 into gamma of n plus 2 is equal to [00:18:28.864] gamma of n plus three. So we can write this term as thirty-two into gamma of n plus three. So therefore y1 is equal to 1 divided by 2 power n into gamma of n plus 1 into x to the power n. [00:18:44.036] Minus 1 divided by 2 square into 2 power n into gamma of n plus 2 into x power n plus 2. [00:18:51.972] Plus 1 divided by 2 into 2 to the power 4. Because we had 32 in the denominator, 32 can be written as 2 to the power 5, it can be written as 2 into 2 to the power 4. [00:19:05.348] into 2 power n into gamma of n plus 3 into x to the power n plus 4. Plus other terms. [00:19:13.316] So it is equal to 1 divided by 2 power n into gamma of n plus 1 into x power n minus 1 by uh 2 square into 2 power n can be written as 2 to the power n plus 2 into gamma of n plus 2 into x power n plus 2. [00:19:32.036] Plus this 1 by 2 we keep it as it is and 2 power 4 into 2 power n can be written as 2 to the power n plus 4 into gamma of n plus 3 into x to the power n plus 4 plus other terms. [00:19:47.748] That is equal to C first term, so we can write it as minus 1 power 0. Minus 1 power 0 is 1. So it is minus 1 power 0 divided by gamma of n plus 1. And consider x power n divided by 2 power n, which can be written as x by 2 whole to the power n.

EV0000082 · EV0000083 · EV0000084 · EV0000085

Transcript block 5 (source interval 001200-001499 seconds): [00:20:06.468] Then plus, since there is a negative sign here, we can write it as minus 1 whole to the power 1 divided by gamma of n plus 2 into x power n plus 2 divided by 2 to the power n plus 2, which is x by 2 whole to the power n plus 2. [00:20:23.620] Next term is positive, so you can write it as 1 by 2 into minus 1 whole square in divided by gamma of n plus 3 into x power n plus 4 divided by 2 to the power n plus 4, which is x by 2 whole to the power n plus 4 plus other terms. [00:20:42.820] That is equal to minus 1 power 0 divided by 0 factorial. 0 factorial is 1. Into gamma of n plus 1 into x by 2 whole to whole to the power n can be written as x by 2 whole to the power n plus 2 into 0. [00:20:59.012] Plus minus 1 power n divided by 1 factorial, 1 factorial is 1, into gamma of n plus 2 into x by 2 whole to the power n plus 2 can be written as x by two whole to the power n plus 2 into 1. [00:21:14.308] Plus, 1 by 2 can be written as 1 by 2 factorial and minus 1 whole square divided by gamma of n plus 3 into x by 2 whole to the power n plus 2 into 2. That is nothing but x by 2 whole to the power n plus 4 plus other terms. [00:21:32.804] We can write it in the summation form as y1 is equal to summation r from 0 to infinity minus 1 to the power r into x by 2 whole to the power n plus 2 r into 1 divided by gamma of n plus r plus 1. [00:21:50.464] into r factorial. This is because when r is equal to 0, you get minus 1 power 0 x by 2 whole to the power n plus 2 into 0 into 1 divided by gamma of n plus 1 into [00:22:06.596] One factorial, uh zero factorial. That is what the first term here. [00:22:11.300] When you put r is equal to one, you get minus one to the power one into x by two whole to the power n plus two into one into one divided by gamma of n plus two into one factorial. When you put r is equal to two, [00:22:28.048] You get minus one whole square into x by two whole to the power n plus two into two into one divided by gamma of n plus two plus one that is gamma of n plus three into two factorial that is what the third term here. [00:22:44.868] So we have y1 is equal to this expression. Call it as equation number nine. Equation number nine is called power series solution of basal differential equation. It is called Basel function of first kind of order n. [00:23:03.300] and it is denoted by jn of x. Thus jn of x is equal to summation r from 0 to

infinity minus 1 power r divided by r factorial into gamma of n plus r plus 1 into x by 2 whole to the power n plus 2 r. [00:23:23.396] Remember this expression, it's an important expression. Now consider the second case when k is equal to minus n. [00:23:31.940] Put k is equal to minus n in equation number five. Then we get j minus n of x is equal to summation r from 0 to infinity minus 1 power r x by 2 whole to the power minus n plus 2 r into 1 divided by gamma of minus n plus [00:23:49.280] plus r plus 1 into r factorial. Thus the complete solution of Basel differential equation of order n is given by y is equal to c1 into j n of x plus c2 into j minus n of x, where c1 and c two are arbitrary constants. [00:24:07.108] So you have already studied that for a differential equation of order 2, the general solution involves two arbitrary constants. [00:24:15.940] Basically, Paz's differential equation of order n is a differential equation of order 2. So, the general solution here also involves two arbitrary constants c1 and c. Moreover, we can [00:24:31.172] Uh have a condition that n is a non integer here because we know that for a differential equation of order two, if y one and y two are two linearly independent solutions, [00:24:42.656] Then C1 by one plus c two by two that is the linear combination of those two linearly independent solutions is also a solution of the differential equation. Here jn and j minus c n are two linearly independent solutions only if n is not an integer.

EV0000086 · EV0000087 · EV0000088 · EV0000089

Transcript block 6 (source interval 001500-001799 seconds): [00:25:00.836] If n is an integer, jn and j minus n are not linearly independent. You can prove this by taking the expression for jn of x. Just replace n by minus n. [00:25:14.372] And hence you can prove that j minus n of x is minus 1 to the power n into jn of x. That shows that jn and j minus n are not linearly independent when n is an integer. [00:25:29.540] Thus, if n is non-integer, then we have y is equal to c1 j n of x plus c2 j minus c n of x as the complete solution of the Bessel's differential equation. [00:25:44.548] You can just try this to prove that jn and j minus n are linearly dependent when n is an integer. It is interesting, just take it as an assignment and try it on your own. With this, we will end this session. Thank you.

EV0000090 · EV0000091 · EV0000092 · EV0000093

Full transcript: 3992 normalized words, 104 timestamped segments, 1561.301333 seconds duration. Low-confidence segments flagged: 0.

EV0000070 · EV0000071 · EV0000072 · EV0000073

8 Source slide deck: introduction to Series solution

This entry preserves the complete slide deck titled introduction to Series solution as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_04, to avoid an unsupported match. EV0000307 · EV0000308

Graphs and diagrams



Figure G000072. introduction to Series solution · source slide page 1.

EV0000308

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000073. introduction to Series solution · source slide page 2.

EV0000311

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

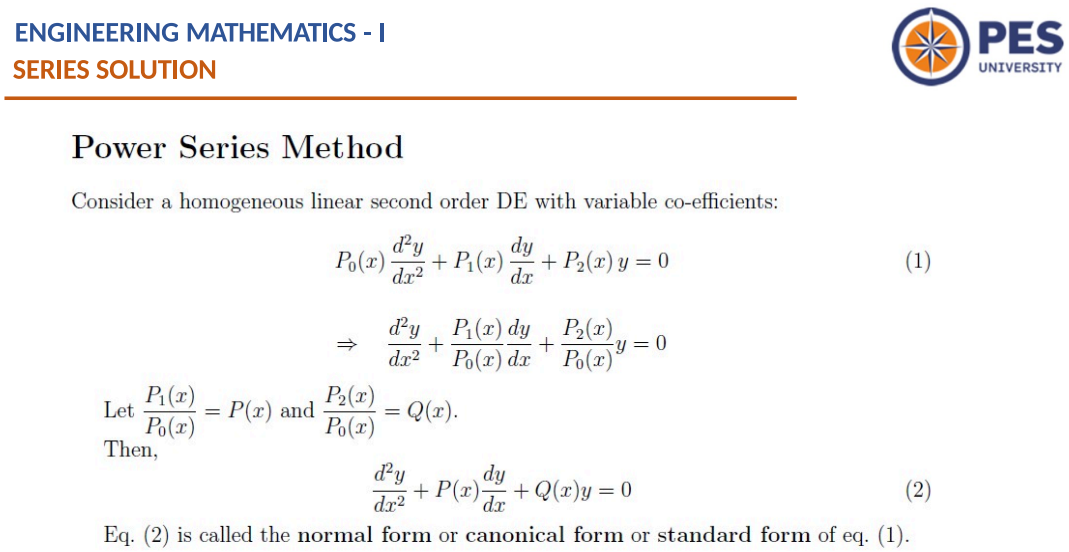


Figure G000074. introduction to Series solution · source slide page 3.

EV0000314

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

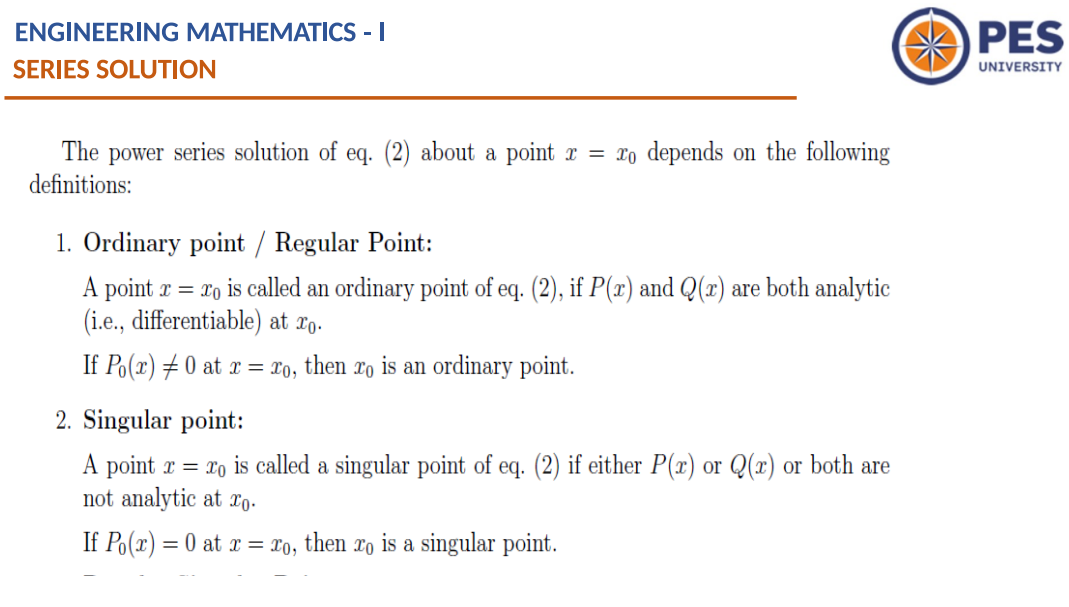


Figure G000075. introduction to Series solution · source slide page 4.

EV0000317

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

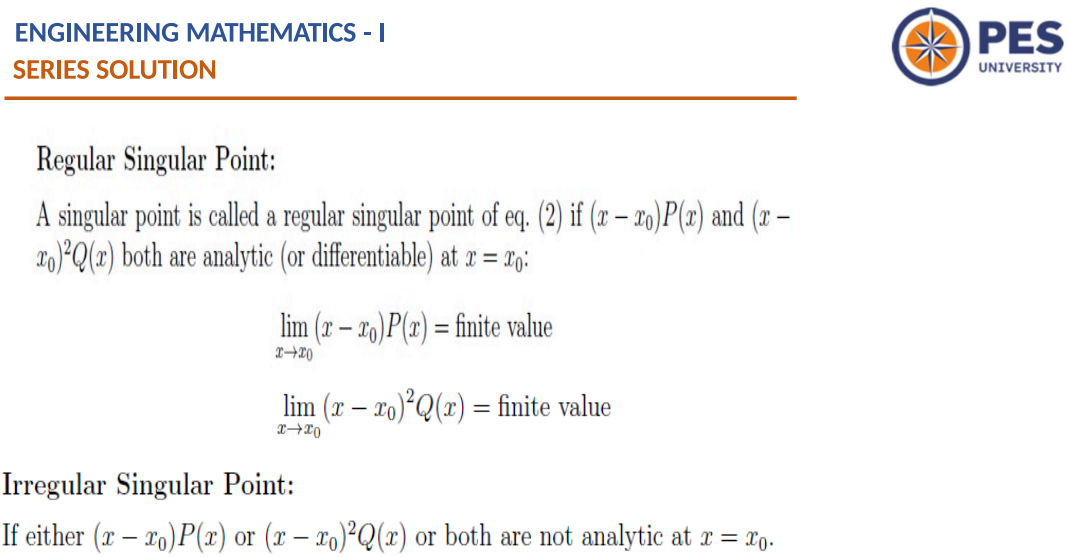


Figure G000076. introduction to Series solution · source slide page 5.

EV0000320

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

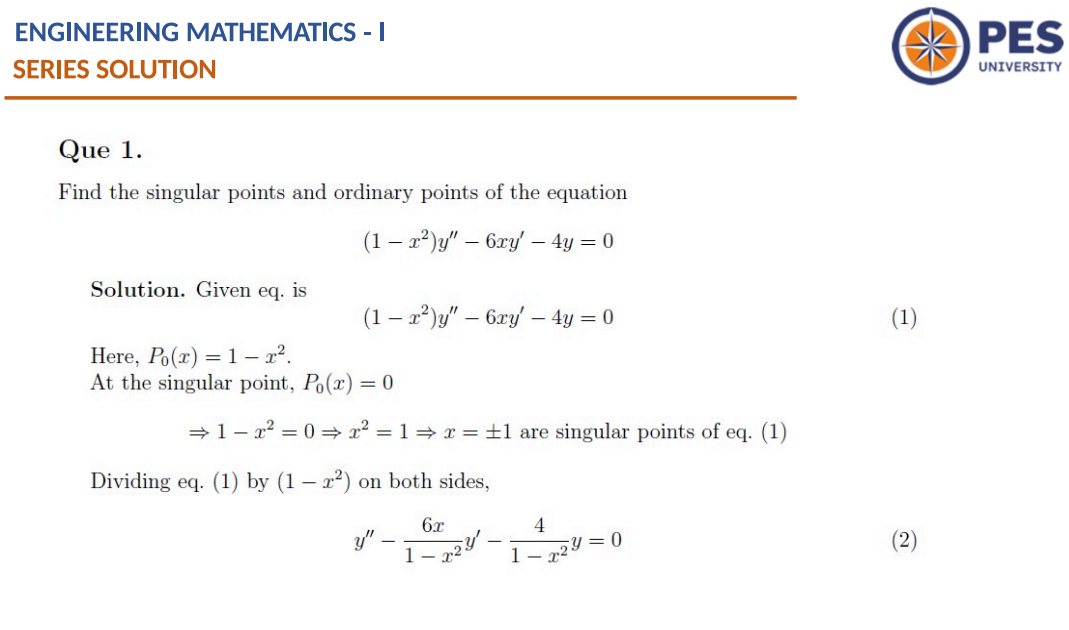


Figure G000077. introduction to Series solution · source slide page 6.

EV0000323

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

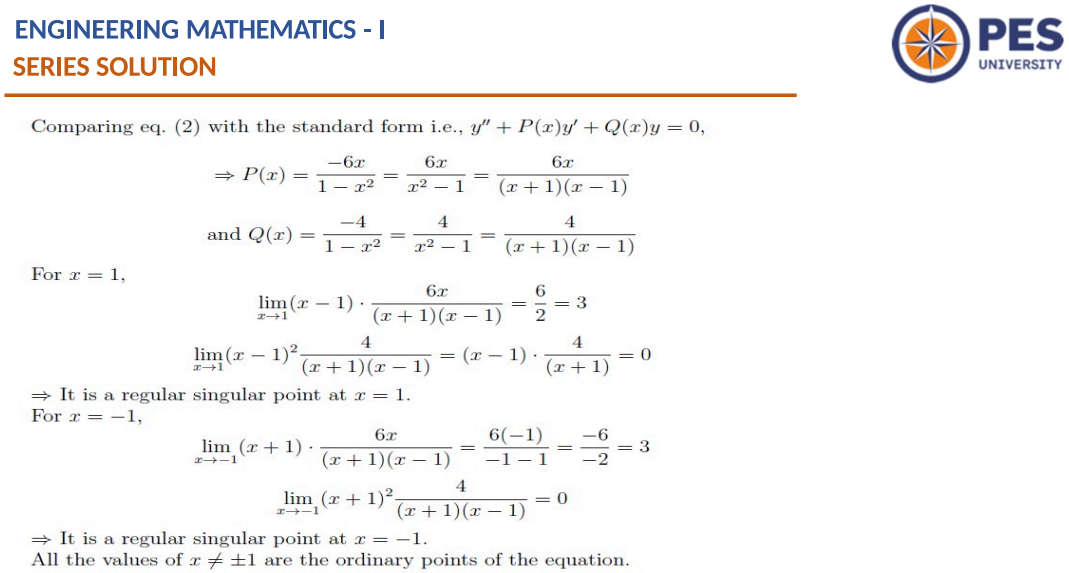
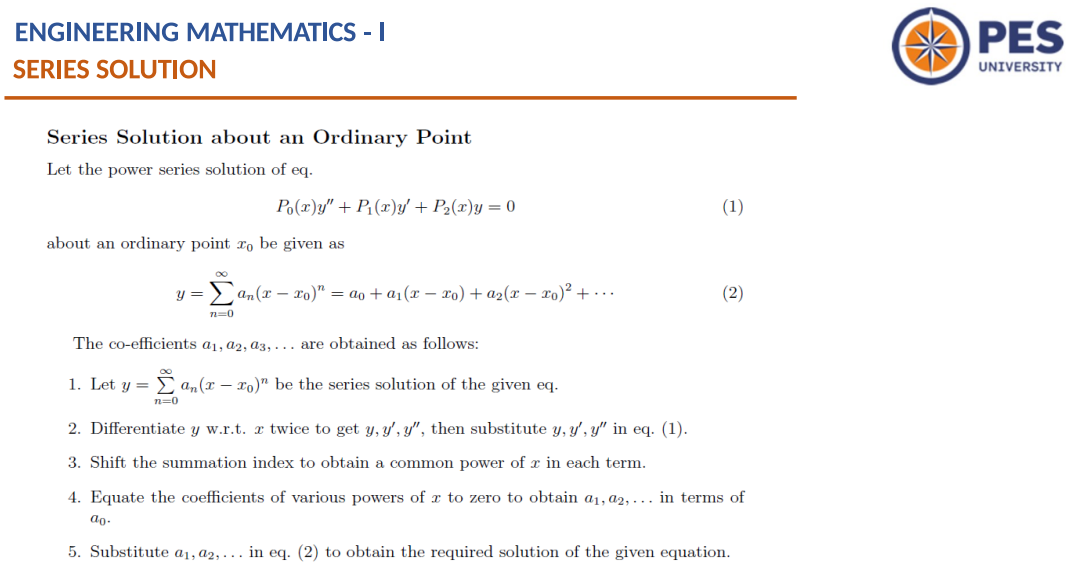


Figure G000078. introduction to Series solution · source slide page 7.

EV0000326

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



ENGINEERING MATHEMATICS - I
SERIES SOLUTION

PES UNIVERSITY

Series Solution about an Ordinary Point

Let the power series solution of eq.

$$P_0(x)y'' + P_1(x)y' + P_2(x)y = 0 \qquad (1)$$

about an ordinary point x_0 be given as

$$y = \sum_{n=0}^{\infty} a_n(x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots \qquad (2)$$

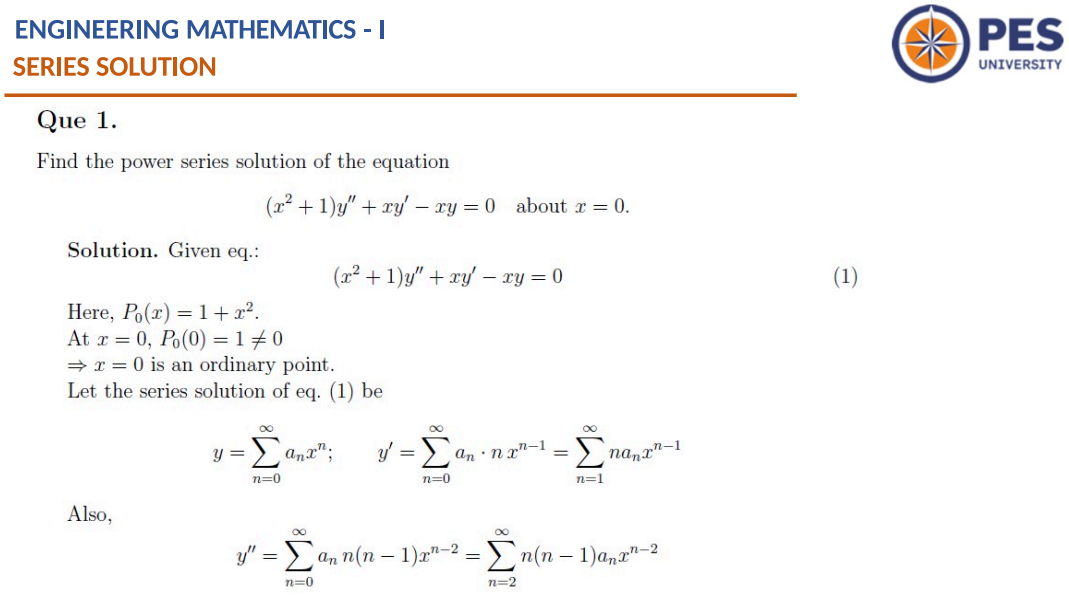
The co-efficients $a_1, a_2, a_3, \ldots$ are obtained as follows:

1. Let $y = \sum_{n=0}^{\infty} a_n(x - x_0)^n$ be the series solution of the given eq.
2. Differentiate y w.r.t. x twice to get y, y', y'', then substitute y, y', y'' in eq. (1).
3. Shift the summation index to obtain a common power of x in each term.
4. Equate the coefficients of various powers of x to zero to obtain $a_1, a_2, \ldots$ in terms of a_0.
5. Substitute $a_1, a_2, \ldots$ in eq. (2) to obtain the required solution of the given equation.

Figure G000079. introduction to Series solution · source slide page 8.

EV0000329

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



ENGINEERING MATHEMATICS - I
SERIES SOLUTION

PES UNIVERSITY

Que 1.

Find the power series solution of the equation

$$(x^2 + 1)y'' + xy' - xy = 0 \quad \text{about } x = 0.$$

Solution. Given eq.:

$$(x^2 + 1)y'' + xy' - xy = 0 \qquad (1)$$

Here, $P_0(x) = 1 + x^2$.
At $x = 0$, $P_0(0) = 1 \neq 0$
$\Rightarrow x = 0$ is an ordinary point.
Let the series solution of eq. (1) be

$$y = \sum_{n=0}^{\infty} a_n x^n; \qquad y' = \sum_{n=0}^{\infty} a_n \cdot n\, x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Also,

$$y'' = \sum_{n=0}^{\infty} a_n\, n(n-1)x^{n-2} = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$$

Figure G000080. introduction to Series solution · source slide page 9.

EV0000332

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

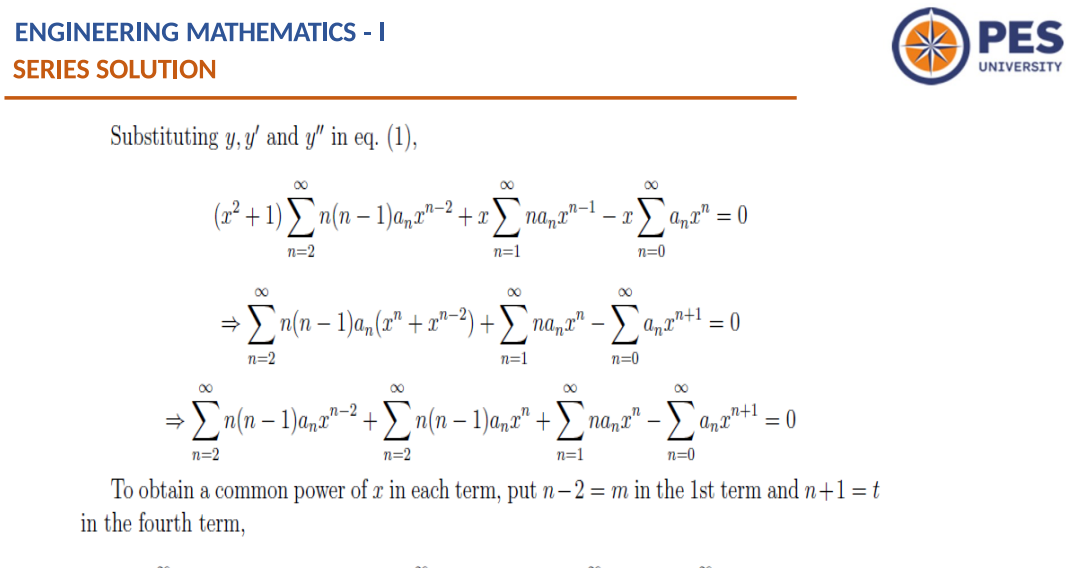


Figure G000081. introduction to Series solution · source slide page 10.

EV0000335

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

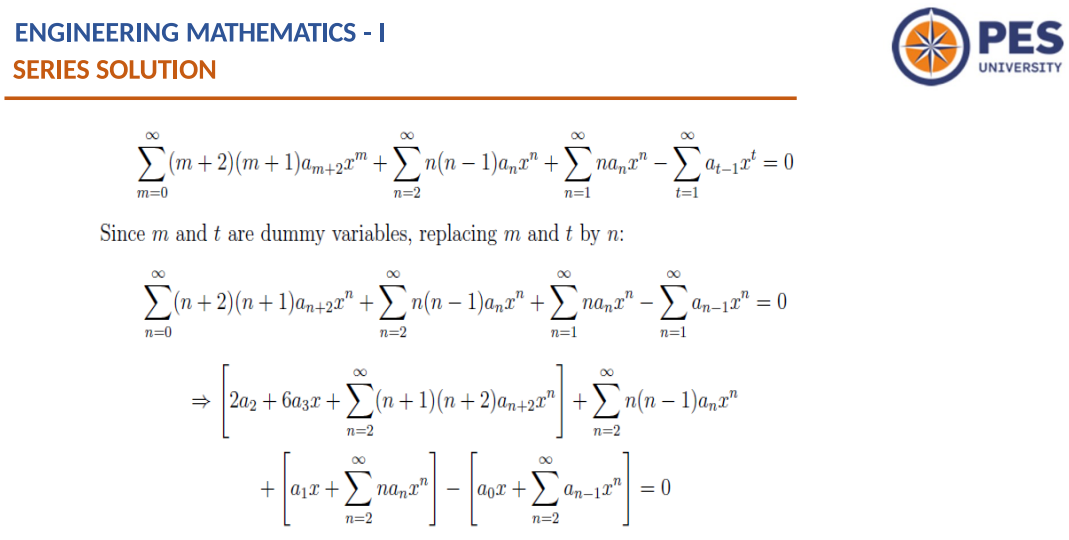


Figure G000082. introduction to Series solution · source slide page 11.

EV0000338

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

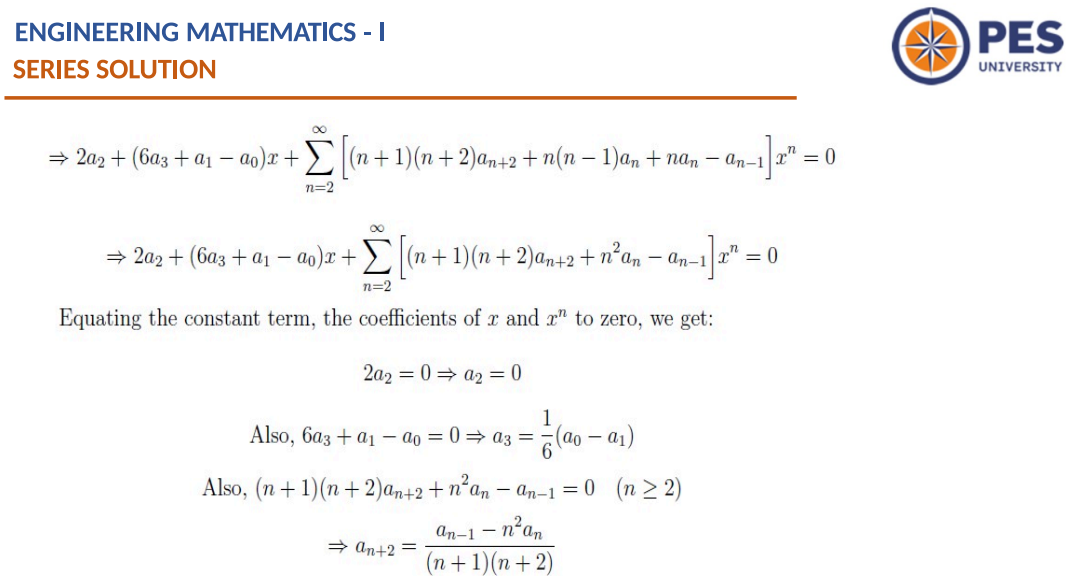


Figure G000083. introduction to Series solution · source slide page 12.

EV0000341

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

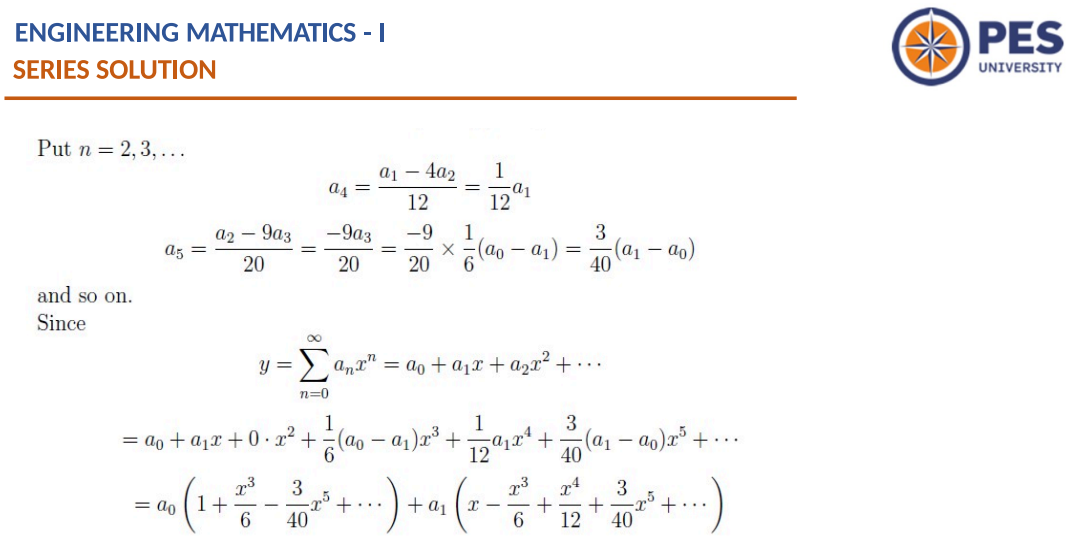


Figure G000084. introduction to Series solution · source slide page 13.

EV0000344

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000085. introduction to Series solution · source slide page 14.

EV0000347

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000307 · EV0000308 · EV0000309
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 4: SPECIAL FUNCTIONS AND SERIES SOLUTION OF DE SESSION: 6 SUB-TOPIC: SERIES SOLUTION Department of Science and Humanities EV0000310 · EV0000311 · EV0000312
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Power Series Method Consider a homogeneous linear second order DE with variable co-efficients: d²y dy Po(c) + Pi(c) + P2(c) y = 0 (1) dx2 dc d²y Pi(c) dy P2(c) dx2 Po(c) dr y=0 Po(c) P2(c) Let = Q(x). Po(c) Po(c) Then, d²y dy + P(c) + Q(x)y = 0 (2) dr2 dr Eq. (2) is called the normal form or canonical form or standard form of eq. (1). EV0000313 · EV0000314 · EV0000315
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] The power series solution of eq. (2) about a point c = co depends on the following definitions: 1. Ordinary point / Regular Point: A point = co is called an ordinary point of eq. (2), if P(c) and Q(c) are both analytic (i.e., differentiable) at To. If Po(c) ≠ O at c = co, then co is an ordinary point. 2. Singular point: A point c = co is called a singular point of eq. (2) if either P(c) or Q(z) or both are not analytic at co. If Po(c) = O at c = co, then co is a singular point. Regular Singular Point: A singular point is called a regular singular point of eq. (2) if (c - co)P(c) and (c - co)²Q(x) both are analytic (or differentiable) at c = Co: lim (c - co)P(c) = finite value C-20 EV0000316 · EV0000317 · EV0000318

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Regular Singular Point: A singular point is called a regular singular point of eq. 1. (2) if (x -co)P(α) and (c c)²Q(x) b both n are analytic (or differentiable) at = o: lim (-o)P() =f finitevalue 2>20 [Image text, OCR] lim (- co)²Q(x) = finite value Irregular r Singular Point: If either (-o)P(x) or r (c - co)²Q(c) or both I are not analytic at r = o. EV0000319 · EV0000320 · EV0000321

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Que 1. Find the singular points and ordinary points of the equation (1 - α²)y" - 6cy' - 4y = 0 Solution. Given eq. is (1 - α²)y" - 6ry' - 4y = 0 (1) Here, Po(x) = 1 - α². At the singular point, Po(c) = 0 ≥ 1 - α² = 0 → α2 = 1 → = ±1 are singular points of eq. (1) Dividing eq. (1) by (1 - r?) on both sides, 6c 4 (2) - x2y = 0 EV0000322 · EV0000323 · EV0000324

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Comparing eq. (2) with the standard form i.e., y" + P(x)y' + Q(x)y = 0, ≥ P(x) = 1- r2 -6c 6c 6c c2 -1 (c + 1)(c - 1) -4 4 and Q(α) = 4 1- x2 c2 -1 (c + 1)(x - 1) For r = 1, 6c lim(r - 1) · 6 3 c→→1 (c + 1)(c - 1) 2 lim(r - 1)2 4 4 = (c - 1) (c + 1)(x - 1) (c + 1) →→1 → It is a regular singular point at c = 1. For c = -1, 6c 6(-1) lim (r + 1) · 6 ⁚ 3 (c + 1)(r - 1) - -1 - 1 -2 4 lim, (c + 1)² (x + 1)(α - 1) =0 = It is a regular singular point at c = -1. All the values of c ≠ ±l are the ordinary points of the equation. EV0000325 · EV0000326 · EV0000327

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Series Solution about an Ordinary Point Let the power series solution of eq. Po(x)y" + Pi(c)y' + P2(x)y = 0 (1) about an ordinary point co be given as 8 > an(z - co)n = ao + a1(x - ro) + a2(c - co)² + ... (2) n=0 The co-efficients a1, @2, a3, .. . are obtained as follows: 1. Let y = 8 ≥ an(r - co)n be the series solution of the given eq. n=0 2. Differentiate y w.r.t. r twice to get y,y',y", then substitute y, y', y" in eq. (1). 3. Shift the summation index to obtain a common power of r in each term. 4. Equate the coefficients of various powers of r to zero to obtain @1, @2, ... in terms of ao· 5. Substitute @1, @2, ... in eq. (2) to obtain the required solution of the given equation. EV0000328 · EV0000329 · EV0000330

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Que 1. Find the power series solution of the equation (x² + 1)y" + ry' - ry = 0 about r = 0. Solution. Given eq.: (x² + 1)y" + ry' - ry = 0 (1) Here, Po(x) = 1 + x2. At = 0, Po(0) = 1 ≠ 0 → r = 0 is an ordinary point. Let the series solution of eq. (1) be y= anz"; an·nrn- nancn-1 n=0 n=0 n=1 Also, an n(n - 1)xn-2 n(n - 1)anxn-2 n=0 n=2 EV0000331 · EV0000332 · EV0000333

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Substituting y,y' and y" in eq. (1), 8 (x² + 1) ≥n(n - 1)anαn-2 + x na,c C anc^ = 0 n=2 n=1 n=0 ∑n(n - 1)an(xn + xn-2) + nanr' =0 n=2 n=1 n=0 ∑n(n - 1)anxn-2 +n(n - 1)an2" + "nanr" anT n=2 n=2 n=1 n=0 To obtain a common power of c in each term, put n - 2 = m in the 1st term and n +1 = t in the fourth term, (m + 2)(m + 1)am+2αm + n(n - 1)anα"n + ≥ 8 8 nanx" > at-1c = 0 m=0 n=2 n=1 t=1 Since m and t are dummy variables, replacing m and t by n: 8 8 8 8 (n + 2)(n + 1)an+2x" + n(n - 1)anxn + nanr" > an-1c^ = 0 n=0 n=2 n=1 n=1 2a2 + 6a3 + +(n + 1)(n + 2)an+2n +∑n(n - 1)anxn n=2 n=2 ax+nanc- aoc + an-1x" n=2 EV0000334 · EV0000335 · EV0000336

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Substituting y,y' and y" in eq. (1), 8 (x² + 1) ≥n(n - 1)anαn-2 + x na,c C anc^ = 0 n=2 n=1 n=0 ∑n(n - 1)an(xn + xn-2) + nanr' =0 n=2 n=1 n=0 ∑n(n - 1)anxn-2 +n(n - 1)an2" + "nanr" anT n=2 n=2 n=1 n=0 To obtain a common power of c in each term, put n - 2 = m in the 1st term and n +1 = t in the fourth term, (m + 2)(m + 1)am+2αm + n(n - 1)anα"n + ≥ 8 8 nanx" > at-1c = 0 m=0 n=2 n=1 t=1 Since m and t are dummy variables, replacing m and t by n: 8 8 8 8 (n + 2)(n + 1)an+2x" + n(n - 1)anxn + nanr" > an-1c^ = 0 n=0 n=2 n=1 n=1 2a2 + 6a3 + +(n + 1)(n + 2)an+2n +∑n(n - 1)anxn n=2 n=2 ax+nanc- aoc + an-1x" n=2 EV0000337 · EV0000338 · EV0000339
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] → 2a2 + (6a3 + a1 -ao)c + ∑(n + 1)(n + 2)an+2 + n(n - 1)an + nan - an-1 ”=0 n=2 → 2a2 + (6a3 + a1 - ao)c + |(n + 1)(n + 2)an+2 + n²an - an-1 c^ = 0 n=2 Equating the constant term, the coefficients of r and r^ to zero, we get: 2a2 = 0 =→ a2 = 0 Also, 6a3 + a1 - ao = 0 = a3 = (ao - a1) 6 Also, (n + 1)(n + 2)an+2 + n²an - an-1 = 0 (n ≥ 2) an-1 - n²an →αn+2= (n+ 1)(n + 2) EV0000340 · EV0000341 · EV0000342
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Put n = 2,3,.. @1 - 4a2 a4= 12 12 a @2 - 9a3 -9a3 -9 3 @5= a (a1 -ao) 20 20 20 6 40 and so on. Since anz" = ao + az + a2x? + .. n=0 3 = ao+a+0·² (ao - a1)x² + ar 6 12 40 2.3 3 3 24 3 ao 6 40 6 12 40 EV0000343 · EV0000344 · EV0000345
14. Source slide page 14 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000346 · EV0000347 · EV0000348

SHORT-NOTE RECAP

Source deck preserved: introduction to Series solution. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000307

9 Source slide deck: Power series Continued

This entry preserves the complete slide deck titled Power series Continued as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_04, to avoid an unsupported match. EV0000349 · EV0000350

Graphs and diagrams



Figure G000086. Power series Continued · source slide page 1.

EV0000350

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000087. Power series Continued · source slide page 2.

EV0000353

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

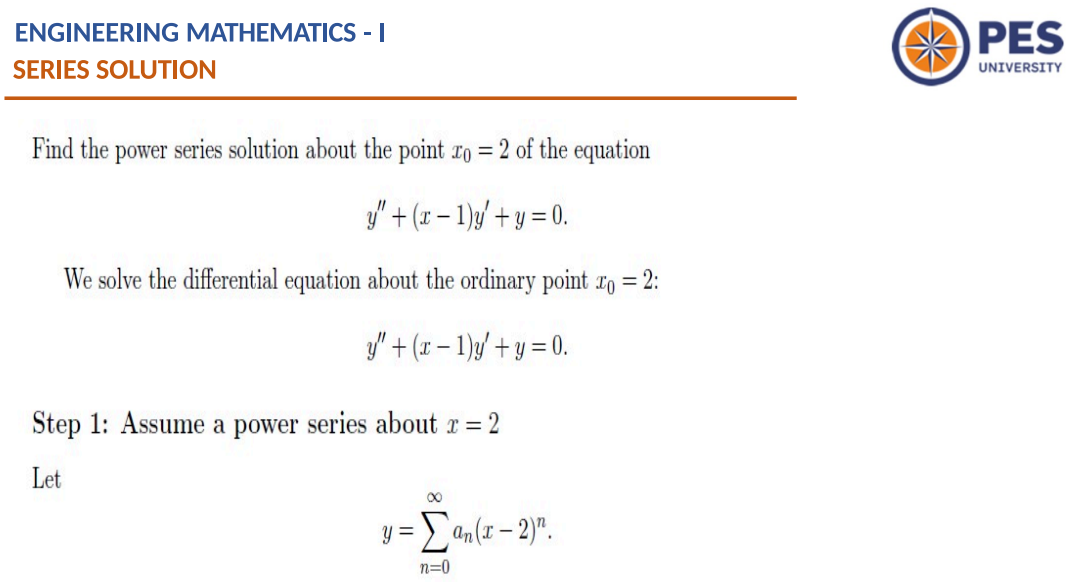


Figure G000088. Power series Continued · source slide page 3.

EV0000356

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

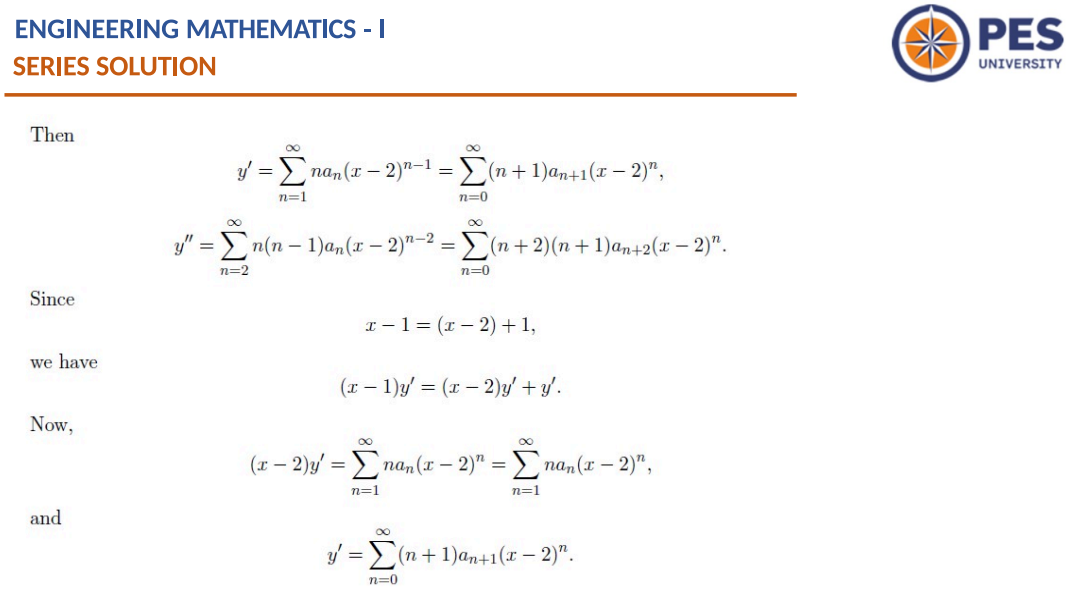


Figure G000089. Power series Continued · source slide page 4.

EV0000359

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

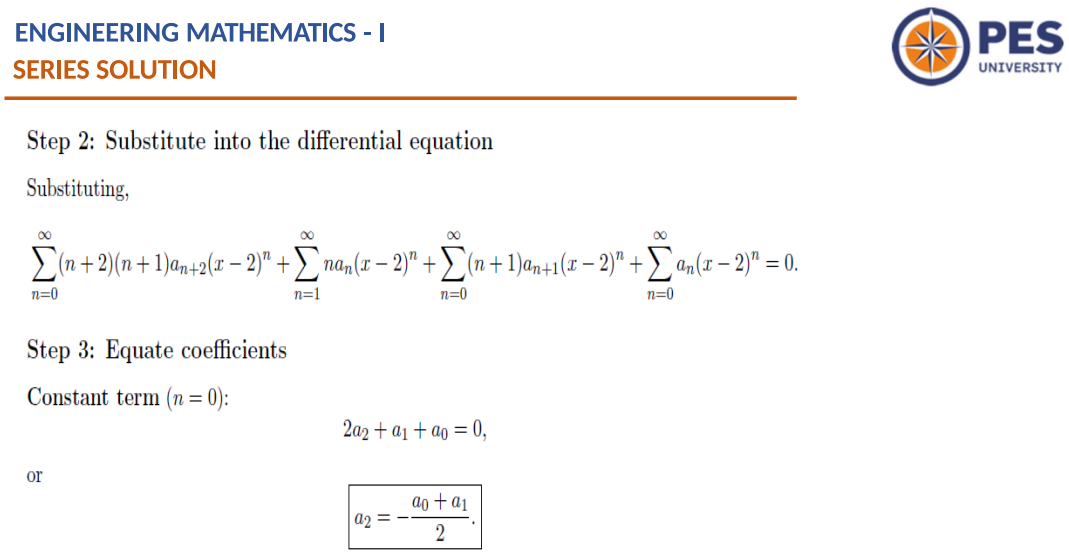


Figure G000090. Power series Continued · source slide page 5.

EV0000362

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
SERIES SOLUTION

PES UNIVERSITY

General term $(n \geq 1)$:

$$(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} + (n+1)a_n = 0.$$

Hence,

$$\boxed{a_{n+2} = -\frac{a_{n+1} + a_n}{n+2}, \qquad n \geq 1.}$$

Step 4: Compute the coefficients

Let

$$a_0 = A, \qquad a_1 = B.$$

Then

$$a_2 = -\frac{A+B}{2},$$

$$a_3 = -\frac{a_2 + a_1}{3} = \frac{A-B}{6},$$

Figure G000091. Power series Continued · source slide page 6.

EV0000365

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
SERIES SOLUTION

PES UNIVERSITY

$$a_4 = -\frac{a_3 + a_2}{4} = \frac{A+2B}{12},$$

$$a_5 = -\frac{a_4 + a_3}{5} = -\frac{3A+B}{60},$$

$$a_6 = -\frac{a_5 + a_4}{6} = -\frac{A+3B}{360},$$

and so on.

Final Answer

$$\boxed{\begin{aligned} y(x) = A\left[1 - \frac{(x-2)^2}{2} + \frac{(x-2)^3}{6} + \frac{(x-2)^4}{12} - \frac{(x-2)^5}{20} - \frac{(x-2)^6}{360} + \cdots\right] \\ + B\left[(x-2) - \frac{(x-2)^2}{2} - \frac{(x-2)^3}{6} + \frac{(x-2)^4}{6} - \frac{(x-2)^5}{60} - \frac{(x-2)^6}{120} + \cdots\right]. \end{aligned}}$$

Figure G000092. Power series Continued · source slide page 7.

EV0000368

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000093. Power series Continued · source slide page 8.

EV0000371

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000349 · EV0000350 · EV0000351
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 4: SPECIAL FUNCTIONS AND SERIES SOLUTION OF DE SESSION: 7 SUB-TOPIC: POWER SERIES SOLUTION_continued Department of Science and Humanities EV0000352 · EV0000353 · EV0000354
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Find the power series solution about the point co = 2 of the equation y" + (x - 1)y' + y = 0. We solve the differential equation about the ordinary point To = 2: y"+(c -1)y' + y = 0. Step 1: Assume a power series about r = 2 Let y= > an(r - 2)n. n=0 EV0000355 · EV0000356 · EV0000357
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Then y=> nan(x - 2)n-1 = (n + 1)an+1(x - 2)n, n=1 n=0 n(n - 1)an(z - 2)n-2 = >(n + 2)(n + 1)an+2(c - 2)n. n=2 n=0 Since -1=(-2)+l, we have (c - 1)y' = (π - 2)y' + y'. Now, (c - 2)y' = ∑nan(x - 2)n = nan(r - 2)n, n=1 n=1 and y' =(n + 1)an+1(x - 2)n. n=0 EV0000358 · EV0000359 · EV0000360

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] Step 2: Substitute into the differential equation Substituting, (n+ 2)(n + 1)an+2(c - 2)n + > nan(c - 2)n + ≥(n + 1)an+1(x - 2)n + > an(c -2)n = 0. n=0 n=1 n=0 n=0 Step 3:Equate coefficients Constant term (n = 0): 2a2+ a1 + ao = 0, or ao+ a1 @2= 2 EV0000361 · EV0000362 · EV0000363
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] General term (n ≥ 1): (n + 2)(n + 1)an+2 + (n + 1)an+1 + (n + 1)an = 0. Hence, an+1 + an @n+2=- n≥1. n+2 Step 4: Compute the coefficients Let ao = A, @1 = B. Then A+B @2=- 2 a2+a1 A-B a3 = 3 6 EV0000364 · EV0000365 · EV0000366
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I SERIES SOLUTION [Image text, OCR] a3+a2 A+2B a4=: 4 12 a4+a3 3A+ B a5 = 5 60 a5+a4 A+3B a6= 6 360 and so on. Final Answer (c-2)2 (c - 2)3 (c - 2)4 (x - 2)5 (c - 2)6 y(c) = A + + 2 6 12 20 360 (c -2)2 (c - 2)3 (c -2)4 (c - 2)5 (c - 2)6 +B (c-2) 2 6 6 60 120 EV0000367 · EV0000368 · EV0000369
8. Source slide page 8 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000370 · EV0000371 · EV0000372

SHORT-NOTE RECAP

Source deck preserved: Power series Continued. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000349

10 Source slide deck: Frobenius Method

This entry preserves the complete slide deck titled Frobenius Method as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_04, to avoid an unsupported match. EV0000373 · EV0000374

Graphs and diagrams



Figure G000094. Frobenius Method · source slide page 1.

EV0000374

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

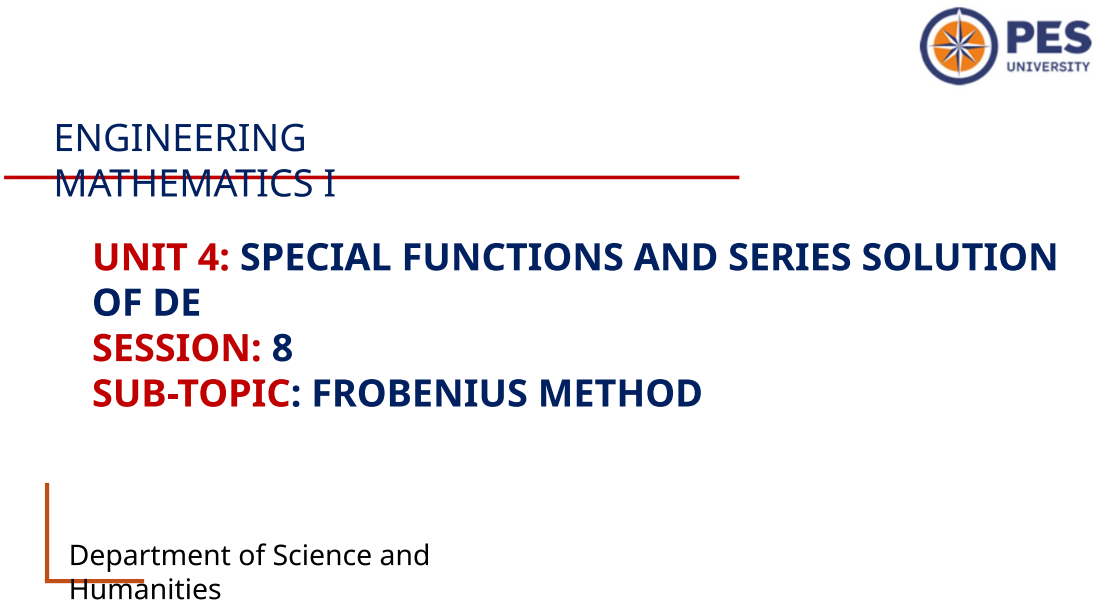


Figure G000095. Frobenius Method · source slide page 2.

EV0000377

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

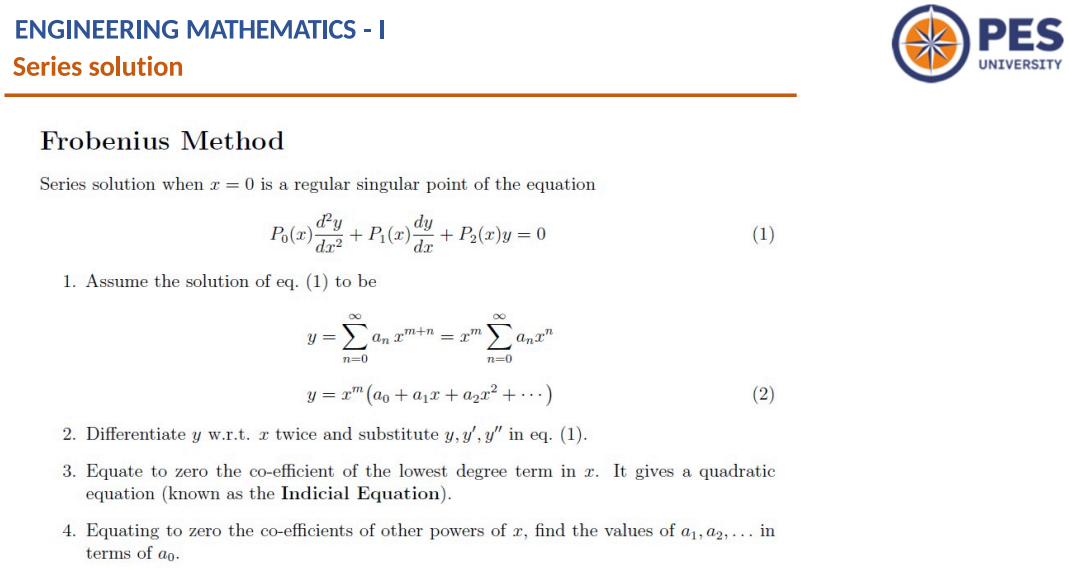
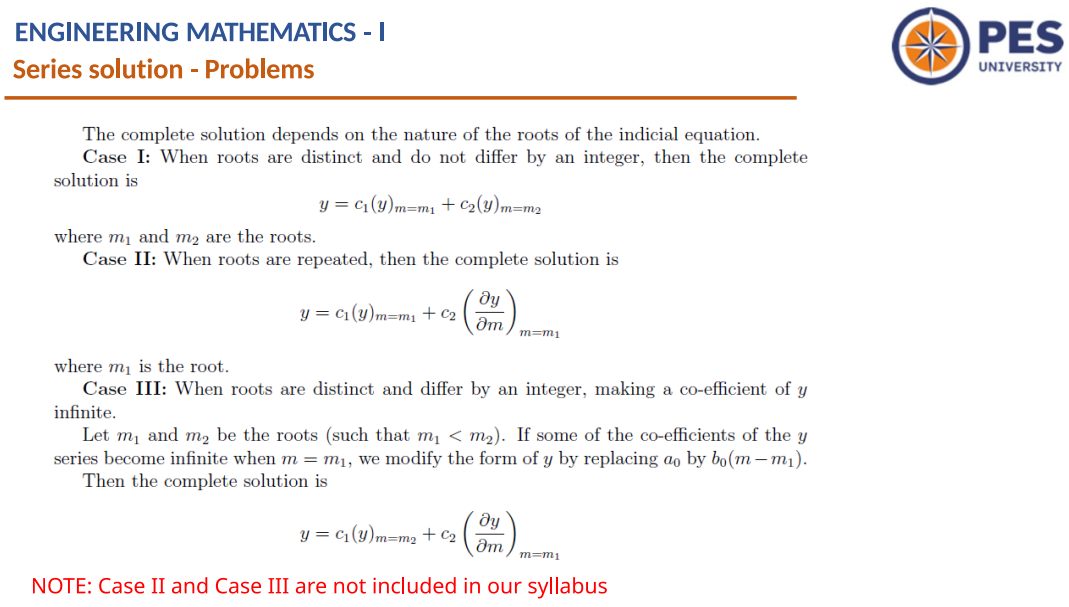


Figure G000096. Frobenius Method · source slide page 3.

EV0000380

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



ENGINEERING MATHEMATICS - I
Series solution - Problems

PES UNIVERSITY

The complete solution depends on the nature of the roots of the indicial equation.

Case I: When roots are distinct and do not differ by an integer, then the complete solution is

$$y = c_1(y)_{m=m_1} + c_2(y)_{m=m_2}$$

where m_1 and m_2 are the roots.

Case II: When roots are repeated, then the complete solution is

$$y = c_1(y)_{m=m_1} + c_2\left(\frac{\partial y}{\partial m}\right)_{m=m_1}$$

where m_1 is the root.

Case III: When roots are distinct and differ by an integer, making a co-efficient of y infinite.

Let m_1 and m_2 be the roots (such that $m_1 < m_2$). If some of the co-efficients of the y series become infinite when $m = m_1$, we modify the form of y by replacing a_0 by $b_0(m - m_1)$.

Then the complete solution is

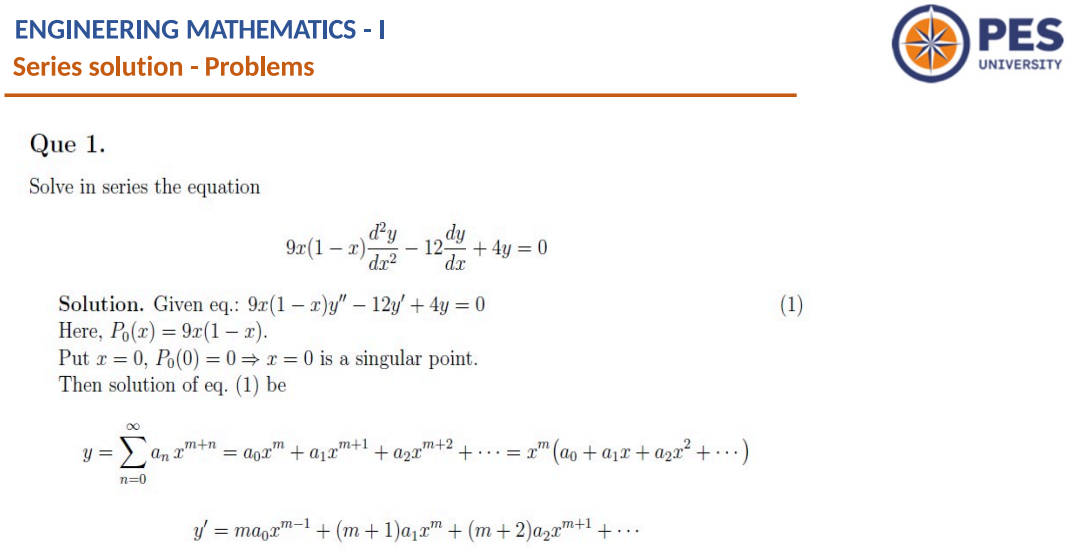
$$y = c_1(y)_{m=m_2} + c_2\left(\frac{\partial y}{\partial m}\right)_{m=m_1}$$

NOTE: Case II and Case III are not included in our syllabus

Figure G000097. Frobenius Method · source slide page 4.

EV0000383

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



ENGINEERING MATHEMATICS - I
Series solution - Problems

PES UNIVERSITY

Que 1.

Solve in series the equation

$$9x(1-x)\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 4y = 0$$

Solution. Given eq.: $9x(1-x)y'' - 12y' + 4y = 0$ (1)

Here, $P_0(x) = 9x(1-x)$.

Put $x = 0$, $P_0(0) = 0 \Rightarrow x = 0$ is a singular point.

Then solution of eq. (1) be

$$y = \sum_{n=0}^{\infty} a_n x^{m+n} = a_0x^m + a_1x^{m+1} + a_2x^{m+2} + \cdots = x^m\left(a_0 + a_1x + a_2x^2 + \cdots\right)$$

$$y' = ma_0x^{m-1} + (m+1)a_1x^m + (m+2)a_2x^{m+1} + \cdots$$

Figure G000098. Frobenius Method · source slide page 5.

EV0000386

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Series solution - Problems

PES UNIVERSITY

$$y'' = m(m-1)a_0x^{m-2} + (m+1)ma_1x^{m-1} + (m+2)(m+1)a_2x^m + \cdots$$

Substituting y, y', y'' in eq. (1),

$$9x(1-x)\left[m(m-1)a_0x^{m-2} + (m+1)ma_1x^{m-1} + (m+2)(m+1)a_2x^m + \cdots\right]$$

$$-12\left[ma_0x^{m-1} + (m+1)a_1x^m + (m+2)a_2x^{m+1} + \cdots\right] + 4\left[a_0x^m + a_1x^{m+1} + a_2x^{m+2} + \cdots\right] = 0$$

$$\Rightarrow \left[9m(m-1)a_0x^{m-1} + 9(m+1)ma_1x^m + 9(m+2)(m+1)a_2x^{m+1} + \cdots\right.$$

$$\left. -9m(m-1)a_0x^m - 9(m+1)ma_1x^{m+1} + 9(m+2)(m+1)a_2x^{m+2} - \cdots\right]$$

$$+\left[-12ma_0x^{m-1} - 12(m+1)a_1x^m - 12(m+2)a_2x^{m+1} - \cdots\right]$$

$$+\left[4a_0x^m + 4a_1x^{m+1} + 4a_2x^{m+2} + \cdots\right] = 0$$

Figure G000099. Frobenius Method · source slide page 6.

EV0000389

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Series solution - Problems

PES UNIVERSITY

The lowest power of x is $m-1$. So equating its coefficient to zero (coefficient of a_0):

$$9m(m-1)a_0 - 12ma_0 = 0 \Rightarrow a_0\left[9m^2 - 9m - 12m\right] = 0 \Rightarrow a_0\left[9m^2 - 21m\right] = 0$$

$$\Rightarrow 3m(3m-7) = 0 \Rightarrow m = 0, \frac{7}{3}$$

i.e., roots are distinct but do not differ by an integer.

The coefficient of x^m equated to zero:

$$9m(m+1)a_1 - 9m(m-1)a_0 - 12(m+1)a_1 + 4a_0 = 0$$

$$\Rightarrow a_1\left[9m^2 + 9m - 12m - 12\right] = a_0\left[9m^2 - 9m - 4\right]$$

$$\Rightarrow 3a_1(3m^2 - m - 4) = a_0\left[9m^2 + 3m - 12m - 4\right]$$

Figure G000100. Frobenius Method · source slide page 7.

EV0000392

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

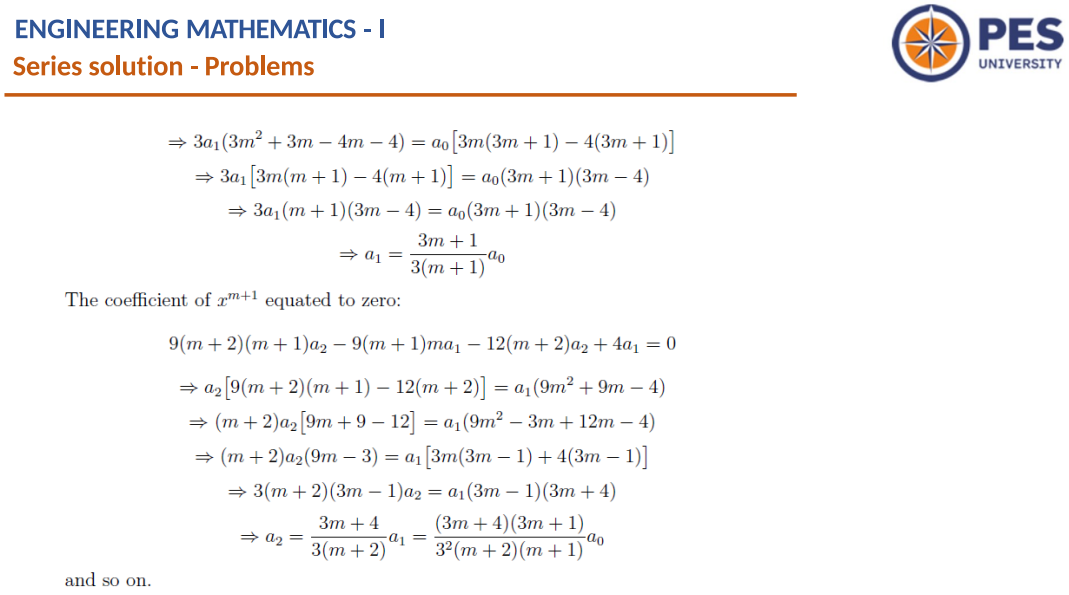


Figure G000101. Frobenius Method · source slide page 8.

EV0000395

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

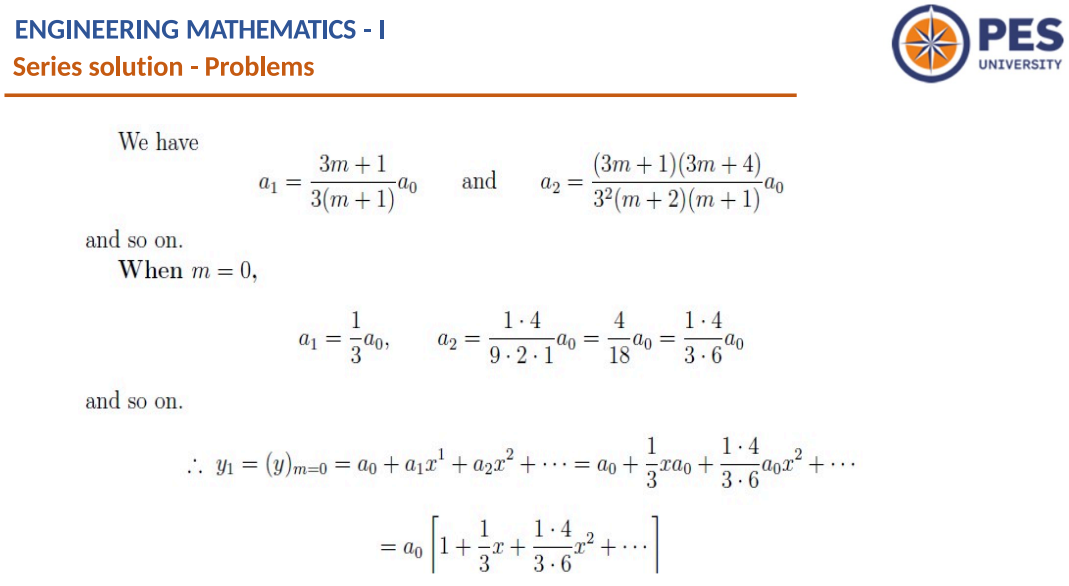


Figure G000102. Frobenius Method · source slide page 9.

EV0000398

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

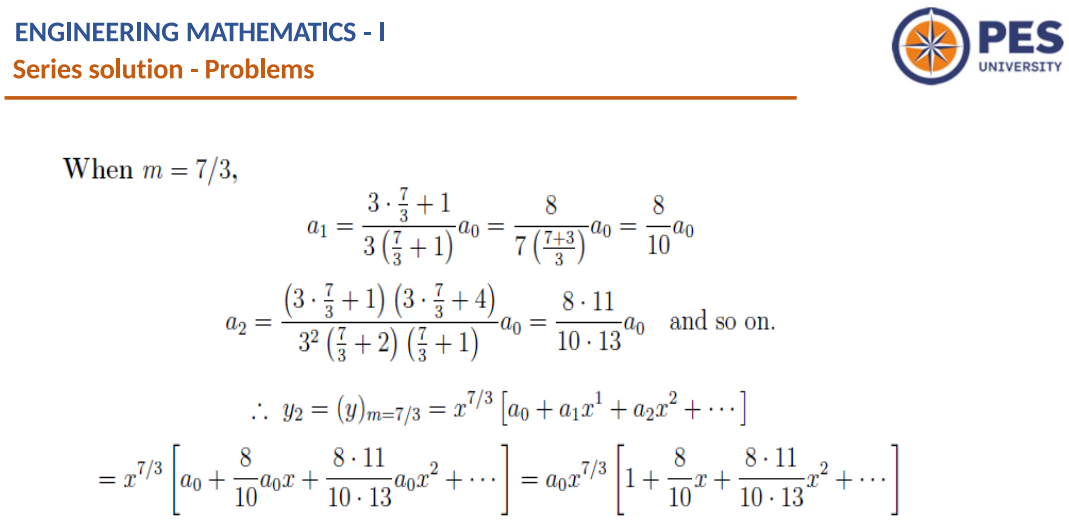


Figure G000103. Frobenius Method · source slide page 10.

EV0000401

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

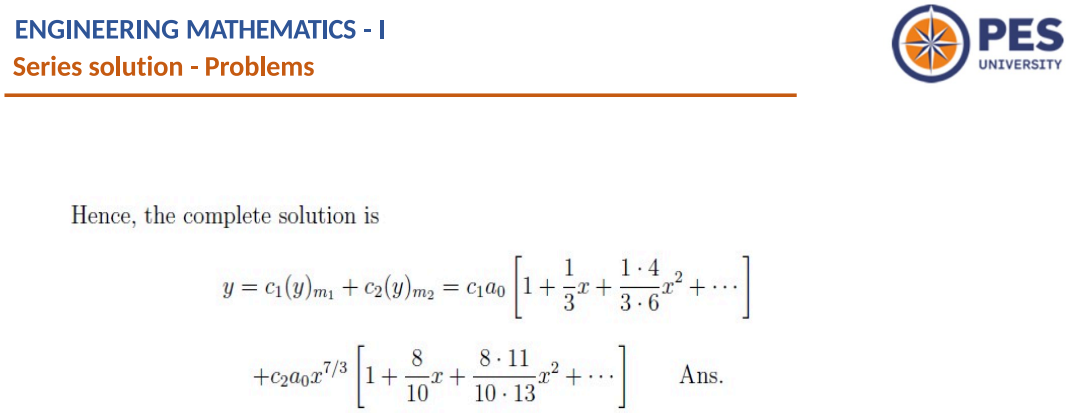


Figure G000104. Frobenius Method · source slide page 11.

EV0000404

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000105. Frobenius Method · source slide page 12.

EV0000407

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000373 · EV0000374 · EV0000375
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 4: SPECIAL FUNCTIONS AND SERIES SOLUTION OF DE SESSION: 8 SUB-TOPIC: frobenius method Department of Science and Humanities EV0000376 · EV0000377 · EV0000378
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Series solution [Image text, OCR] Frobenius Method Series solution when c = O is a regular singular point of the equation d²y + Pi(c) dy Po(x) + P2(x)y = 0 (1) dr2 dr 1. Assume the solution of eq. (1) to be y=≥ an rm+n m anzn n=0 n=0 y = rm(ao + ac + a2r² + .. . (2) 2. Differentiate y w.r.t. r twice and substitute y,y',y" in eq. (1). 3. Equate to zero the co-efficient of the lowest degree term in c. It gives a quadratic equation (known as the Indicial Equation). 4. Equating to zero the co-efficients of other powers of r, find the values of a1, @2,... in terms of ao. EV0000379 · EV0000380 · EV0000381

4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] The complete solution depends on the nature of the roots of the indicial equation. Case I: When roots are distinct and do not differ by an integer, then the complete solution is y = C1(y)m=m + C2(y)m=m2 where m1 and m2 are the roots. Case II: When roots are repeated, then the complete solution is ay y = C1(y)m=m + C2 0m m=m1 where mi is the root. Case III: When roots are distinct and differ by an integer, making a co-efficient of y infinite. Let m1 and m2 be the roots (such that m1 < m2). If some of the co-efficients of the y series become infinite when m = m1, we modify the form of y by replacing ao by bo(m - m1). Then the complete solution is dy y = C1(y)m=m2 + C2 0m m=m1 NOTE: Case II and Case III are not included in our syllabus EV0000382 · EV0000383 · EV0000384

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] Que 1. Solve in series the equation d²y dy 9x(1 -c) 12 +4y= 0 dr2 dc Solution. Given eq.: 9c(1 - c)y" - 12y' + 4y = 0 (1) Here, Po(r) = 9c(1 - c). Put r = 0, Po(O) = 0 = = 0 is a singular point. Then solution of eq. (1) be y=∑ n=0 y' = maoαm-1 + (m + 1)a2m + (m + 2)a2xm+1 + ... EV0000385 · EV0000386 · EV0000387

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] y" = m(m - 1)aoxm-2 + (m + 1)mazm-1 + (m + 2)(m + 1)a2xm + .. - Substituting y,y',y" in eq. (1), 9z(1 - x)|m(m - 1)aoxm-2 + (m + 1)ma1cm-1 + (m + 2)(m + 1)a2αm -12|maoαm-1 + (m+ 1)a1α"m + (m + 2)a2xm+1 + .. +4 aoxm+a12m+1 + a2xm+2 9m(m - 1)aozm-1 + 9(m + 1)ma1α"m + 9(m + 2)(m + 1)a2xm+1 + .. -9m(m - 1)aoxm - 9(m + 1)ma12m+1 + 9(m + 2)(m + 1)a2xm+2 12maocm-1 - 12(m + 1)acm - 12(m + 2)a2xm+1 +|4aoxm + 4a1cm+1 + 4a2xm+2 + .. = 0 EV0000388 · EV0000389 · EV0000390

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] The lowest power of r is m - 1. So equating its coefficient to zero (coefficient of ao): 9m(m - 1)ao - 12mao = 0 → ao [9m² - 9m - 12m] = 0 ≥ ao [9m² - 21m] = 0 ≥ 3m(3m - 7) = 0 ≥ m = 0,3 i.e., roots are distinct but do not differ by an integer. The coefficient of r"m equated to zero: 9m(m + 1)a1 - 9m(m - 1)ao - 12(m + 1)a + 4ao = 0 → a1 [9m² + 9m - 12m - 12] = ao [9m² - 9m - 4] → 3a(3m² - m - 4) = ao [9m² + 3m - 12m - 4 EV0000391 · EV0000392 · EV0000393

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] → 3a(3m² + 3m - 4m - 4) = ao [3m(3m + 1) - 4(3m + 1)] > 3a1 [3m(m + 1) - 4(m + 1)] = ao(3m + 1)(3m - 4) → 3a(m + 1)(3m - 4) = ao(3m + 1)(3m - 4) 3m + 1 →a1=3 3(m + 1) ao The coeffcient of rm+1 equated to zero: 9(m + 2)(m + 1)a2 - 9(m + 1)ma1 - 12(m + 2)a2 + 4a = 0 → a2 [9(m + 2)(m + 1) - 12(m + 2)] = a1(9m² + 9m - 4) → (m + 2)a2[9m + 9 - 12] = a1(9m² - 3m + 12m - 4) → (m + 2)a2(9m - 3) = a1 [3m(3m - 1) + 4(3m - 1)] → 3(m + 2)(3m - 1)a2 = a1(3m - 1)(3m + 4) 3m + 4 (3m + 4)(3m + 1) →2= 3 3(m + 2) @1 32(m + 2)(m + 1) ao and so on EV0000394 · EV0000395 · EV0000396

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] We have 3m + 1 (3m + 1)(3m + 4) a1= ao and a2= 3(m + 1) 32(m + 2)(m + 1) ao and so on. When m = 0, 1.4 4 1.4 -a0; @2= ao ao ao 3 9.2.1 18 3.6 and so on. . y1 = (y)m=0 = ao + a1x + a2x² + ..· = ao + 1 1.4 --cao+ 3.6 aor 3 1.4 =ao 1+ 3 3.6 EV0000397 · EV0000398 · EV0000399

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] When m = 7/3, 3. +1 3 8 a1= ao ao = -ao 3 7(7+3 10 (3·+1)(3·+4) 8.11 a2 = ao ao and so on. 3² (3+ 2) (+ 1) 10·13 y2= (y)m=7/3 = α7/3 ao + a1x + a2x² + 8 8 . 11 8 8.11 = r7/3 ao+ aoc+ aoc aox7/3 1+ -c+ 10 10·13 10 10 ·13 EV0000400 · EV0000401 · EV0000402
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] Hence, the complete solution is 1.4 y = C1(y)m + C2(y)m2 =C1a0 1+ ⋮ -C┼ 3 3.6 C 8 8 . 11 +C2aox7/3 1+ 2 Ans. 10 10.13 EV0000403 · EV0000404 · EV0000405
12. Source slide page 12 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000406 · EV0000407 · EV0000408

SHORT-NOTE RECAP

Source deck preserved: Frobenius Method. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000373

11 Source slide deck: Frobenius method Continued

This entry preserves the complete slide deck titled Frobenius method Continued as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_04, to avoid an unsupported match. EV0000409 · EV0000410

Graphs and diagrams



Figure G000106. Frobenius method Continued · source slide page 1.

EV0000410

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

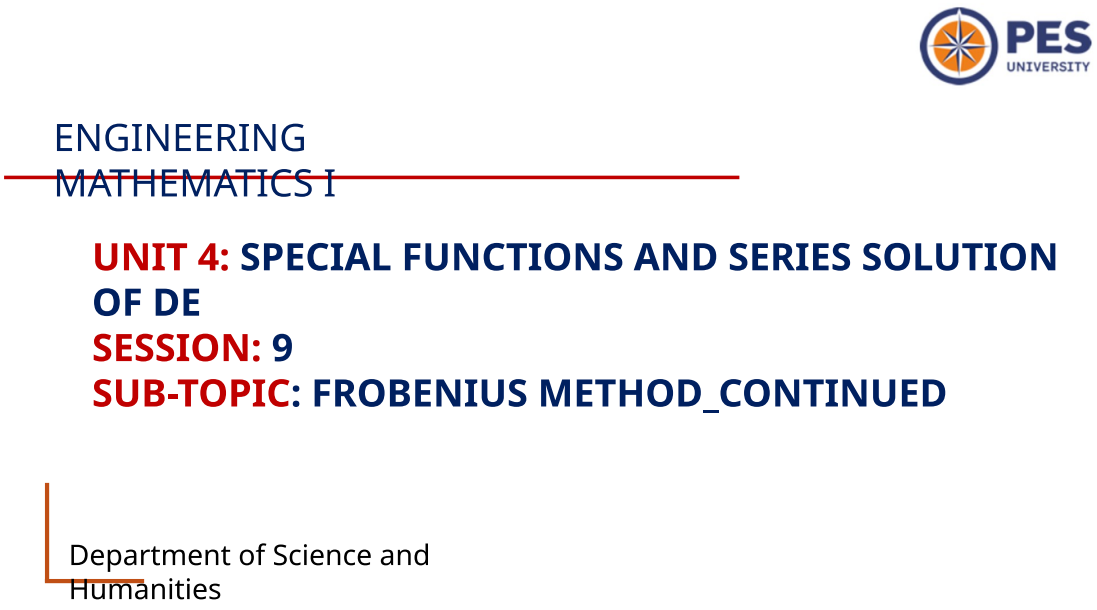


Figure G000107. Frobenius method Continued · source slide page 2.

EV0000413

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

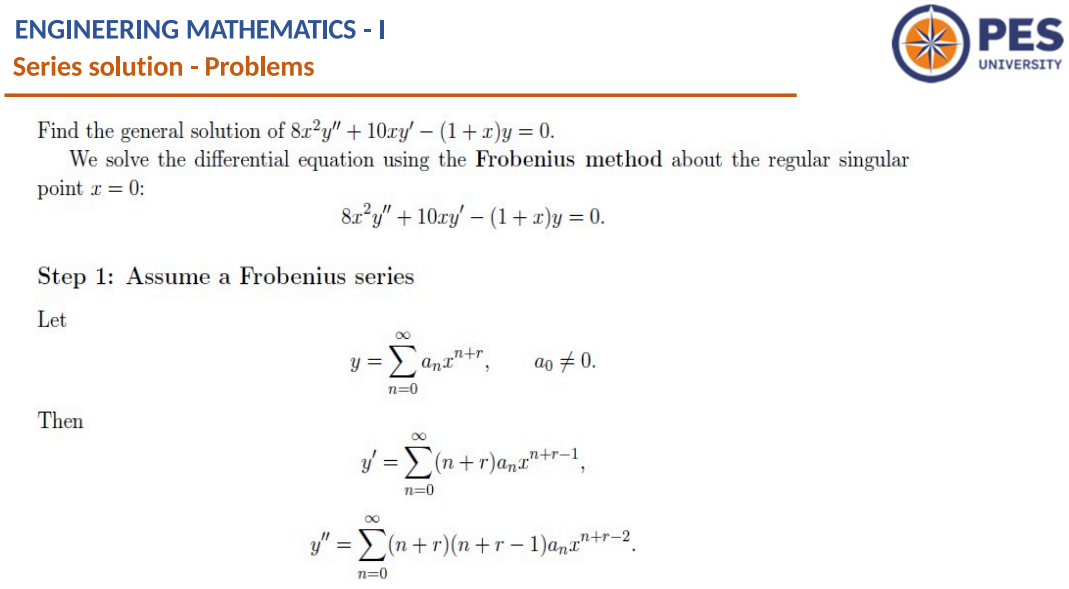


Figure G000108. Frobenius method Continued · source slide page 3.

EV0000416

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Series solution - Problems

PES UNIVERSITY

Step 2: Substitute into the differential equation

Substituting,

$$8x^2y'' + 10xy' - (1+x)y = 0,$$

gives

$$8\sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r} + 10\sum_{n=0}^{\infty}(n+r)a_nx^{n+r} - \sum_{n=0}^{\infty}a_nx^{n+r} - \sum_{n=0}^{\infty}a_nx^{n+r+1} = 0.$$

Shift the last summation:

$$\sum_{n=1}^{\infty}a_{n-1}x^{n+r}.$$

Hence,

$$[8r(r-1)+10r-1]a_0 + \sum_{n=1}^{\infty}\Big[\big(8(n+r)(n+r-1)+10(n+r)-1\big)a_n - a_{n-1}\Big]x^{n+r} = 0.$$

Figure G000109. Frobenius method Continued · source slide page 4.

EV0000419

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Series solution - Problems

PES UNIVERSITY

Step 3: Indicial equation

The coefficient of x^r gives

$$8r(r-1)+10r-1=0,$$

$$8r^2+2r-1=0.$$

Hence

$$\boxed{r=\frac{1}{4},\ -\frac{1}{2}.}$$

Figure G000110. Frobenius method Continued · source slide page 5.

EV0000422

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

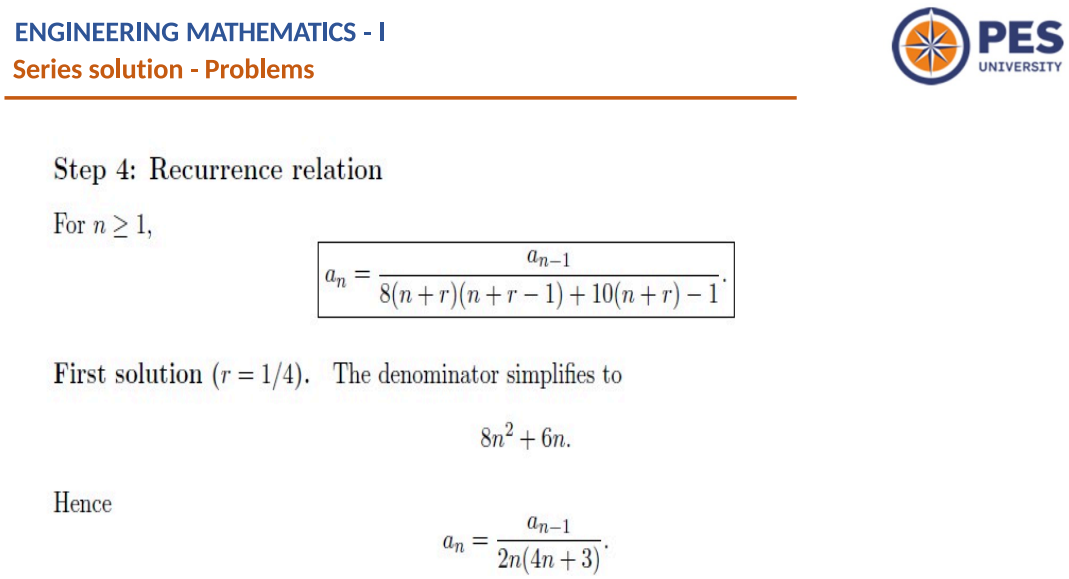


Figure G000111. Frobenius method Continued · source slide page 6.

EV0000425

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

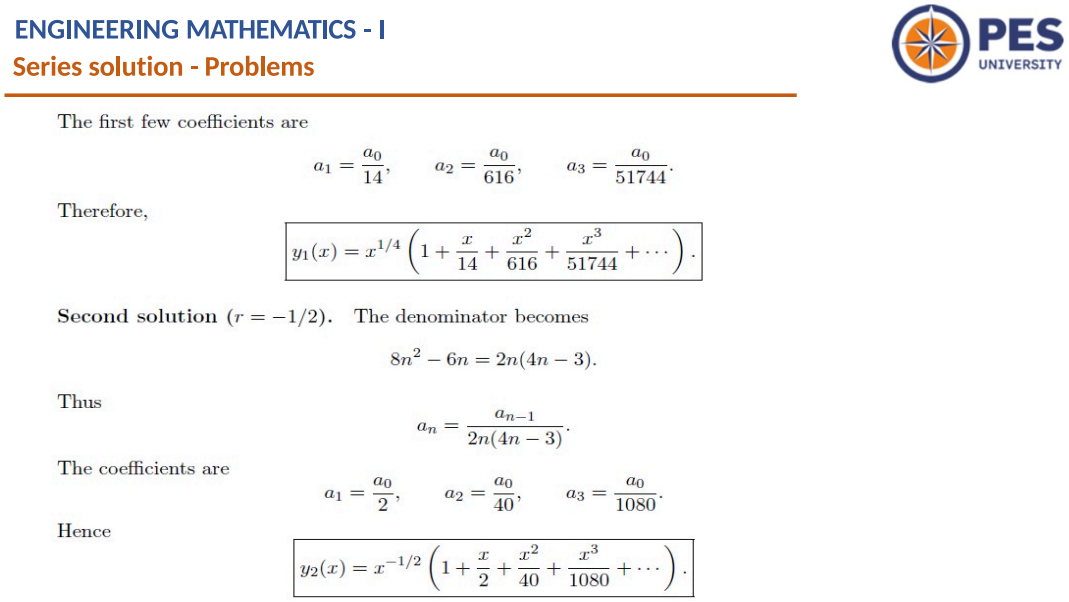


Figure G000112. Frobenius method Continued · source slide page 7.

EV0000428

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Series solution - Problems

PES UNIVERSITY

General Solution

$$y(x) = C_1 x^{1/4}\left(1 + \frac{x}{14} + \frac{x^2}{616} + \frac{x^3}{51744} + \cdots\right) + C_2 x^{-1/2}\left(1 + \frac{x}{2} + \frac{x^2}{40} + \frac{x^3}{1080} + \cdots\right).$$

where C_1 and C_2 are arbitrary constants.

Figure G000113. Frobenius method Continued · source slide page 8.

EV0000431

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PES UNIVERSITY

Thank you

Department of Science and Humanities

Figure G000114. Frobenius method Continued · source slide page 9.

EV0000434

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000409 · EV0000410 · EV0000411
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 4: SPECIAL FUNCTIONS AND SERIES SOLUTION OF DE SESSION: 9 SUB-TOPIC: frobenius method_continued Department of Science and Humanities EV0000412 · EV0000413 · EV0000414
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] Find the general solution of 8r2y" + 10ry' - (1 + c)y = 0. We solve the differential equation using the Frobenius method about the regular singular point c = 0: 8.r²y" + 10cy' - (1 + c)y = 0. Step 1: Assume a Frobenius series Let M y anzn+r. ao≠ 0. n=0 Then y' = (n+r)anzn+r-1 n=0 y" = (n + r)(n +r - 1)anxn+r-2. n=0 EV0000415 · EV0000416 · EV0000417
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] Step 2: Substitute into the differential equation Substituting, 8x²y" + 10y′ -(1 + c)y = 0, gives ∑(n + r)(n + r - 1)anzn+r + 10 (n + r)anzn+r anzn+r n=0 n=0 n=0 n=0 Shift the last summation: an-1c^+r n=1 Hence, [8r(r - 1) + 10r - 1]ao + > (8(n + r)(n + r - 1) + 10(n + r) - 1)an - an-1 = 0. n= EV0000418 · EV0000419 · EV0000420
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] Step 3: Indicial equation The coefficient of r" gives 8r(r - 1) + 10r - 1 = 0, 8r2 + 2r -1 = 0. Hence EV0000421 · EV0000422 · EV0000423
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] Step 4:Recurrence relation For n ≥ 1, an-1 an 8(n + r)(n + r - 1) + 10(n + r) - 1 First solution (r = 1/4). The denominator simplifies to 8n2 + 6n. Hence an-1 an 2n(4n+3) EV0000424 · EV0000425 · EV0000426
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] The first few coefficients are @o ao a0 a1 @2= 616' Q3= 14' 51744 Therefore, 3 y1(c) = x1/4 C 616 51744 Second solution (r = -1/2). The denominator becomes 8n² - 6n = 2n(4n - 3). Thus @n-1 an= 2n(4n - 3) The coefficients are ao ao ao a1 a2 40' @3= 2 1080 Hence y2(c) = z-1/2 T 40 1080 EV0000427 · EV0000428 · EV0000429
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Series solution - Problems [Image text, OCR] General Solution T 3 y(z) = Cx1/4 1+ 22 14 616 ┼ 51744 1+ T + C2c-1/2 2.2 23 -2 ┼ 40 1080 where Ci and C2 :are arbitrary constants. EV0000430 · EV0000431 · EV0000432
9. Source slide page 9 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000433 · EV0000434 · EV0000435

SHORT-NOTE RECAP

Source deck preserved: Frobenius method Continued. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000409

12 Source slide deck: Bessel functions of order n and n

This entry preserves the complete slide deck titled Bessel functions of order n and n as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_04, to avoid an unsupported match. EV0000487 · EV0000488

Graphs and diagrams



Figure G000132. Bessel functions of order n and n · source slide page 1.

EV0000488

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

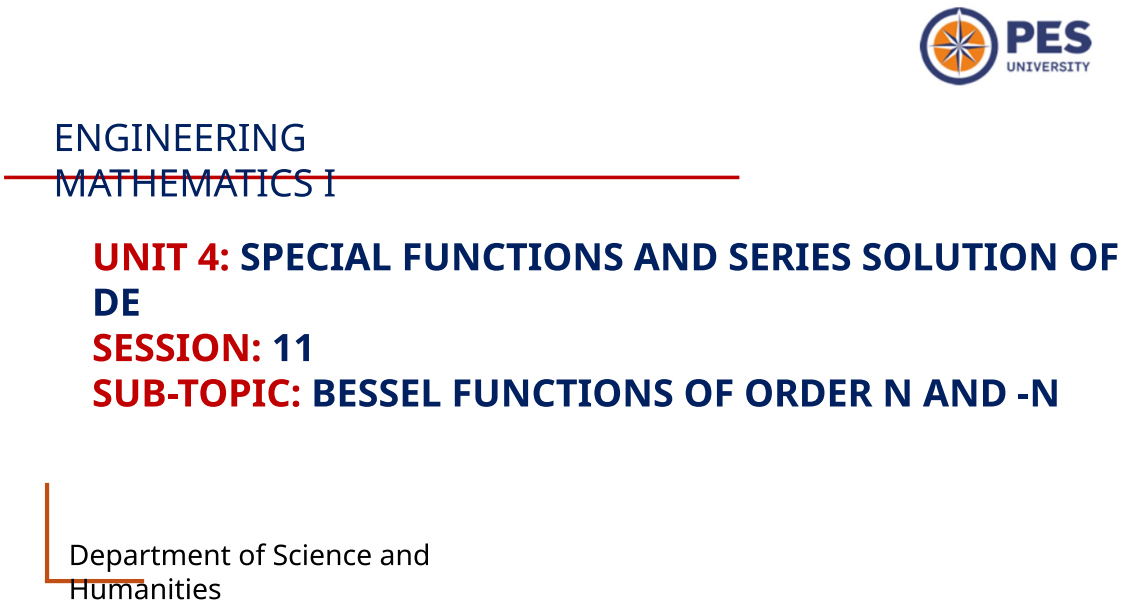


Figure G000133. Bessel functions of order n and n · source slide page 2.

EV0000491

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

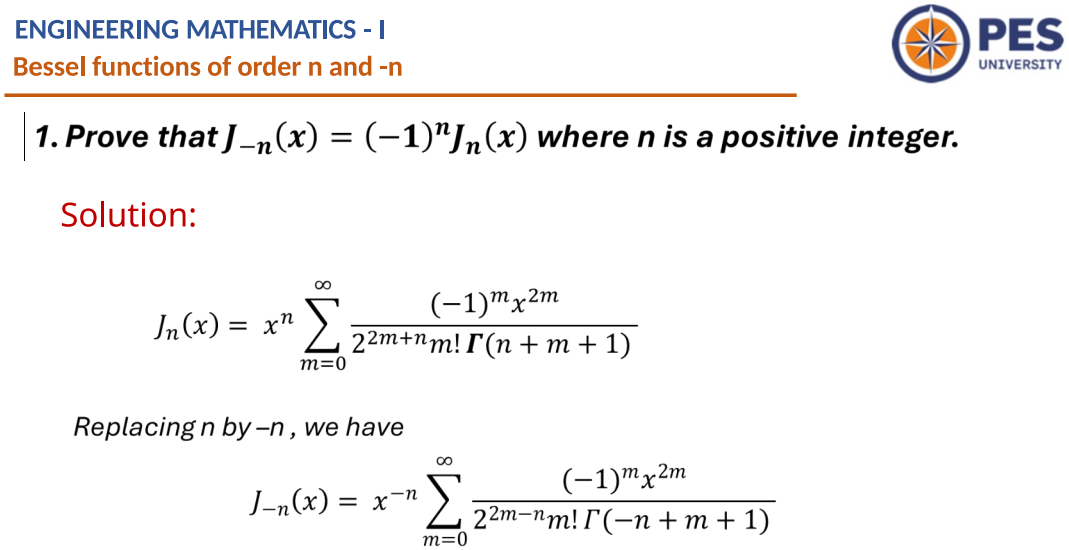


Figure G000134. Bessel functions of order n and n · source slide page 3.

EV0000494

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

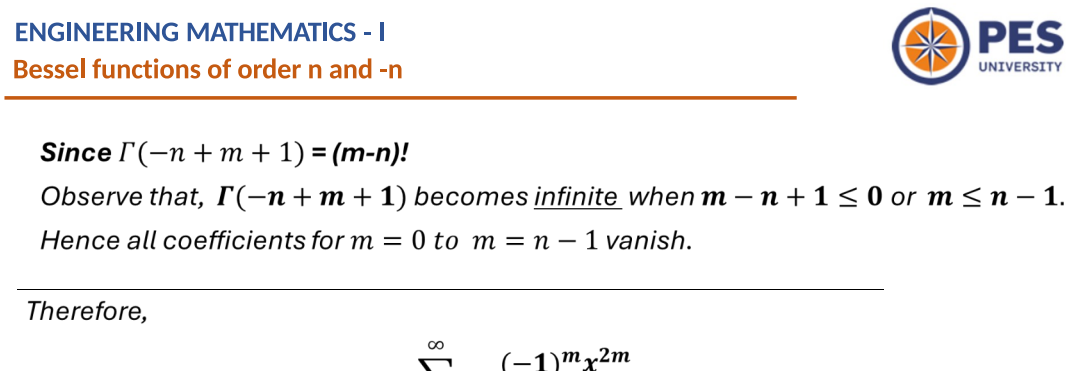


Figure G000135. Bessel functions of order n and n · source slide page 4.

EV0000497

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

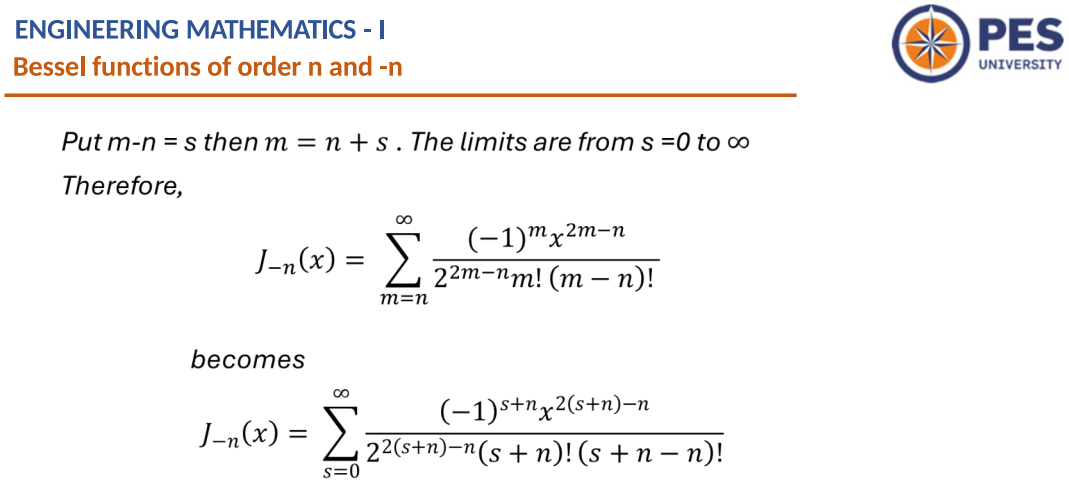


Figure G000136. Bessel functions of order n and n · source slide page 5.

EV0000500

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

Bessel functions of order n and -n

PES UNIVERSITY

$$J_{-n}(x) = \sum_{s=0}^{\infty} \frac{(-1)^{s+n} x^{2(s+n)-n}}{2^{2(s+n)-n}(s+n)!\,(s+n-n)!}$$

$$= (-1)^n \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+n}}{2^{2s+n}(s+n)!\,(s)!}$$

$$= (-1)^n \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+n}}{2^{2s+n}(s)!\Gamma(s+n+1)}$$

$$= (-1)^n J_n(x)$$

$$\boldsymbol{Therefore, J_{-n}(x) = (-1)^n J_n(x)}$$

Figure G000137. Bessel functions of order n and n · source slide page 6.

EV0000503

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PES UNIVERSITY

Thank you

Department of Science and Humanities

Figure G000138. Bessel functions of order n and n · source slide page 7.

EV0000506

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000487 · EV0000488 · EV0000489
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 4: SPECIAL FUNCTIONS AND SERIES SOLUTION OF DE SESSION: 11 SUB-TOPIC: BESSEL FUNCTIONS of order n and -n Department of Science and Humanities EV0000490 · EV0000491 · EV0000492
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Bessel functions of order n and -n Solution: EV0000493 · EV0000494 · EV0000495
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Bessel functions of order n and -n EV0000496 · EV0000497 · EV0000498
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Bessel functions of order n and -n EV0000499 · EV0000500 · EV0000501
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Bessel functions of order n and -n EV0000502 · EV0000503 · EV0000504
7. Source slide page 7 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000505 · EV0000506 · EV0000507

short-note recap

Source deck preserved: Bessel functions of order n and n. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000487

13 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Engineering Mathematics - I, Unit_04 (Unit 4) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000508 · EV0000509

Important points and why

1. Archive page 2 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.5833; source text/OCR may be partial. Class content ☒Series Solution of Bessel's Differential Equation ENGINEERING MATHEMATICS I ☒Bessel differential equation of order n ☒Bessel Function EV0000508

2. Archive page 4 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation 0 t,) e 2 L (r k r r y a x ☒ + = = −−−−− ☒ 0 where, is a positive real number ' are constants and k 0 ra a s ☒ ( )( ) 2 2 2 0 1 (4 an) d r k r r d y a r k r k x dx ☒ + − = = + + − −−−−− ☒ ( ) 1 0 T en, (3 h) r k r r dy a r k x dx ☒ + − = = + −−−−− ☒ ENGINEERING MATHEMATICS I EV0000509

3. Archive page 5 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation Substituting equation and in equati (2), (3) (4) (1) on we get, ( )( ) 2 2 0 1 r k r r x a r k r k x ☒ + − = + + − ☒ + ( ) 1 0 r k r r x a r k x ☒ + − = + ☒ + ( ) 2 2 0 r k r r x n a x ☒ + = − ☒☒ ( )( ) 0 1 r k r r a r k r k x ☒ + = + + − ☒ + ( ) 0 r k r r a r k x ☒ + = + ☒ + 2 0 r k r r a x ☒ + + =☒ ENGINEERING MATHEMATICS I 0 = 2 0 r k r r n a x ☒ + = −☒ 0 = EV0000510

4. Archive page 6 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ☒ ( )☒☒ 0 1 1 r k r r a r k r k x ☒ + = + + −+ ☒ + 2 0 0 r k r r a x ☒ + + = = ☒ ENGINEERING MATHEMATICS I 2 0 r k r r n a x ☒ + = −☒☒ ( ) 2 0 r k r r a r k x ☒ + = + ☒ 2 0 r k r r n a x ☒ + = −☒ + 2 0 0 r k r r a x ☒ + + = = ☒☒ ( ) ☒☒ 2 2 0 r k r r a r k n x ☒ + = + − ☒ + 2 0 0 r k r r a x ☒ + + = = ☒☒ ( ) ☒☒ 2 2 0 r k r r a r k n x ☒ + = + − ☒ + 2 2 0 r k r r a x ☒ + − = = ☒ EV0000511

5. Archive page 7 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ENGINEERING MATHEMATICS I ☒☒☒ 2 2 0 k a k n x − + ( ) ☒☒ 2 2 1 1 1 k a k n x + + − ( ) ☒☒ 2 2 2 r k r r a r k n x ☒ + = + − ☒ + + 2 2 0 r k r r a x ☒ + − = = ☒☒☒☒ 2 2 0 k a k n x − + ( ) ☒☒ 2 2 1 1 1 k a k n x + + − ( ) ☒☒ 2 2 2 2 r k r r r a r k n a x ☒ + − = ☒☒ + − + ☒☒☒ + 0 = Equating the co-efficients of to zero, we , get kx 2 2 0( ) 0 a k n − = 2 2 0 k n ☒ − = ☒ ( ) 0 0 a ☒ k n = ☒ 1 Equating the co-efficient s of to zero, kx + we get, ( ) ( ) 2 2 1 1 0 a k n + − = ☒ By equating the higher x powers of to zero, we get, ( ) 2 2 2 r r a a r k n − = − + − 1 0 a = (5) −−−−−−−− for 2,3,4,5..... r = EV0000512

6. Archive page 8 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ENGINEERING MATHEMATICS I Let us consider the two cases: case ( ) : i k n = Put in equation 5 ( ) k n = Then we get, ( ) 2 2 2 r r a a r k n − ☒☒ = − ☒☒☒☒ + − ☒☒ (6) −−−−−−− 2,3,4,..... Put in (6) we . , . get r = 0 2 2 2 4 a a n = − + (7) −−−−−−−− ( ) 0 2 2 1 a n = − + ( ) 0 2 2 1 a n = − + 2 a ☒ 2 2 2 r r a a r nr − = − + EV0000513

7. Archive page 9 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ENGINEERING MATHEMATICS I 2 4 2 4 8 a a n = − + 1 3 2 3 6 a a n = − + 0 = ( ) 1 0 a = 2 2 we have 2 , r r a a r nr − ☒☒ = − ☒☒ + ☒☒ 0 4 32(1)(2) a a n n ☒ = + + ( ) 0 2 1 8 2 2 (1) a n n ☒☒ = − − ☒☒ + + ☒☒ (8) −−−−−−−− 0 2 2 2 (1) a a n ☒☒ = − ☒☒ + ☒☒ EV0000514

8. Archive page 10 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation 3 5 2 5 10 a a n = − + ENGINEERING MATHEMATICS I 0 = . . . ' substituting s (2), in weget, ra 1 2 3 4 0 1 2 3 4 ..... k k k k k y a x a x a x a x a x + + + + = + + + + We have, by (2), 1 0 n y a x = 3 0 n x + + ☒ 1 0 n x + + ☒ ( ) 2 0 4 1 n a x n + − + 4 0 32(1)(2) n a x n n + + + + ... + EV0000515

9. Archive page 11 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ENGINEERING MATHEMATICS I i.e. ( ) 2 4 1 0 1 1 ..... 4 1 32(1)(2) n n n y a x x x n n n + + ☒ ☒ ☒ ☒ = − + + ☒ ☒ + + + ☒ ☒ ☒ ☒ To standardize the result ,let us take, 0 1 2 (1) n a n = ☒ + then we get , ( ) 2 4 1 1 1 1 ..... 2 (1) 4 1 32(1)(2) n n n n y x x x n n n n + + ☒ ☒ ☒ ☒ = − + + ☒ ☒ ☒ + + + + ☒ ☒ ☒ ☒ ( ) 2 4 1 1 1 ..... 2 (1) 4 1 2 (1) 32(1)(2)2 (1) n n n n n n x x x n n n n n n + + = − + + ☒ + + ☒ + + + ☒ + EV0000516

10. Archive page 12 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ( ) ( ) 2 1 0 1 1 2 1 ! n r r r x y n r r + ☒ = ☒ ☒ = − ☒ ☒ ☒ + + ☒ ☒ ☒ 1 1 2 (1) n n y x n = ☒ + ENGINEERING MATHEMATICS I 2 4 2 4 1 1 1 1 .... 2 (1) 2 (2) 2 2 (3) n n n n n n n x x x n n n + + + + = − + + ☒ + ☒ + ☒ + ( ) 0 1 (1) 2 n x n − ☒ ☒ = ☒ ☒ ☒ + ☒ ☒ ( ) 0 2 0 1 0! (1) 2 n x n + ☒ − ☒ ☒ = ☒ ☒ ☒ + ☒ ☒ (9) −−−−−− 2 2 1 2 2 (2) n n x n + − ☒ ☒ + 4 4 1 2 2 2 (3) n n x n + + ☒ ☒ ☒ + ... + ( ) 1 2 1 (2) 2 n x n + − ☒ ☒ + ☒ ☒ ☒ + ☒ ☒ ( ) 2 4 1 1 2 (3) 2 n x n + − ☒ ☒ + ☒ ☒ ☒ + ☒ ☒ .... + ( ) 1 2 1 1 1! (2) 2 n x n + ☒ − ☒ ☒ + ☒ ☒ ☒ + ☒ ☒ ( ) 2 2 2 1 1 2! (3) 2 n x n + ☒ − ☒ ☒ + ☒ ☒ ☒ + ☒ ☒ .... + EV0000517

11. Archive page 13 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Series Solution of Bessel Differential Equation Thus, ☒It is called Bessel function of first kind of order n. ☒Equation (9) Is called power series solution of Bessel differential equation. ☒It is denoted by ( ) n J x ( ) 2 0 1 ( ) ! (1) 2 r n r n r x J x r n r + ☒ = − ☒ ☒ = ☒ ☒ ☒ + + ☒ ☒ ☒ ENGINEERING MATHEMATICS I EV0000518

12. Archive page 14 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation case ( ) : ii k n = − Put in equatio n (5) k n = − ( ) ( ) 2 0 1 (!) 1 2 1 n r r n r x J x n r r −+ ☒ − = ☒ ☒ = − ☒ ☒ ☒−+ + ☒ ☒ ☒ Then we get, ENGINEERING MATHEMATICS I EV0000519

13. Archive page 15 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Thus the complete solution of Bessel differential equation is given by 1 2 ( ) ( ) n n y C J x C J x − = + provided n is a non integer. ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation ( ) 1 2 and are arbitraryconstants C C EV0000520

SHORT-NOTE RECAP

13 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000508

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Gamma functions - definition, graph and properties transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gamma functions - definition, graph and properties transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gamma functions - definition, graph and properties transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Gamma functions - definition, graph and properties transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Definition of Beta function and its properties transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Definition of Beta function and its properties transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Definition of Beta function and its properties transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Definition of Beta function and its properties transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Relation between Beta and Gamma functions, Duplication formula transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Relation between Beta and Gamma functions, Duplication formula transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Relation between Beta and Gamma functions, Duplication formula transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.
Relation between Beta and Gamma functions, Duplication formula transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta an Gamma functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta an Gamma functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta an Gamma functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Beta an Gamma functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Series solution of Besselâ??s differential equation, Bessel functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Series solution of Besselâ??s differential equation, Bessel functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Series solution of Besselâ??s differential equation, Bessel functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Series solution of Besselâ??s differential equation, Bessel functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Gamma Function	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Gamma Function	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Gamma Function (converted PDF; original hash fb0f3bbeb4bc9e91a815c62874ef14de7f58b751a004ad6c54b652bccfab9429)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Beta Function	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Beta Function	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Beta Function (converted PDF; original hash 9d6a1d1f962fe15e6c957ef3379b68b52d9d2a8aa9803bdc9d95ac791fef02dd)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Relation between beta and gamma functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Relation between beta and gamma functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Relation between beta and gamma functions (converted PDF; original hash 11134e6e9a904408513e6f892e1b5a00a4af0ddd2603e89ad1e2912a6055ad27)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on Beta and Gamma Functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems on Beta and Gamma Functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems on Beta and Gamma Functions (converted PDF; original hash 75162fc20072e4d69545d66f5bc6ddc7262b679c4cddd51270d7e317f40f1b81)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on Beta and Gamma Functions Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems on Beta and Gamma Functions Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Problems on Beta and Gamma Functions Continued (converted PDF; original hash ccf0075124a4efd6b73292a676cf072c47bf12005afe28a69b35c36f8a30baa2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: introduction to Series solution	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: introduction to Series solution	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for introduction to Series solution (converted PDF; original hash bb598f809b2c77ec29e57e49ea30074f6b2d59877db910c8835dddd52ec124ab)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Power series Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Power series Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Power series Continued (converted PDF; original hash af23c8f1ad5d56206fe5c548a8779ae8daf6acf2554a18d766e8a359588d9772)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Frobenius Method	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Frobenius Method	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Frobenius Method (converted PDF; original hash d59bdb13f4eaaaeb79bf4f66bf3429b0091d457a548b6aa4396550239ca8365a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Frobenius method Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Frobenius method Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Frobenius method Continued (converted PDF; original hash 8f64f3331f5de673a7f2e637c4c81f4b3b6bcec79f1343e30794d0bbafbd866e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Series solution of Bessel differential equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Series solution of Bessel differential equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Series solution of Bessel differential equation (converted PDF; original hash 76969bce20da67eb569a4e523bf3d9be3e0ec9c57606847ce60e42285d3178b0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Bessel functions of order n and n	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Bessel functions of order n and n	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Bessel functions of order n and n (converted PDF; original hash 8908775f3114e13979f0e25f05c9130615d83c425c6155065017c2168b4a4929)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: Series Solution of Bessel__1GK7DgclHD.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Math 1/Extra/Extra Questions/Series Solution of Bessel__1GK7DgclHD.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV0000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV0000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000014	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000015	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000016	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000017	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000018	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000019	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000020	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000021	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000022	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000023	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000025	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000030	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000032	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000036	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000037	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000038	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000039	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000040	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000041	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000043	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000044	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000046	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000048	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000049	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000050	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000051	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000053	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00014/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00015/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00016/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000057	SRC00017/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000058	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000060	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000065	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000066	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000067	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000068	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000069	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000070	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000071	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000072	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000074	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000075	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000076	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000079	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000080	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000081	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000082	SRC00022/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00023/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000085	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000086	SRC00022/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00023/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000088	SRC00024/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0100. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000089	SRC00025/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000090	SRC00022/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0104. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000091	SRC00023/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0104. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000092	SRC00024/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0104. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000093	SRC00025/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0104. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000094	SRC00026/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000095	SRC00028/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000096	SRC00027/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000097	SRC00026/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000098	SRC00028/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000099	SRC00027/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000100	SRC00026/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000101	SRC00028/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000102	SRC00027/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000103	SRC00026/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000104	SRC00028/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000105	SRC00027/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000106	SRC00026/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000107	SRC00028/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000108	SRC00027/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000109	SRC00026/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000110	SRC00028/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000111	SRC00027/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000112	SRC00026/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000113	SRC00028/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000114	SRC00027/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000115	SRC00026/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000116	SRC00028/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000117	SRC00027/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000118	SRC00026/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000119	SRC00028/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000120	SRC00027/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000121	SRC00026/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000122	SRC00028/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000123	SRC00027/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000124	SRC00026/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000125	SRC00028/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000126	SRC00027/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000127	SRC00026/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000128	SRC00028/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000129	SRC00027/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00001130	SRC00026/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001131	SRC00028/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001132	SRC00027/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV00001133	SRC00026/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001134	SRC00028/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001135	SRC00027/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00001136	SRC00026/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00001137	SRC00028/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001138	SRC00027/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV00001139	SRC00026/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001140	SRC00028/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001141	SRC00027/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV00001142	SRC00029/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001143	SRC00031/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000144	SRC00030/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000145	SRC00029/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000146	SRC00031/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000147	SRC00030/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000148	SRC00029/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000149	SRC00031/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000150	SRC00030/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000151	SRC00029/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000152	SRC00031/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000153	SRC00030/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000154	SRC00029/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000155	SRC00031/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000156	SRC00030/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000157	SRC00029/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00001158	SRC00031/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001159	SRC00030/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00001160	SRC00029/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001161	SRC00031/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001162	SRC00030/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00001163	SRC00029/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001164	SRC00031/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00001 65	SRC00030/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00001 66	SRC00029/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 67	SRC00031/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001 68	SRC00030/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00001 69	SRC00029/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 70	SRC00031/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001 71	SRC00030/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000172	SRC00029/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000173	SRC00031/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000174	SRC00030/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000175	SRC00029/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000176	SRC00031/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000177	SRC00030/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000178	SRC00029/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00001 79	SRC00031/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001 80	SRC00030/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV00001 81	SRC00029/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 82	SRC00031/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001 83	SRC00030/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00001 84	SRC00029/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 85	SRC00031/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000186	SRC00030/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000187	SRC00029/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000188	SRC00031/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000189	SRC00030/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000190	SRC00032/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000191	SRC00034/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000192	SRC00033/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000193	SRC00032/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000194	SRC00034/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000195	SRC00033/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000196	SRC00032/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000197	SRC00034/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000198	SRC00033/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000199	SRC00032/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000200	SRC00034/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000201	SRC00033/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000202	SRC00032/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000203	SRC00034/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000204	SRC00033/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000205	SRC00032/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000206	SRC00034/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000207	SRC00033/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000208	SRC00032/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000209	SRC00034/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000210	SRC00033/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000211	SRC00032/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000212	SRC00034/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000213	SRC00033/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000214	SRC00032/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000215	SRC00034/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000216	SRC00033/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002217	SRC00032/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002218	SRC00034/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002219	SRC00033/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00002220	SRC00032/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002221	SRC00034/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002222	SRC00033/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000223	SRC00035/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000224	SRC00037/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000225	SRC00036/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000226	SRC00035/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000227	SRC00037/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000228	SRC00036/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000229	SRC00035/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000230	SRC00037/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000231	SRC00036/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000232	SRC00035/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000233	SRC00037/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000234	SRC00036/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000235	SRC00035/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000236	SRC00037/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000237	SRC00036/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000238	SRC00035/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000239	SRC00037/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000240	SRC00036/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000241	SRC00035/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000242	SRC00037/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000243	SRC00036/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000244	SRC00035/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000245	SRC00037/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000246	SRC00036/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000247	SRC00035/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000248	SRC00037/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000249	SRC00036/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000250	SRC00035/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000251	SRC00037/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000252	SRC00036/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000253	SRC00035/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000254	SRC00037/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000255	SRC00036/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000256	SRC00035/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000257	SRC00037/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000258	SRC00036/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000259	SRC00035/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000260	SRC00037/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000261	SRC00036/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000262	SRC00035/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000263	SRC00037/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000264	SRC00036/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000265	SRC00035/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000266	SRC00037/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000267	SRC00036/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000268	SRC00035/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000269	SRC00037/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000270	SRC00036/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000271	SRC00035/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000272	SRC00037/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000273	SRC00036/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000274	SRC00035/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000275	SRC00037/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000276	SRC00036/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000277	SRC00035/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000278	SRC00037/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000279	SRC00036/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000280	SRC00035/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000281	SRC00037/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000282	SRC00036/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000283	SRC00035/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000284	SRC00037/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000285	SRC00036/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV0000286	SRC00035/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000287	SRC00037/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000288	SRC00036/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000289	SRC00038/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000290	SRC00040/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000291	SRC00039/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000292	SRC00038/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000293	SRC00040/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000294	SRC00039/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 95	SRC00038/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 96	SRC00040/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 97	SRC00039/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 98	SRC00038/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 99	SRC00040/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 00	SRC00039/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 01	SRC00038/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 02	SRC00040/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 03	SRC00039/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 04	SRC00038/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 05	SRC00040/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 06	SRC00039/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000307	SRC00041/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000308	SRC00043/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000309	SRC00042/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000310	SRC00041/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000311	SRC00043/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000312	SRC00042/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000313	SRC00041/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000314	SRC00043/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000315	SRC00042/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000316	SRC00041/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000317	SRC00043/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000318	SRC00042/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000319	SRC00041/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000320	SRC00043/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000321	SRC00042/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000322	SRC00041/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000323	SRC00043/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000324	SRC00042/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 25	SRC00041/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 26	SRC00043/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 27	SRC00042/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 28	SRC00041/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 29	SRC00043/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 30	SRC00042/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000331	SRC00041/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000332	SRC00043/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000333	SRC00042/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000334	SRC00041/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000335	SRC00043/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000336	SRC00042/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000337	SRC00041/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000338	SRC00043/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000339	SRC00042/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000340	SRC00041/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000341	SRC00043/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000342	SRC00042/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000343	SRC00041/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000344	SRC00043/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000345	SRC00042/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000346	SRC00041/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000347	SRC00043/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000348	SRC00042/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000349	SRC00044/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000350	SRC00046/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000351	SRC00045/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000352	SRC00044/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000353	SRC00046/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000354	SRC00045/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000355	SRC00044/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 56	SRC00046/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 57	SRC00045/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 58	SRC00044/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 59	SRC00046/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 60	SRC00045/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 61	SRC00044/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 62	SRC00046/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000363	SRC00045/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000364	SRC00044/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000365	SRC00046/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000366	SRC00045/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000367	SRC00044/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000368	SRC00046/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000369	SRC00045/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 70	SRC00044/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 71	SRC00046/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 72	SRC00045/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 73	SRC00047/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 74	SRC00049/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 75	SRC00048/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 76	SRC00047/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000377	SRC00049/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000378	SRC00048/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000379	SRC00047/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000380	SRC00049/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000381	SRC00048/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000382	SRC00047/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000383	SRC00049/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000384	SRC00048/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000385	SRC00047/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000386	SRC00049/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000387	SRC00048/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000388	SRC00047/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000389	SRC00049/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000390	SRC00048/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000391	SRC00047/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000392	SRC00049/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000393	SRC00048/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000394	SRC00047/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000395	SRC00049/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000396	SRC00048/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000397	SRC00047/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 98	SRC00049/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 99	SRC00048/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 00	SRC00047/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 01	SRC00049/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 02	SRC00048/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 03	SRC00047/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 04	SRC00049/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000405	SRC00048/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000406	SRC00047/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000407	SRC00049/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000408	SRC00048/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000409	SRC00050/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000410	SRC00052/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000411	SRC00051/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000412	SRC00050/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000413	SRC00052/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000414	SRC00051/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000415	SRC00050/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000416	SRC00052/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000417	SRC00051/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000418	SRC00050/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000419	SRC00052/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000420	SRC00051/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000421	SRC00050/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000422	SRC00052/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000423	SRC00051/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000424	SRC00050/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000425	SRC00052/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000426	SRC00051/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000427	SRC00050/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000428	SRC00052/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000429	SRC00051/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000430	SRC00050/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000431	SRC00052/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000432	SRC00051/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000433	SRC00050/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000434	SRC00052/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000435	SRC00051/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000436	SRC00053/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000437	SRC00055/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000438	SRC00054/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000439	SRC00053/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000440	SRC00055/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000441	SRC00054/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000442	SRC00053/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000443	SRC00055/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000444	SRC00054/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000445	SRC00053/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000446	SRC00055/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000447	SRC00054/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000448	SRC00053/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000449	SRC00055/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000450	SRC00054/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000451	SRC00053/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000452	SRC00055/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000453	SRC00054/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000454	SRC00053/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000455	SRC00055/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000456	SRC00054/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000457	SRC00053/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000458	SRC00055/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000459	SRC00054/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 60	SRC00053/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 61	SRC00055/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 62	SRC00054/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 63	SRC00053/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 64	SRC00055/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 65	SRC00054/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000466	SRC00053/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000467	SRC00055/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000468	SRC00054/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000469	SRC00053/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000470	SRC00055/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000471	SRC00054/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000472	SRC00053/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000473	SRC00055/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000474	SRC00054/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000475	SRC00053/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000476	SRC00055/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000477	SRC00054/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004078	SRC00053/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004079	SRC00055/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004080	SRC00054/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV00004081	SRC00053/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004082	SRC00055/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004083	SRC00054/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000484	SRC00053/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000485	SRC00055/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000486	SRC00054/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000487	SRC00056/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000488	SRC00058/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000489	SRC00057/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000490	SRC00056/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000491	SRC00058/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000492	SRC00057/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000493	SRC00056/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000494	SRC00058/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000495	SRC00057/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 96	SRC00056/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 97	SRC00058/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 98	SRC00057/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 99	SRC00056/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 00	SRC00058/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 01	SRC00057/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000502	SRC00056/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000503	SRC00058/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000504	SRC00057/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000505	SRC00056/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000506	SRC00058/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000507	SRC00057/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000508	SRC00059/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000509	SRC00059/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV0000510	SRC00059/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000511	SRC00059/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000512	SRC00059/q7	covered	Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
EV0000513	SRC00059/q8	covered	Original archive PDF page 8; candidate unit association is text-based and unconfirmed.
EV0000514	SRC00059/q9	covered	Original archive PDF page 9; candidate unit association is text-based and unconfirmed.
EV0000515	SRC00059/q10	covered	Original archive PDF page 10; candidate unit association is text-based and unconfirmed.
EV0000516	SRC00059/q11	covered	Original archive PDF page 11; candidate unit association is text-based and unconfirmed.
EV0000517	SRC00059/q12	covered	Original archive PDF page 12; candidate unit association is text-based and unconfirmed.
EV0000518	SRC00059/q13	covered	Original archive PDF page 13; candidate unit association is text-based and unconfirmed.
EV0000519	SRC00059/q14	covered	Original archive PDF page 14; candidate unit association is text-based and unconfirmed.
EV0000520	SRC00059/q15	covered	Original archive PDF page 15; candidate unit association is text-based and unconfirmed.

C Revision and retrieval practice